Long Island Clustering Workflow v2.0

Corrected and Strengthened — Based on Audit Report

AMS 597 Project Team

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1 Phase 1: Environment and Reproducibility Setup

This file is the main reproducible workflow for the Long Island clustering analysis. It is meant to run from a clean R session and use `li_accidents_clean.csv` as the only project dataset input.

```
required_packages <- c(
  "data.table",
  "dplyr",
  "lubridate",
  "clustMixType",
```

```r
  "dbscan",
  "cluster",
  "fpc",
  "ggplot2",
  "ggrepel",
  "geosphere",
  "factoextra",
  "knitr",
  "rmarkdown"
)

optional_packages <- c("plotly", "htmlwidgets")

missing_packages <- required_packages[
  !vapply(required_packages, requireNamespace, logical(1), quietly = TRUE)
]

if (length(missing_packages) > 0) {
  stop(
    paste(
      "Install these packages before running the full workflow:",
      paste(missing_packages, collapse = ", ")
    )
  )
}

invisible(lapply(required_packages, library, character.only = TRUE))

set.seed(42)

input_path <- "../li_accidents_clean.csv"
output_dir <- "local_artifacts/output"
run_profile <- "consensus_16gb"

if (!run_profile %in% c("local_safe", "consensus_16gb", "full_srs")) {
  stop("run_profile must be one of 'local_safe', 'consensus_16gb', or 'full_srs'.")
}

phase6_validation_mode <- dplyr::case_when(
  run_profile == "full_srs" ~ "full",
  run_profile == "consensus_16gb" ~ "consensus_sampled",
  TRUE ~ "sampled"
)
phase6_validation_sample_n <- dplyr::case_when(
  run_profile == "full_srs" ~ NA_integer_,
  run_profile == "consensus_16gb" ~ 4000L,
  TRUE ~ 5000L
)
phase6_validation_repeats <- dplyr::case_when(
  run_profile == "consensus_16gb" ~ 4L,
  TRUE ~ 1L
)
```

```r
phase6_stability_mode <- dplyr::case_when(
  run_profile == "full_srs" ~ "full",
  run_profile == "consensus_16gb" ~ "consensus_sampled",
  TRUE ~ "sampled"
)
phase6_stability_sample_n <- dplyr::case_when(
  run_profile == "full_srs" ~ NA_integer_,
  run_profile == "consensus_16gb" ~ 2500L,
  TRUE ~ 3000L
)
phase6_stability_repeats <- dplyr::case_when(
  run_profile == "consensus_16gb" ~ 3L,
  TRUE ~ 1L
)
phase6_stability_bootstraps <- dplyr::case_when(
  run_profile == "full_srs" ~ 50L,
  run_profile == "consensus_16gb" ~ 10L,
  TRUE ~ 15L
)
phase9_interactive_mode <- "auto"

interactive_exports_enabled <- dplyr::case_when(
  phase9_interactive_mode == "off" ~ FALSE,
  phase9_interactive_mode == "on" ~ all(
    vapply(optional_packages, requireNamespace, logical(1), quietly = TRUE)
  ),
  phase9_interactive_mode == "auto" ~ all(
    vapply(optional_packages, requireNamespace, logical(1), quietly = TRUE)
  ),
  TRUE ~ FALSE
)

if (!dir.exists(output_dir)) {
  dir.create(output_dir, recursive = TRUE)
}

if (!file.exists(input_path)) {
  stop("Expected input file not found: ", input_path)
}

phase_1_check <- data.frame(
  item = c(
    "Input dataset",
    "Output directory",
    "Seed",
    "Run profile",
    "Phase 6 validation mode",
    "Phase 6 sample size",
    "Phase 6 validation repeats",
    "Phase 6 stability mode",
    "Phase 6 stability sample size",
    "Phase 6 stability repeats",
    "Phase 6 stability bootstraps",
```

```r
    "Phase 9 interactive mode",
    "Working directory"
  ),
  value = c(
    normalizePath(input_path, winslash = "/"),
    normalizePath(output_dir, winslash = "/"),
    "42",
    run_profile,
    phase6_validation_mode,
    ifelse(is.na(phase6_validation_sample_n), "Full data", as.character(phase6_validation_sample_n)),
    as.character(phase6_validation_repeats),
    phase6_stability_mode,
    ifelse(is.na(phase6_stability_sample_n), "Full data", as.character(phase6_stability_sample_n)),
    as.character(phase6_stability_repeats),
    as.character(phase6_stability_bootstraps),
    ifelse(
      interactive_exports_enabled,
      "Enabled",
      ifelse(phase9_interactive_mode == "off", "Off for safe local runs", "Packages missing")
    ),
    normalizePath(".", winslash = "/")
  )
)

knitr::kable(phase_1_check, caption = "Phase 1 setup check")
```

Table 1: Phase 1 setup check

item	value
Input dataset	/Users/survaghasiya/Documents/AMS 597 Statistical computing Project/li_accidents_clean.csv
Output directory	/Users/survaghasiya/Documents/AMS 597 Statistical computing Project/topic3_nyc_clustering/local_artifacts/output
Seed	42
Run profile	consensus_16gb
Phase 6 validation mode	consensus_sampled
Phase 6 sample size	4000
Phase 6 validation repeats	4
Phase 6 stability mode	consensus_sampled
Phase 6 stability sample size	2500
Phase 6 stability repeats	3
Phase 6 stability bootstraps	10
Phase 9 interactive mode	Packages missing
Working directory	/Users/survaghasiya/Documents/AMS 597 Statistical computing Project/topic3_nyc_clustering

```r
df_raw <- data.table::fread(input_path, showProgress = FALSE)

data_overview <- data.frame(
  rows = nrow(df_raw),
  columns = ncol(df_raw),
  source_file = basename(input_path)
)

knitr::kable(data_overview, caption = "Input file loaded successfully")
```

Table 2: Input file loaded successfully

rows	columns	source_file
33375	48	li_accidents_clean.csv

2 Phase 2: Data Ingestion and Raw Profiling

This phase stays read-only. The goal is to confirm that the cleaned Long Island dataset matches the expectations in the execution plan before any feature engineering starts.

```r
expected_rows <- 33375L
expected_cols <- 48L

if (nrow(df_raw) != expected_rows || ncol(df_raw) != expected_cols) {
  stop(
    paste0(
      "Unexpected dataset dimensions. Expected ",
      expected_rows, " rows and ", expected_cols, " columns, got ",
      nrow(df_raw), " rows and ", ncol(df_raw), " columns."
    )
  )
}

phase_2_dimensions <- data.frame(
  metric = c("Rows", "Columns"),
  expected = c(expected_rows, expected_cols),
  observed = c(nrow(df_raw), ncol(df_raw))
)

knitr::kable(phase_2_dimensions, caption = "Phase 2 dimension check")
```

Table 3: Phase 2 dimension check

metric	expected	observed
Rows	33375	33375
Columns	48	48

```r
county_profile <- as.data.table(df_raw)[, .N, by = County][order(match(County, c("Nassau", "Suffolk", "
severity_profile <- as.data.table(df_raw)[, .N, by = Severity][order(as.integer(Severity))]

expected_counties <- data.table(
  County = c("Nassau", "Suffolk", "Queens"),
  expected_n = c(19390L, 13838L, 147L)
)

expected_severity <- data.table(
  Severity = c(1L, 2L, 3L, 4L),
  expected_n = c(161L, 28357L, 3938L, 919L)
)

county_check <- merge(expected_counties, county_profile, by = "County", all.x = TRUE)
severity_check <- merge(
  expected_severity,
  severity_profile[, Severity := as.integer(Severity)],
  by = "Severity",
  all.x = TRUE
)

if (any(county_check$expected_n != county_check$N)) {
  stop("County counts do not match the execution plan expectations.")
}

if (any(severity_check$expected_n != severity_check$N)) {
  stop("Severity counts do not match the execution plan expectations.")
}

fwrite(county_profile, file.path(output_dir, "phase2_county_profile.csv"))
fwrite(severity_profile, file.path(output_dir, "phase2_severity_profile.csv"))

knitr::kable(county_check, caption = "County profile check")
```

Table 4: County profile check

County	expected_n	N
Nassau	19390	19390
Queens	147	147
Suffolk	13838	13838

```r
knitr::kable(severity_check, caption = "Severity profile check")
```

Table 5: Severity profile check

Severity	expected_n	N
1	161	161
2	28357	28357
3	3938	3938
4	919	919

```r
missingness_profile <- data.table(
  column = names(df_raw),
  missing_n = vapply(df_raw, function(column) sum(is.na(column)), integer(1)),
  missing_pct = round(vapply(df_raw, function(column) mean(is.na(column)) * 100, numeric(1)), 2)
)[order(-missing_pct, column)]

missingness_focus <- missingness_profile[
  column %in% c("End_Lat", "End_Lng", "Precipitation(in)", "Wind_Chill(F)")
]

fwrite(missingness_profile, file.path(output_dir, "phase2_missingness_profile.csv"))

knitr::kable(head(missingness_profile, 12), caption = "Top missingness profile")
```

Table 6: Top missingness profile

column	missing_n	missing_pct
End_Lat	13085	39.21
End_Lng	13085	39.21
Precipitation(in)	6724	20.15
Wind_Chill(F)	5011	15.01
Airport_Code	0	0.00
Amenity	0	0.00
Astronomical_Twilight	0	0.00
Bump	0	0.00
City	0	0.00
Civil_Twilight	0	0.00
Country	0	0.00
County	0	0.00

```r
knitr::kable(missingness_focus, caption = "Critical missingness checks")
```

Table 7: Critical missingness checks

column	missing_n	missing_pct
End_Lat	13085	39.21
End_Lng	13085	39.21
Precipitation(in)	6724	20.15
Wind_Chill(F)	5011	15.01

```r
binary_flag_profile <- data.table(
  flag = c("Bump", "Roundabout", "Turning_Loop", "Traffic_Calming", "No_Exit"),
  true_count = vapply(
    df_raw[, c("Bump", "Roundabout", "Turning_Loop", "Traffic_Calming", "No_Exit")],
    function(column) sum(as.logical(column), na.rm = TRUE),
    integer(1)
  )
)
```

```r
binary_flag_profile[, true_pct := round(100 * true_count / nrow(df_raw), 4)]

fwrite(binary_flag_profile, file.path(output_dir, "phase2_binary_flag_profile.csv"))

knitr::kable(binary_flag_profile, caption = "Near-zero binary flag checks")
```

Table 8: Near-zero binary flag checks

flag	true_count	true_pct
Bump	0	0.000
Roundabout	0	0.000
Turning_Loop	0	0.000
Traffic_Calming	4	0.012
No_Exit	5	0.015

```r
duration_values <- as.numeric(df_raw$Duration_Minutes)
distance_values <- as.numeric(df_raw[["Distance(mi)"]])

duration_profile <- data.table(
  metric = c(
    "Median duration",
    "Mean duration",
    "Max duration",
    "Rows above 1 day",
    "Rows above 1 week",
    "99th percentile"
  ),
  value = c(
    median(duration_values, na.rm = TRUE),
    mean(duration_values, na.rm = TRUE),
    max(duration_values, na.rm = TRUE),
    sum(duration_values > 1440, na.rm = TRUE),
    sum(duration_values > 10080, na.rm = TRUE),
    as.numeric(quantile(duration_values, 0.99, na.rm = TRUE))
  )
)

distance_profile <- data.table(
  metric = c("Zero-distance rows", "Zero-distance pct", "Max distance"),
  value = c(
    sum(distance_values == 0, na.rm = TRUE),
    100 * mean(distance_values == 0, na.rm = TRUE),
    max(distance_values, na.rm = TRUE)
  )
)

fwrite(duration_profile, file.path(output_dir, "phase2_duration_profile.csv"))
fwrite(distance_profile, file.path(output_dir, "phase2_distance_profile.csv"))

knitr::kable(duration_profile, caption = "Duration anomaly profile")
```

Table 9: Duration anomaly profile

metric	value
Median duration	7.783333e+01
Mean duration	4.485395e+02
Max duration	2.236406e+06
Rows above 1 day	2.100000e+01
Rows above 1 week	6.000000e+00
99th percentile	4.805410e+02

```
knitr::kable(distance_profile, caption = "Distance profile")
```

Table 10: Distance profile

metric	value
Zero-distance rows	11724.00000
Zero-distance pct	35.12809
Max distance	35.62700

```
wind_direction_profile <- as.data.table(df_raw)[, .N, by = Wind_Direction][order(-N, Wind_Direction)]
weather_profile <- as.data.table(df_raw)[, .N, by = Weather_Condition][order(-N, Weather_Condition)]

wind_focus <- wind_direction_profile[
  Wind_Direction %in% c("N", "North", "S", "South", "E", "East", "W", "West")
]

fwrite(wind_direction_profile, file.path(output_dir, "phase2_wind_direction_profile.csv"))
fwrite(weather_profile, file.path(output_dir, "phase2_weather_profile.csv"))

knitr::kable(wind_focus, caption = "Wind direction duplicate label check")
```

Table 11: Wind direction duplicate label check

Wind_Direction	N
S	2787
N	1930
W	1540
E	836
South	757
West	675
North	625
East	319

```
knitr::kable(head(weather_profile, 12), caption = "Top weather labels")
```

Table 12: Top weather labels

Weather_Condition	N
Fair	10224
Mostly Cloudy	5724
Cloudy	4310
Partly Cloudy	3217
Clear	2288
Light Rain	1930
Overcast	1718
Scattered Clouds	691
Fog	577
Rain	447
Light Snow	358
Fair / Windy	288

```r
profiling_summary <- data.table(
  finding = c(
    "Dimensions match the execution plan",
    "County counts match the Long Island scope",
    "Severity counts match the execution plan",
    "End_Lat and End_Lng remain heavily missing",
    "Precipitation and Wind_Chill remain too sparse for core clustering use",
    "Five binary roadway flags carry no signal",
    "Duration still shows extreme right-tail outliers",
    "Distance includes many zero values",
    "Wind direction labels need normalization",
    "Weather labels still need grouping before clustering"
  ),
  status = c(
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed",
    "Confirmed"
  )
)

fwrite(profiling_summary, file.path(output_dir, "phase2_profiling_summary.csv"))

knitr::kable(profiling_summary, caption = "Phase 2 profiling summary")
```

Table 13: Phase 2 profiling summary

finding	status
Dimensions match the execution plan	Confirmed

finding	status
County counts match the Long Island scope	Confirmed
Severity counts match the execution plan	Confirmed
End_Lat and End_Lng remain heavily missing	Confirmed
Precipitation and Wind_Chill remain too sparse for core clustering use	Confirmed
Five binary roadway flags carry no signal	Confirmed
Duration still shows extreme right-tail outliers	Confirmed
Distance includes many zero values	Confirmed
Wind direction labels need normalization	Confirmed
Weather labels still need grouping before clustering	Confirmed

3 Phase 3: Feature Engineering

This phase derives the analysis-ready fields needed for the clustering inputs while keeping the original cleaned dataset unchanged in `df_raw`.

```
df_phase3 <- as.data.table(copy(df_raw))
```

```
df_phase3[, Start_Time_Parsed := lubridate::ymd_hms(Start_Time, quiet = TRUE)]

df_phase3[, hour := lubridate::hour(Start_Time_Parsed)]
df_phase3[, month := lubridate::month(Start_Time_Parsed)]
df_phase3[, weekday := factor(
  weekdays(as.Date(Start_Time_Parsed)),
  levels = ordered_weekdays,
  ordered = TRUE
)]
df_phase3[, weekend := weekdays(as.Date(Start_Time_Parsed)) %in% c("Saturday", "Sunday")]
df_phase3[, hour_band := dplyr::case_when(
  hour >= 0 & hour <= 5 ~ "Early_AM",
  hour >= 6 & hour <= 9 ~ "AM_Rush",
  hour >= 10 & hour <= 14 ~ "Midday",
  hour >= 15 & hour <= 18 ~ "PM_Rush",
  hour >= 19 & hour <= 21 ~ "Evening",
  hour >= 22 & hour <= 23 ~ "Late_Night",
  TRUE ~ NA_character_
)]
df_phase3[, hour_band := factor(
  hour_band,
  levels = c("Early_AM", "AM_Rush", "Midday", "PM_Rush", "Evening", "Late_Night"),
  ordered = TRUE
)]
df_phase3[, season := dplyr::case_when(
  month %in% c(12L, 1L, 2L) ~ "Winter",
  month %in% c(3L, 4L, 5L) ~ "Spring",
  month %in% c(6L, 7L, 8L) ~ "Summer",
  month %in% c(9L, 10L, 11L) ~ "Fall",
  TRUE ~ NA_character_
)]
df_phase3[, season := factor(
  season,
```

```r
  levels = c("Winter", "Spring", "Summer", "Fall"),
  ordered = TRUE
)]
df_phase3[, rush_hour := hour %in% c(6L, 7L, 8L, 9L, 15L, 16L, 17L, 18L)]

temporal_na_check <- data.table(
  feature = c("Start_Time_Parsed", "hour", "month", "weekday", "weekend", "hour_band", "season", "rush_h
  missing_n = c(
    sum(is.na(df_phase3$Start_Time_Parsed)),
    sum(is.na(df_phase3$hour)),
    sum(is.na(df_phase3$month)),
    sum(is.na(df_phase3$weekday)),
    sum(is.na(df_phase3$weekend)),
    sum(is.na(df_phase3$hour_band)),
    sum(is.na(df_phase3$season)),
    sum(is.na(df_phase3$rush_hour))
  )
)

if (any(temporal_na_check$missing_n > 0)) {
  stop("Phase 3 temporal feature derivation created unexpected missing values.")
}

fwrite(temporal_na_check, file.path(output_dir, "phase3_temporal_check.csv"))

knitr::kable(temporal_na_check, caption = "Phase 3 temporal feature check")
```

Table 14: Phase 3 temporal feature check

feature	missing_n
Start_Time_Parsed	0
hour	0
month	0
weekday	0
weekend	0
hour_band	0
season	0
rush_hour	0

```r
duration_values <- as.numeric(df_phase3$Duration_Minutes)
duration_cap_99 <- as.numeric(quantile(duration_values, 0.99, na.rm = TRUE))
df_phase3[, Duration_cap := pmin(duration_values, duration_cap_99)]
df_phase3[, log_Duration := log1p(Duration_cap)]

duration_engineering_summary <- data.table(
  metric = c("99th percentile cap", "Median log_Duration", "Max log_Duration"),
  value = c(
    duration_cap_99,
    median(df_phase3$log_Duration, na.rm = TRUE),
    max(df_phase3$log_Duration, na.rm = TRUE)
  )
)
```

```r
fwrite(duration_engineering_summary, file.path(output_dir, "phase3_duration_engineering.csv"))

knitr::kable(duration_engineering_summary, caption = "Phase 3 duration treatment")
```

Table 15: Phase 3 duration treatment

metric	value
99th percentile cap	480.541000
Median log_Duration	4.367336
Max log_Duration	6.176991

```r
weather_counts <- as.data.table(df_phase3)[, .N, by = Weather_Condition]
rare_weather <- weather_counts[N < 30, Weather_Condition]

df_phase3[, weather_group := dplyr::case_when(
  Weather_Condition %in% weather_group_map$Clear       ~ "Clear",
  Weather_Condition %in% weather_group_map$Cloudy      ~ "Cloudy",
  Weather_Condition %in% weather_group_map$Rain        ~ "Rain",
  Weather_Condition %in% weather_group_map$Snow_Ice    ~ "Snow_Ice",
  Weather_Condition %in% weather_group_map$Fog_Haze    ~ "Fog_Haze",
  # v2.0: explicit lookup first; grepl catches novel variants not in the map
  Weather_Condition %in% weather_group_map$Thunderstorm ~ "Thunderstorm",
  grepl("Thunder", Weather_Condition, ignore.case = TRUE) ~ "Thunderstorm",
  Weather_Condition %in% rare_weather ~ "Other",
  TRUE ~ "Other"
)]

df_phase3[, weather_group := factor(
  weather_group,
  levels = c("Clear", "Cloudy", "Rain", "Snow_Ice", "Fog_Haze", "Thunderstorm", "Other")
)]

weather_group_profile <- as.data.table(df_phase3)[, .N, by = weather_group][order(-N)]
fwrite(weather_group_profile, file.path(output_dir, "phase3_weather_group_profile.csv"))

knitr::kable(weather_group_profile, caption = "Phase 3 weather group profile")
```

Table 16: Phase 3 weather group profile

weather_group	N
Cloudy	16193
Clear	12800
Rain	2849
Fog_Haze	624
Snow_Ice	483
Other	267
Thunderstorm	159

```r
df_phase3[, wind_dir_norm := normalize_wind_direction(Wind_Direction)]

wind_direction_summary <- as.data.table(df_phase3)[, .N, by = wind_dir_norm][order(-N)]

if (length(levels(df_phase3$wind_dir_norm)) > 9) {
  stop("Phase 3 wind direction normalization produced too many levels.")
}

fwrite(wind_direction_summary, file.path(output_dir, "phase3_wind_direction_normalized.csv"))

knitr::kable(wind_direction_summary, caption = "Phase 3 normalized wind direction profile")
```

Table 17: Phase 3 normalized wind direction profile

wind_dir_norm	N
Variable_Calm	16225
S	3544
N	2555
NW	2414
SW	2386
W	2215
NE	1880
E	1155
SE	1001

```r
zero_variance_flags <- c("Bump", "Roundabout", "Turning_Loop", "Traffic_Calming", "No_Exit")
near_zero_flags <- c("Give_Way", "Station", "Railway", "Amenity")
retained_flags <- c("Crossing", "Junction", "Traffic_Signal", "Stop")

binary_audit <- data.table(
  flag = c(zero_variance_flags, near_zero_flags, retained_flags),
  true_count = vapply(
    df_phase3[, c(zero_variance_flags, near_zero_flags, retained_flags), with = FALSE],
    function(column) sum(as.logical(column), na.rm = TRUE),
    integer(1)
  )
)

binary_audit[, true_pct := round(100 * true_count / nrow(df_phase3), 4)]
binary_audit[, decision := c(
  rep("Exclude: zero variance", length(zero_variance_flags)),
  rep("Exclude: near-zero variance", length(near_zero_flags)),
  rep("Retain for scenario matrix", length(retained_flags))
)]

for (flag_name in retained_flags) {
  # v2.0: use native FALSE/TRUE factor levels (no string labels)
  df_phase3[, (flag_name) := factor(
    as.logical(get(flag_name)),
    levels = c(FALSE, TRUE)
  )]
}
```

```r
fwrite(binary_audit, file.path(output_dir, "phase3_binary_audit.csv"))

knitr::kable(binary_audit, caption = "Phase 3 binary flag audit")
```

Table 18: Phase 3 binary flag audit

flag	true_count	true_pct	decision
Bump	0	0.0000	Exclude: zero variance
Roundabout	0	0.0000	Exclude: zero variance
Turning_Loop	0	0.0000	Exclude: zero variance
Traffic_Calming	4	0.0120	Exclude: zero variance
No_Exit	5	0.0150	Exclude: zero variance
Give_Way	304	0.9109	Exclude: near-zero variance
Station	233	0.6981	Exclude: near-zero variance
Railway	50	0.1498	Exclude: near-zero variance
Amenity	112	0.3356	Exclude: near-zero variance
Crossing	3374	10.1094	Retain for scenario matrix
Junction	5332	15.9760	Retain for scenario matrix
Traffic_Signal	3186	9.5461	Retain for scenario matrix
Stop	347	1.0397	Retain for scenario matrix

```r
phase_3_summary <- data.table(
  feature_group = c(
    "Temporal features",
    "Duration treatment",
    "Weather grouping",
    "Wind normalization",
    "Binary flag audit"
  ),
  status = c(
    "Added hour, hour_band, weekday, weekend, month, season, rush_hour",
    "Added Duration_cap and log_Duration",
    "Added weather_group with 7 levels",
    "Added wind_dir_norm with at most 9 levels",
    "Marked zero and near-zero flags for exclusion; retained core road-context flags"
  )
)

fwrite(phase_3_summary, file.path(output_dir, "phase3_summary.csv"))

knitr::kable(phase_3_summary, caption = "Phase 3 feature engineering summary")
```

Table 19: Phase 3 feature engineering summary

feature_group	status
Temporal features	Added hour, hour_band, weekday, weekend, month, season, rush_hour
Duration treatment	Added Duration_cap and log_Duration
Weather grouping	Added weather_group with 7 levels
Wind normalization	Added wind_dir_norm with at most 9 levels
Binary flag audit	Marked zero and near-zero flags for exclusion; retained core road-context flags

feature_group	status

4 Phase 4: Build Clustering Input Matrices

This phase constructs the final matrices that will be used by the clustering methods. Nothing outside these matrices should enter the clustering algorithms.

```
df_scenario_base <- as.data.table(copy(df_phase3))

# v2.0: native FALSE/TRUE levels, no string labels
df_scenario_base[, weekend   := factor(weekend,   levels = c(FALSE, TRUE))]
df_scenario_base[, rush_hour := factor(rush_hour, levels = c(FALSE, TRUE))]
df_scenario_base[, Sunrise_Sunset := factor(Sunrise_Sunset)]

scenario_factor_cols <- c(
  "hour_band", "weekday", "weekend", "season", "rush_hour",
  "Sunrise_Sunset", "weather_group", "Crossing", "Junction",
  "Traffic_Signal", "Stop"
)

scenario_numeric_cols <- c(
  "Temperature(F)", "Humidity(%)", "Pressure(in)",
  "Visibility(mi)", "Wind_Speed(mph)"
)

if (mean(is.na(df_scenario_base$log_Duration)) < 0.05) {
  scenario_numeric_cols <- c(scenario_numeric_cols, "log_Duration")
}

scenario_columns <- c(scenario_factor_cols, scenario_numeric_cols)

df_scenario <- df_scenario_base[, ..scenario_columns]

excluded_columns <- c(
  "Severity", "ID", "Source", "Description", "Street", "City", "Zipcode", "nzip",
  "Start_Lat", "Start_Lng", "End_Lat", "End_Lng", "County", "Distance(mi)",
  "Precipitation(in)", "Wind_Chill(F)", "Bump", "Roundabout", "Turning_Loop",
  "Traffic_Calming", "No_Exit", "Give_Way", "Station", "Railway", "Amenity"
)

unexpected_excluded <- intersect(names(df_scenario), excluded_columns)
if (length(unexpected_excluded) > 0) {
  stop(
    paste(
      "Excluded columns were found in df_scenario:",
      paste(unexpected_excluded, collapse = ", ")
    )
  )
}

numeric_imputation_log <- data.table(
```

```r
  column = character(),
  missing_before = integer(),
  missing_pct = numeric(),
  imputation = character()
)

for (column_name in scenario_numeric_cols) {
  missing_before <- sum(is.na(df_scenario[[column_name]]))
  if (!missing_before) {
    next
  }

  missing_pct <- 100 * mean(is.na(df_scenario[[column_name]]))
  median_value <- median(df_scenario[[column_name]], na.rm = TRUE)
  df_scenario[is.na(get(column_name)), (column_name) := median_value]

  numeric_imputation_log <- rbind(
    numeric_imputation_log,
    data.table(
      column = column_name,
      missing_before = missing_before,
      missing_pct = round(missing_pct, 4),
      imputation = "Median"
    )
  )
}

factor_imputation_log <- data.table(
  column = character(),
  missing_before = integer(),
  rule_applied = character()
)

for (column_name in scenario_factor_cols) {
  missing_before <- sum(is.na(df_scenario[[column_name]]))
  if (!missing_before) {
    next
  }

  if (missing_before < 10) {
    fill_value <- mode_value(df_scenario[[column_name]])
    df_scenario[is.na(get(column_name)), (column_name) := fill_value]
    rule_applied <- "Mode"
  } else {
    df_scenario[[column_name]] <- addNA(df_scenario[[column_name]])
    levels(df_scenario[[column_name]])[is.na(levels(df_scenario[[column_name]]))] <- "Unknown"
    rule_applied <- "Unknown level"
  }

  factor_imputation_log <- rbind(
    factor_imputation_log,
    data.table(
      column = column_name,
```

```r
      missing_before = missing_before,
      rule_applied = rule_applied
    )
  )
}

scenario_na_total <- sum(is.na(df_scenario))
if (scenario_na_total > 0) {
  stop("df_scenario still contains missing values after Phase 4 imputations.")
}

scenario_column_profile <- data.table(
  column = names(df_scenario),
  type = vapply(df_scenario, function(column) class(column)[1], character(1))
)

scenario_matrix_summary <- data.table(
  metric = c("Rows", "Columns", "Factor columns", "Numeric columns", "Total NAs"),
  value = c(
    nrow(df_scenario),
    ncol(df_scenario),
    length(scenario_factor_cols),
    length(scenario_numeric_cols),
    scenario_na_total
  )
)

fwrite(scenario_column_profile, file.path(output_dir, "phase4_scenario_columns.csv"))
fwrite(numeric_imputation_log, file.path(output_dir, "phase4_numeric_imputation_log.csv"))
fwrite(factor_imputation_log, file.path(output_dir, "phase4_factor_imputation_log.csv"))
fwrite(scenario_matrix_summary, file.path(output_dir, "phase4_scenario_matrix_summary.csv"))

knitr::kable(scenario_matrix_summary, caption = "Phase 4 scenario matrix summary")
```

Table 20: Phase 4 scenario matrix summary

metric	value
Rows	33375
Columns	17
Factor columns	11
Numeric columns	6
Total NAs	0

```r
knitr::kable(scenario_column_profile, caption = "Phase 4 scenario column list")
```

Table 21: Phase 4 scenario column list

column	type
hour_band	ordered
weekday	ordered

column	type
weekend	factor
season	ordered
rush_hour	factor
Sunrise_Sunset	factor
weather_group	factor
Crossing	factor
Junction	factor
Traffic_Signal	factor
Stop	factor
Temperature(F)	numeric
Humidity(%)	integer
Pressure(in)	numeric
Visibility(mi)	numeric
Wind_Speed(mph)	numeric
log_Duration	numeric

```r
if (any(is.na(df_phase3$Start_Lat)) || any(is.na(df_phase3$Start_Lng))) {
  stop("Start_Lat or Start_Lng contains missing values, so df_spatial cannot be built.")
}

lat0 <- 40.8
lng0 <- -73.2

df_spatial <- data.table(
  x_km = (df_phase3$Start_Lng - lng0) * 111.32 * cos(lat0 * pi / 180),
  y_km = (df_phase3$Start_Lat - lat0) * 110.574
)

spatial_extent <- data.table(
  metric = c("Rows", "Columns", "x_km min", "x_km max", "x_km span", "y_km min", "y_km max", "y_km span
  value = c(
    nrow(df_spatial),
    ncol(df_spatial),
    min(df_spatial$x_km),
    max(df_spatial$x_km),
    diff(range(df_spatial$x_km)),
    min(df_spatial$y_km),
    max(df_spatial$y_km),
    diff(range(df_spatial$y_km))
  )
)

if (sum(is.na(df_spatial)) > 0) {
  stop("df_spatial contains missing values after coordinate projection.")
}

fwrite(spatial_extent, file.path(output_dir, "phase4_spatial_extent.csv"))

knitr::kable(spatial_extent, caption = "Phase 4 spatial matrix extent")
```

Table 22: Phase 4 spatial matrix extent

metric	value
Rows	33375.00000
Columns	2.00000
x_km min	-46.06598
x_km max	90.43842
x_km span	136.50441
y_km min	-23.82284
y_km max	37.95928
y_km span	61.78212

```
phase_4_summary <- data.table(
  matrix = c("df_scenario", "df_spatial"),
  rows = c(nrow(df_scenario), nrow(df_spatial)),
  columns = c(ncol(df_scenario), ncol(df_spatial)),
  na_count = c(sum(is.na(df_scenario)), sum(is.na(df_spatial)))
)

fwrite(phase_4_summary, file.path(output_dir, "phase4_summary.csv"))

knitr::kable(phase_4_summary, caption = "Phase 4 matrix readiness summary")
```

Table 23: Phase 4 matrix readiness summary

matrix	rows	columns	na_count
df_scenario	33375	17	0
df_spatial	33375	2	0

```
# v2.0: Reduced feature matrix for sensitivity analysis.
# Drops Pressure(in) and Visibility(mi) - both required median imputation.
# Used in Phase 5 and Phase 6 to test whether removing imputed features
# improves silhouette and Jaccard scores.

reduced_numeric_cols <- setdiff(
  scenario_numeric_cols,
  c("Pressure(in)", "Visibility(mi)")
)

df_scenario_reduced <- df_scenario[
  , c(scenario_factor_cols, reduced_numeric_cols), with = FALSE
]

stopifnot(sum(is.na(df_scenario_reduced)) == 0)
stopifnot(
  ncol(df_scenario_reduced) ==
    length(scenario_factor_cols) + length(reduced_numeric_cols)
)

reduced_matrix_summary <- data.table(
  item = c(
```

```r
    "Rows",
    "Factor columns",
    "Numeric columns",
    "Dropped from full matrix"
  ),
  value = c(
    nrow(df_scenario_reduced),
    length(scenario_factor_cols),
    length(reduced_numeric_cols),
    paste(c("Pressure(in)", "Visibility(mi)"), collapse = ", ")
  )
)

fwrite(
  reduced_matrix_summary,
  file.path(output_dir, "phase4_reduced_matrix_summary.csv")
)

knitr::kable(
  reduced_matrix_summary,
  caption = "Phase 4 reduced scenario matrix (dropped imputed features)"
)
```

Table 24: Phase 4 reduced scenario matrix (dropped imputed features)

item	value
Rows	33375
Factor columns	11
Numeric columns	4
Dropped from full matrix	Pressure(in), Visibility(mi)

5 Phase 5: k-Prototypes Scenario Clustering

This phase runs the candidate mixed-type clustering solutions on `df_scenario` and stores the diagnostics needed for later model selection.

```r
scenario_type_check <- data.table(
  column = names(df_scenario),
  class = vapply(df_scenario, function(column) class(column)[1], character(1))
)

factor_check <- scenario_type_check[column %in% scenario_factor_cols]
numeric_check <- scenario_type_check[column %in% scenario_numeric_cols]

if (!all(vapply(df_scenario[, ..scenario_factor_cols], is.factor, logical(1)))) {
  stop("One or more Phase 5 scenario factor columns are not stored as factors.")
}

if (!all(vapply(df_scenario[, ..scenario_numeric_cols], is.numeric, logical(1)))) {
```

```r
  stop("One or more Phase 5 scenario numeric columns are not stored as numeric values.")
}

fwrite(scenario_type_check, file.path(output_dir, "phase5_scenario_type_check.csv"))

knitr::kable(scenario_type_check, caption = "Phase 5 scenario type check")
```

Table 25: Phase 5 scenario type check

column	class
hour_band	ordered
weekday	ordered
weekend	factor
season	ordered
rush_hour	factor
Sunrise_Sunset	factor
weather_group	factor
Crossing	factor
Junction	factor
Traffic_Signal	factor
Stop	factor
Temperature(F)	numeric
Humidity(%)	integer
Pressure(in)	numeric
Visibility(mi)	numeric
Wind_Speed(mph)	numeric
log_Duration	numeric

```r
candidate_k <- 3:8
kp_results <- list()

for (k_value in candidate_k) {
  set.seed(42)
  kp_results[[as.character(k_value)]] <- clustMixType::kproto(
    x = df_scenario,
    k = k_value,
    lambda = NULL,
    nstart = 10,
    type = "gower",
    keep.data = TRUE,
    verbose = FALSE
  )
}
```

```r
cluster_fit_summary <- rbindlist(
  lapply(names(kp_results), function(k_label) {
    model_fit <- kp_results[[k_label]]
    cluster_sizes <- table(model_fit$cluster)

    data.table(
      k = as.integer(k_label),
```

```r
      tot_withinss = model_fit$tot.withinss,
      lambda = if (is.null(model_fit$lambda)) NA_real_ else as.numeric(model_fit$lambda)[1],
      min_cluster_size = min(cluster_sizes),
      max_cluster_size = max(cluster_sizes),
      n_clusters = length(cluster_sizes),
      potentially_unstable = min(cluster_sizes) < 200
    )
  })
)

# v2.0: Add approximate numeric R² (variance explained) to fit summary.
# Note: tot.withinss in kproto includes both numeric and categorical
# components. This R² is computed on the numeric columns only and should
# be labelled "approximate" in any report table.
total_ss_numeric <- sum(
  sapply(scenario_numeric_cols, function(col) {
    x <- df_scenario[[col]]
    sum((x - mean(x, na.rm = TRUE))^2, na.rm = TRUE)
  })
)

cluster_fit_summary[, variance_explained_pct := round(
  100 * (1 - tot_withinss / total_ss_numeric), 2
)]

cluster_size_table <- rbindlist(
  lapply(names(kp_results), function(k_label) {
    model_fit <- kp_results[[k_label]]
    data.table(
      k = as.integer(k_label),
      cluster = names(table(model_fit$cluster)),
      size = as.integer(table(model_fit$cluster))
    )
  })
)

dominance_checks <- rbindlist(
  lapply(names(kp_results), function(k_label) {
    model_fit <- kp_results[[k_label]]
    clustered_data <- copy(df_scenario)
    clustered_data[, cluster := factor(model_fit$cluster)]

    rbindlist(lapply(scenario_factor_cols, function(column_name) {
      clustered_data[, .(
        dominant_share = round(dominant_share(get(column_name)), 4)
      ), by = cluster][
        , `:=`(
          k = as.integer(k_label),
          feature = column_name
        )
      ]
    }), fill = TRUE)
  }),
```

```r
  fill = TRUE
)

dominance_flags <- dominance_checks[
  dominant_share >= 0.9
][order(k, cluster, feature)]

fwrite(cluster_fit_summary, file.path(output_dir, "phase5_kproto_fit_summary.csv"))
fwrite(cluster_size_table, file.path(output_dir, "phase5_kproto_cluster_sizes.csv"))
fwrite(dominance_checks, file.path(output_dir, "phase5_kproto_dominance_checks.csv"))
fwrite(dominance_flags, file.path(output_dir, "phase5_kproto_dominance_flags.csv"))

knitr::kable(cluster_fit_summary, caption = "Phase 5 k-prototypes fit summary")
```

Table 26: Phase 5 k-prototypes fit summary

k	tot_withinss	lambda	min_cluster_size	max_cluster_size	n_clusters	potentially_unstable	variance_explained_pct
3	85386.21	NA	9312	13470	3	FALSE	99.65
4	80677.86	NA	5126	10584	4	FALSE	99.67
5	78654.23	NA	2319	10509	5	FALSE	99.68
6	75186.66	NA	2562	8994	6	FALSE	99.69
7	72539.36	NA	2089	9126	7	FALSE	99.70
8	69535.25	NA	2137	8543	8	FALSE	99.72

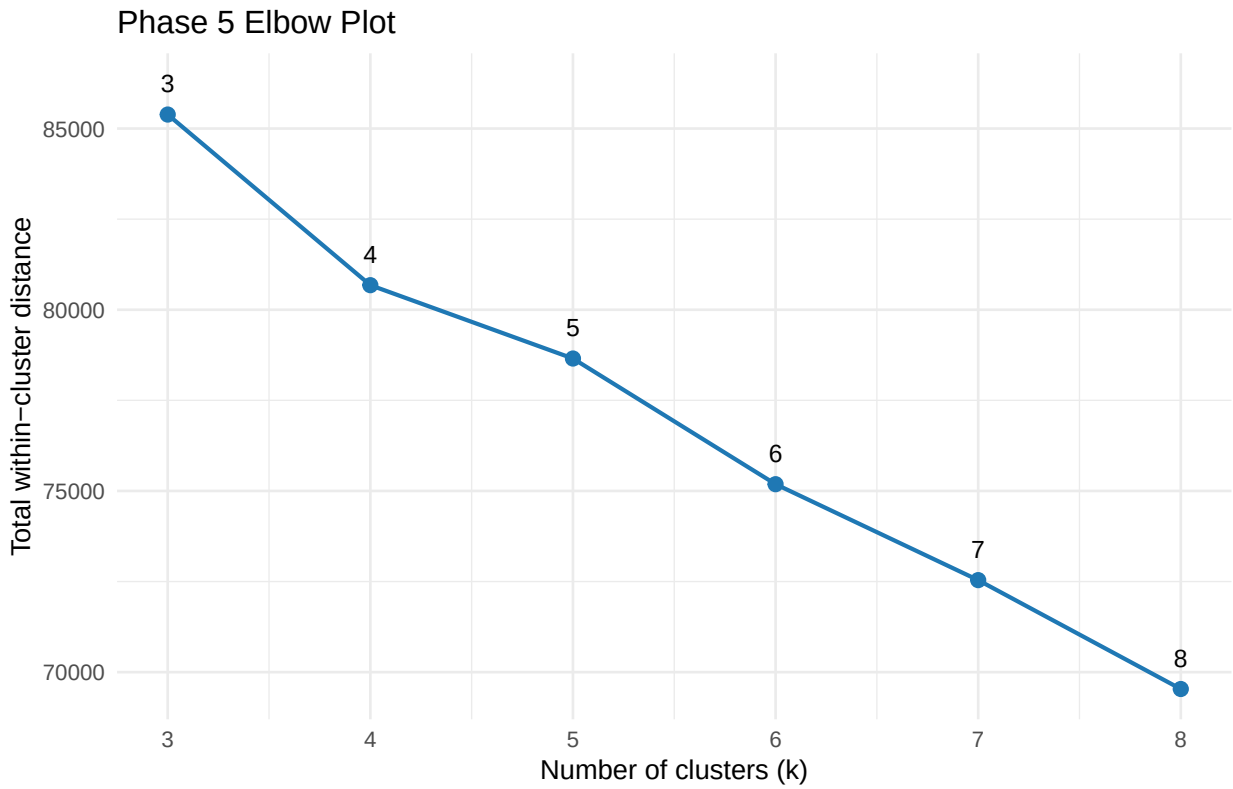
```r
knitr::kable(cluster_size_table, caption = "Phase 5 cluster sizes by k")
```

Table 27: Phase 5 cluster sizes by k

k	cluster	size
3	1	10593
3	2	13470
3	3	9312
4	1	10584
4	2	10353
4	3	7312
4	4	5126
5	1	10389
5	2	3857
5	3	10509
5	4	6301
5	5	2319
6	1	6439
6	2	8994
6	3	3800
6	4	2932
6	5	8648
6	6	2562
7	1	3516
7	2	9126
7	3	6395

k	cluster	size
7	4	6410
7	5	2089
7	6	2803
7	7	3036
8	1	2137
8	2	3566
8	3	4281
8	4	8543
8	5	2293
8	6	4051
8	7	2915
8	8	5589

```r
elbow_plot <- ggplot2::ggplot(
  cluster_fit_summary,
  ggplot2::aes(x = k, y = tot_withinss)
) +
  ggplot2::geom_line(color = "#1f78b4", linewidth = 0.8) +
  ggplot2::geom_point(color = "#1f78b4", size = 2.4) +
  ggplot2::geom_text(
    ggplot2::aes(label = k),
    nudge_y = max(cluster_fit_summary$tot_withinss) * 0.01,
    size = 3.5
  ) +
  ggplot2::labs(
    title = "Phase 5 Elbow Plot",
    x = "Number of clusters (k)",
    y = "Total within-cluster distance"
  ) +
  ggplot2::theme_minimal()

print(elbow_plot)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase5_kproto_elbow_plot.png"),
  plot = elbow_plot,
  width = 7,
  height = 4.5,
  dpi = 300
)
```

```
phase_5_summary <- data.table(
  checkpoint = c(
    "Candidate solutions created",
    "k range covered",
    "Elbow plot saved",
    "Cluster sizes recorded",
    "Dominance checks recorded"
  ),
  status = c(
    length(kp_results) == 6,
    paste(candidate_k, collapse = ", "),
    file.exists(file.path(output_dir, "phase5_kproto_elbow_plot.png")),
    nrow(cluster_size_table) > 0,
    nrow(dominance_checks) > 0
  )
)

fwrite(phase_5_summary, file.path(output_dir, "phase5_summary.csv"))

knitr::kable(phase_5_summary, caption = "Phase 5 readiness summary")
```

Table 28: Phase 5 readiness summary

checkpoint	status
Candidate solutions created	TRUE
k range covered	3, 4, 5, 6, 7, 8
Elbow plot saved	TRUE
Cluster sizes recorded	TRUE
Dominance checks recorded	TRUE

```r
# v2.0: Run k-prototypes on the reduced feature matrix (no Pressure/Visibility).
# Results compared against full-feature runs in Phase 6.

kp_results_reduced <- list()
for (k_val in candidate_k) {
  set.seed(42)
  kp_results_reduced[[as.character(k_val)]] <- clustMixType::kproto(
    x         = df_scenario_reduced,
    k         = k_val,
    lambda    = NULL,
    nstart    = 10,
    type      = "gower",
    keep.data = TRUE,
    verbose   = FALSE
  )
}

reduced_fit_summary <- rbindlist(
  lapply(names(kp_results_reduced), function(k_label) {
    model_fit    <- kp_results_reduced[[k_label]]
    cluster_sizes <- table(model_fit$cluster)
    data.table(
      k             = as.integer(k_label),
      tot_withinss  = model_fit$tot.withinss,
      min_cluster_size = min(cluster_sizes),
      max_cluster_size = max(cluster_sizes),
      feature_set   = "reduced"
    )
  })
)

full_fit_labeled <- copy(
  cluster_fit_summary[, .(k, tot_withinss, min_cluster_size, max_cluster_size)]
)[, feature_set := "full"]

feature_comparison_fits <- rbind(full_fit_labeled, reduced_fit_summary)
setorder(feature_comparison_fits, k, feature_set)

fwrite(
  feature_comparison_fits,
  file.path(output_dir, "phase5_feature_comparison_fits.csv")
```

```
)

knitr::kable(
  feature_comparison_fits,
  caption = "Phase 5: within-cluster distance - full vs reduced feature set"
)
```

Table 29: Phase 5: within-cluster distance — full vs reduced feature set

k	tot_withinss	min_cluster_size	max_cluster_size	feature_set
3	85386.21	9312	13470	full
3	81533.86	9304	13475	reduced
4	80677.86	5126	10584	full
4	76820.19	5113	10932	reduced
5	78654.23	2319	10509	full
5	74681.36	5849	8009	reduced
6	75186.66	2562	8994	full
6	71332.52	2560	9028	reduced
7	72539.36	2089	9126	full
7	68140.15	2026	8326	reduced
8	69535.25	2137	8543	full
8	65781.80	2097	8182	reduced

6 Phase 6: Cluster Validation and Stability

This phase uses internal validation, bootstrap stability, and cluster profile checks to choose a defensible value of `k`.

```
validation_input <- copy(df_scenario)
validation_rows <- nrow(df_scenario)
validation_repeat_results <- NULL

if (identical(phase6_validation_mode, "consensus_sampled")) {
  validation_rows <- min(phase6_validation_sample_n, nrow(df_scenario))
  validation_repeat_results <- rbindlist(
    lapply(seq_len(phase6_validation_repeats), function(repeat_id) {
      set.seed(42 + repeat_id)
      sampled_index <- sort(sample.int(nrow(df_scenario), validation_rows))
      validation_input_repeat <- copy(df_scenario[sampled_index])

      val_repeat <- clustMixType::validation_kproto(
        method = "silhouette",
        data = validation_input_repeat,
        k = candidate_k,
        nstart = 5,
        type = "gower",
        kp_obj = "all",
        verbose = FALSE
      )
```

```
      data.table(
        repeat_id = repeat_id,
        validation_rows = validation_rows,
        repeat_k_opt = as.integer(val_repeat$k_opt),
        k = as.integer(names(val_repeat$indices)),
        silhouette = as.numeric(val_repeat$indices)
      )
    }),
    fill = TRUE
  )

  validation_optimal_counts <- unique(
    validation_repeat_results[, .(repeat_id, repeat_k_opt)]
  )[, .(optimal_repeat_count = .N), by = .(k = repeat_k_opt)]

  validation_table <- validation_repeat_results[, .(
    silhouette = mean(silhouette),
    sd_silhouette = stats::sd(silhouette),
    median_silhouette = stats::median(silhouette)
  ), by = k]

  validation_table <- merge(
    validation_table,
    validation_optimal_counts,
    by = "k",
    all.x = TRUE
  )
  validation_table[is.na(optimal_repeat_count), optimal_repeat_count := 0L]
  validation_table[, optimal_repeat_share := round(optimal_repeat_count / phase6_validation_repeats, 4)]
  validation_table <- validation_table[order(-optimal_repeat_count, -silhouette, k)]
  validation_optimal_k <- validation_table$k[1]
} else {
  if (identical(phase6_validation_mode, "sampled")) {
    set.seed(42)
    validation_rows <- min(phase6_validation_sample_n, nrow(df_scenario))
    sampled_index <- sort(sample.int(nrow(df_scenario), validation_rows))
    validation_input <- copy(df_scenario[sampled_index])
  }

  val_result <- clustMixType::validation_kproto(
    method = "silhouette",
    data = validation_input,
    k = candidate_k,
    nstart = 5,
    type = "gower",
    kp_obj = "all",
    verbose = FALSE
  )

  validation_table <- data.table(
    k = as.integer(names(val_result$indices)),
    silhouette = as.numeric(val_result$indices),
    sd_silhouette = NA_real_,
```

```r
    median_silhouette = as.numeric(val_result$indices),
    optimal_repeat_count = 1L,
    optimal_repeat_share = 1
  )[order(-silhouette, k)]
  validation_optimal_k <- as.integer(val_result$k_opt)
}

validation_metadata <- data.table(
  setting = c("validation_mode", "validation_rows", "full_rows", "sample_fraction", "validation_repeats
  value = c(
    phase6_validation_mode,
    validation_rows,
    nrow(df_scenario),
    round(validation_rows / nrow(df_scenario), 4),
    phase6_validation_repeats
  )
)

top_two_k <- head(validation_table$k, 2)

fwrite(validation_table, file.path(output_dir, "phase6_validation_indices.csv"))
fwrite(validation_metadata, file.path(output_dir, "phase6_validation_metadata.csv"))
if (!is.null(validation_repeat_results)) {
  fwrite(validation_repeat_results, file.path(output_dir, "phase6_validation_repeats.csv"))
}

knitr::kable(validation_table, caption = "Phase 6 internal validation indices")
```

Table 30: Phase 6 internal validation indices

k	silhouette	sd_silhouette	median_silhouette	optimal_repeat_count	optimal_repeat_share
6	0.1569816	0.0192620	0.1631837	2	0.50
3	0.1666649	0.0262799	0.1565771	1	0.25
7	0.1530481	0.0316865	0.1564251	1	0.25
4	0.1548585	0.0051952	0.1561717	0	0.00
8	0.1472494	0.0126613	0.1524335	0	0.00
5	0.1377024	0.0139478	0.1350989	0	0.00

```r
knitr::kable(validation_metadata, caption = "Phase 6 validation run settings")
```

Table 31: Phase 6 validation run settings

setting	value
validation_mode	consensus_sampled
validation_rows	4000
full_rows	33375
sample_fraction	0.1199
validation_repeats	4

```r
if (!is.null(validation_repeat_results)) {
  knitr::kable(validation_repeat_results, caption = "Phase 6 validation repeat detail")
}
```

Table 32: Phase 6 validation repeat detail

repeat_id	validation_rows	repeat_k_opt	k	silhouette
1	4000	6	3	0.1480435
1	4000	6	4	0.1558177
1	4000	6	5	0.1568782
1	4000	6	6	0.1631544
1	4000	6	7	0.1574220
1	4000	6	8	0.1534186
2	4000	6	3	0.1594434
2	4000	6	4	0.1565257
2	4000	6	5	0.1331969
2	4000	6	6	0.1632131
2	4000	6	7	0.1111711
2	4000	6	8	0.1284366
3	4000	3	3	0.2054617
3	4000	3	4	0.1474699
3	4000	3	5	0.1237338
3	4000	3	6	0.1288796
3	4000	3	7	0.1554281
3	4000	3	8	0.1514484
4	4000	7	3	0.1537109
4	4000	7	4	0.1596207
4	4000	7	5	0.1370008
4	4000	7	6	0.1726792
4	4000	7	7	0.1881711
4	4000	7	8	0.1556941

```r
stability_input <- copy(df_scenario)
stability_rows <- nrow(df_scenario)
stability_models <- kp_results[as.character(top_two_k)]
stability_repeat_results <- NULL
stability_bootstrap <- data.table()

if (identical(phase6_stability_mode, "consensus_sampled")) {
  stability_rows <- min(phase6_stability_sample_n, nrow(df_scenario))

  consensus_stability_outputs <- lapply(seq_len(phase6_stability_repeats), function(repeat_id) {
      set.seed(420 + repeat_id)
      stability_index <- sort(sample.int(nrow(df_scenario), stability_rows))
      stability_input_repeat <- copy(df_scenario[stability_index])

      repeat_models <- lapply(top_two_k, function(k_value) {
        set.seed(420 + repeat_id)
        clustMixType::kproto(
          x = stability_input_repeat,
          k = k_value,
          lambda = NULL,
```

```r
        nstart = 5,
        type = "gower",
        keep.data = TRUE,
        verbose = FALSE
      )
    })
    names(repeat_models) <- as.character(top_two_k)

    repeat_stability_outputs <- lapply(top_two_k, function(k_value) {
      stab <- clustMixType::stability_kproto(
        object = repeat_models[[as.character(k_value)]],
        method = c("rand", "jaccard"),
        B = phase6_stability_bootstraps,
        verbose = FALSE,
        nstart = 5
      )

      bootstrap_dt <- as.data.table(stab[[2]])
      bootstrap_dt[, `:=`(k = k_value, repeat_id = repeat_id)]

      list(
        summary_dt = data.table(
          repeat_id = repeat_id,
          k = k_value,
          rand = as.numeric(stab[[1]]["R"]),
          jaccard = as.numeric(stab[[1]]["J"])
        ),
        bootstrap_dt = bootstrap_dt
      )
    })

    list(
      summary_dt = rbindlist(lapply(repeat_stability_outputs, `[[`, "summary_dt"), fill = TRUE),
      bootstrap_dt = rbindlist(lapply(repeat_stability_outputs, `[[`, "bootstrap_dt"), fill = TRUE)
    )
  })

stability_repeat_results <- rbindlist(
  lapply(consensus_stability_outputs, `[[`, "summary_dt"),
  fill = TRUE
)
stability_bootstrap <- rbindlist(
  lapply(consensus_stability_outputs, `[[`, "bootstrap_dt"),
  fill = TRUE
)

stability_summary <- stability_repeat_results[, .(
  rand = mean(rand),
  jaccard = mean(jaccard),
  sd_rand = stats::sd(rand),
  sd_jaccard = stats::sd(jaccard),
  stability_repeats = .N,
  stable_repeat_share = round(mean(jaccard >= 0.6), 4),
```

```r
    strong_repeat_share = round(mean(jaccard >= 0.75), 4),
    stable_jaccard = mean(jaccard >= 0.6) >= 0.5,
    strong_jaccard = mean(jaccard >= 0.75) >= 0.5
  ), by = k][order(-jaccard, -rand, k)]
} else {
  if (identical(phase6_stability_mode, "sampled")) {
    set.seed(42)
    stability_rows <- min(phase6_stability_sample_n, nrow(df_scenario))
    stability_index <- sort(sample.int(nrow(df_scenario), stability_rows))
    stability_input <- copy(df_scenario[stability_index])

    stability_models <- lapply(top_two_k, function(k_value) {
      set.seed(42)
      clustMixType::kproto(
        x = stability_input,
        k = k_value,
        lambda = NULL,
        nstart = 5,
        type = "gower",
        keep.data = TRUE,
        verbose = FALSE
      )
    })
    names(stability_models) <- as.character(top_two_k)
  }

  stability_results <- lapply(top_two_k, function(k_value) {
    set.seed(42)
    stab <- clustMixType::stability_kproto(
      object = stability_models[[as.character(k_value)]],
      method = c("rand", "jaccard"),
      B = phase6_stability_bootstraps,
      verbose = FALSE,
      nstart = 5
    )

    list(
      k = k_value,
      summary = stab[[1]],
      bootstrap = as.data.table(stab[[2]])
    )
  })

  stability_summary <- rbindlist(
    lapply(stability_results, function(result) {
      data.table(
        k = result$k,
        rand = as.numeric(result$summary["R"]),
        jaccard = as.numeric(result$summary["J"]),
        sd_rand = NA_real_,
        sd_jaccard = NA_real_,
        stability_repeats = 1L,
        stable_repeat_share = as.numeric(result$summary["J"]) >= 0.6,
```

```r
        strong_repeat_share = as.numeric(result$summary["J"]) >= 0.75,
        stable_jaccard = as.numeric(result$summary["J"]) >= 0.6,
        strong_jaccard = as.numeric(result$summary["J"]) >= 0.75
      )
    })
  )[order(-jaccard, -rand, k)]

  stability_bootstrap <- rbindlist(
    lapply(stability_results, function(result) {
      bootstrap_dt <- copy(result$bootstrap)
      bootstrap_dt[, `:=`(k = result$k, repeat_id = 1L)]
      bootstrap_dt
    }),
    fill = TRUE
  )
}

stability_metadata <- data.table(
  setting = c("stability_mode", "stability_rows", "full_rows", "sample_fraction", "bootstrap_reps", "sta
  value = c(
    phase6_stability_mode,
    stability_rows,
    nrow(df_scenario),
    round(stability_rows / nrow(df_scenario), 4),
    phase6_stability_bootstraps,
    phase6_stability_repeats
  )
)

fwrite(stability_summary, file.path(output_dir, "phase6_stability_summary.csv"))
fwrite(stability_bootstrap, file.path(output_dir, "phase6_stability_bootstrap.csv"))
fwrite(stability_metadata, file.path(output_dir, "phase6_stability_metadata.csv"))
if (!is.null(stability_repeat_results)) {
  fwrite(stability_repeat_results, file.path(output_dir, "phase6_stability_repeats.csv"))
}

knitr::kable(stability_summary, caption = "Phase 6 stability summary")
```

Table 33: Phase 6 stability summary

k	rand	jaccard	sd_rand	sd_jaccard	stability_repeats	stable_repeat_share	strong_repeat_share	stable_jaccard	strong_jaccard
3	0.578070	0.249628	0.017966	0.017403	3	0	0	FALSE	FALSE
6	0.745007	0.172025	0.014269	0.005329	3	0	0	FALSE	FALSE

```r
knitr::kable(stability_metadata, caption = "Phase 6 stability run settings")
```

Table 34: Phase 6 stability run settings

setting	value
stability_mode	consensus_sampled

setting	value
stability_rows	2500
full_rows	33375
sample_fraction	0.0749
bootstrap_reps	10
stability_repeats	3

```r
if (!is.null(stability_repeat_results)) {
  knitr::kable(stability_repeat_results, caption = "Phase 6 stability repeat detail")
}
```

Table 35: Phase 6 stability repeat detail

repeat_id	k	rand	jaccard
1	6	0.7287476	0.1659738
1	3	0.5612865	0.2305071
2	6	0.7554483	0.1760175
2	3	0.5970218	0.2538351
3	6	0.7508265	0.1740848
3	3	0.5759017	0.2645418

```r
# v2.0: Compute Jaccard mean, SD, and range across bootstrap repeats.
# Under consensus_16gb (3 repeats) this gives a variance estimate.
# Under local_safe (1 repeat) this falls back gracefully.
if (!is.null(stability_repeat_results) &&
    nrow(stability_repeat_results) > 0 &&
    "jaccard" %in% names(stability_repeat_results)) {

  jaccard_ci <- stability_repeat_results[, .(
    jaccard_mean = round(mean(jaccard, na.rm = TRUE), 4),
    jaccard_sd   = round(sd(jaccard,   na.rm = TRUE), 4),
    jaccard_min  = round(min(jaccard,  na.rm = TRUE), 4),
    jaccard_max  = round(max(jaccard,  na.rm = TRUE), 4),
    n_repeats    = .N
  ), by = k]

  fwrite(jaccard_ci, file.path(output_dir, "phase6_jaccard_ci.csv"))
  knitr::kable(
    jaccard_ci,
    caption = "Phase 6 Jaccard stability with variance across bootstrap repeats"
  )
} else {
  cat("Single bootstrap repeat --- Jaccard CI not available under local_safe.\n")
}
```

Table 36: Phase 6 Jaccard stability with variance across bootstrap repeats

k	jaccard_mean	jaccard_sd	jaccard_min	jaccard_max	n_repeats
6	0.1720	0.0053	0.1660	0.1760	3
3	0.2496	0.0174	0.2305	0.2645	3

```r
# v2.0: Verbose k-selection with explicit diagnostics.
# The original silent fallback could select a k without any trace of
# which conditions fired or whether NA extraction caused a bypass.

validation_optimal_k_raw <- validation_table$k[1]
silhouette_at_optimal    <- validation_table$silhouette[1]

cat("=== Phase 6 k-Selection Diagnostics ===\n")
```

```
## === Phase 6 k-Selection Diagnostics ===
```

```r
cat("Silhouette-optimal k        :", validation_optimal_k_raw, "\n")
```

```
## Silhouette-optimal k        : 6
```

```r
cat("Silhouette at optimal k     :", round(silhouette_at_optimal, 4), "\n")
```

```
## Silhouette at optimal k     : 0.157
```

```r
cat("Silhouette threshold (0.30) :",
    ifelse(silhouette_at_optimal >= 0.30, "MET", "NOT MET"), "\n\n")
```

```
## Silhouette threshold (0.30) : NOT MET
```

```r
cat("Stability available for k   :",
    paste(stability_summary$k, collapse = ", "), "\n")
```

```
## Stability available for k   : 3, 6
```

```r
selected_stability <- stability_summary[
  k == validation_optimal_k_raw, jaccard
]

cat("Jaccard at silhouette-optimal k :",
    ifelse(length(selected_stability) == 1 && !is.na(selected_stability),
           round(selected_stability, 4), "NOT FOUND / NA"), "\n")
```

```
## Jaccard at silhouette-optimal k : 0.172
```

```r
cat("Jaccard threshold (0.40)        :",
    ifelse(length(selected_stability) == 1 &&
             !is.na(selected_stability) &&
             selected_stability >= 0.40, "MET", "NOT MET"), "\n\n")
```

```
## Jaccard threshold (0.40)        : NOT MET
```

```r
fallback_triggered <- FALSE
if (length(selected_stability) == 1 &&
    !is.na(selected_stability) &&
    selected_stability < 0.5 &&
    nrow(stability_summary) > 1) {

  alt_k       <- stability_summary[order(-jaccard, -rand, k)][1, k]
  alt_jaccard <- stability_summary[order(-jaccard, -rand, k)][1, jaccard]

  cat("Fallback condition met. Stability-optimal k:", alt_k,
      "| Jaccard:", round(alt_jaccard, 4), "\n")

  if (!is.na(alt_jaccard) && alt_jaccard > selected_stability) {
    selected_k         <- alt_k
    fallback_triggered <- TRUE
    cat("Fallback ACTIVATED: selected_k changed to", selected_k, "\n\n")
  } else {
    selected_k <- validation_optimal_k_raw
    cat("Fallback NOT activated: no better Jaccard available.",
        "Retaining k =", selected_k, "\n\n")
  }
} else {
  selected_k <- validation_optimal_k_raw
  cat("Fallback condition not met. Using silhouette-optimal k =",
      selected_k, "\n\n")
}
```

```
## Fallback condition met. Stability-optimal k: 3 | Jaccard: 0.2496
## Fallback ACTIVATED: selected_k changed to 3
```

```r
cat("FINAL selected_k :", selected_k, "\n")
```

```
## FINAL selected_k : 3
```

```r
cat("==========================================\n")
```

```
## ==========================================
```

```r
selected_model <- kp_results[[as.character(selected_k)]]
selected_jaccard <- if (selected_k %in% stability_summary$k) {
  stability_summary[k == selected_k, round(jaccard, 4)]
} else {
  NA_real_
```

```r
}

# v2.0: Explicit quality gate - fires a warning and builds a status table
# whenever silhouette or Jaccard fall below published thresholds.
selected_silhouette <- validation_table[k == selected_k, silhouette]
selected_jaccard_v2 <- if (selected_k %in% stability_summary$k) {
  stability_summary[k == selected_k, jaccard]
} else {
  NA_real_
}

sil_pass     <- !is.na(selected_silhouette) && selected_silhouette >= 0.30
jaccard_pass <- !is.na(selected_jaccard_v2) && selected_jaccard_v2 >= 0.40

quality_gate <- data.table(
  metric = c("Silhouette width", "Jaccard stability"),
  observed = c(
    round(selected_silhouette, 4),
    round(selected_jaccard_v2, 4)
  ),
  threshold = c(0.30, 0.40),
  pass = c(sil_pass, jaccard_pass),
  interpretation = c(
    ifelse(sil_pass,
           "Acceptable cluster separation",
           "Weak separation - exploratory only"),
    ifelse(jaccard_pass,
           "Acceptable bootstrap stability",
           "Poor stability - exploratory only")
  )
)

fwrite(quality_gate, file.path(output_dir, "phase6_quality_gate.csv"))
knitr::kable(quality_gate, caption = "Phase 6 cluster quality gate (v2.0)")
```

Table 37: Phase 6 cluster quality gate (v2.0)

metric	observed	threshold	pass	interpretation
Silhouette width	0.1667	0.3	FALSE	Weak separation — exploratory only
Jaccard stability	0.2496	0.4	FALSE	Poor stability — exploratory only

```r
if (!sil_pass || !jaccard_pass) {
  warning(paste0(
    "\n\n*** CLUSTER QUALITY WARNING (v2.0) ***\n",
    "  Silhouette = ", round(selected_silhouette, 4),
    " (threshold >= 0.30): ", ifelse(sil_pass, "PASS", "FAIL"), "\n",
    "  Jaccard    = ", round(selected_jaccard_v2, 4),
    " (threshold >= 0.40): ", ifelse(jaccard_pass, "PASS", "FAIL"), "\n",
    "  selected_k = ", selected_k,
    " solution is EXPLORATORY.\n",
    "  Do not present clusters as stable accident archetypes.\n",
    "*************************************************\n"
```

```r
  ))
}

selection_summary <- data.table(
  metric = c(
    "Validation method",
    "Validation mode",
    "Validation rows",
    "Stability mode",
    "Stability rows",
    "Stability bootstraps",
    "Validation-optimal k",
    "Top two k values for stability",
    "Selected k",
    "Selected k Jaccard"
  ),
  value = c(
    "silhouette",
    phase6_validation_mode,
    validation_rows,
    phase6_stability_mode,
    stability_rows,
    phase6_stability_bootstraps,
    validation_optimal_k,
    paste(top_two_k, collapse = ", "),
    selected_k,
    selected_jaccard
  )
)

fwrite(selection_summary, file.path(output_dir, "phase6_selection_summary.csv"))

knitr::kable(selection_summary, caption = "Phase 6 k-selection summary")
```

Table 38: Phase 6 k-selection summary

metric	value
Validation method	silhouette
Validation mode	consensus_sampled
Validation rows	4000
Stability mode	consensus_sampled
Stability rows	2500
Stability bootstraps	10
Validation-optimal k	6
Top two k values for stability	6, 3
Selected k	3
Selected k Jaccard	0.2496

```r
# v2.0: Validate reduced-feature k-prototypes candidates.
# Compares silhouette scores with and without Pressure(in)/Visibility(mi).
# v2.0: Mirror the same sampling strategy used in the main Phase 6 validation
# to stay within memory limits. Uses phase6_validation_sample_n rows.
```

```r
set.seed(42)
reduced_val_rows <- min(phase6_validation_sample_n, nrow(df_scenario_reduced))
reduced_val_input <- if (identical(phase6_validation_mode, "full")) {
  copy(df_scenario_reduced)
} else {
  reduced_sampled_idx <- sort(sample.int(nrow(df_scenario_reduced), reduced_val_rows))
  copy(df_scenario_reduced[reduced_sampled_idx])
}

cat("Reduced-feature validation sample size:", nrow(reduced_val_input), "\n")
```

```
## Reduced-feature validation sample size: 4000
```

```r
validation_reduced_raw <- clustMixType::validation_kproto(
  method  = "silhouette",
  data    = reduced_val_input,
  k       = candidate_k,
  nstart  = 5,
  type    = "gower",
  kp_obj  = "all",
  verbose = FALSE
)

validation_reduced_table <- data.table(
  k          = as.integer(names(validation_reduced_raw$indices)),
  silhouette = as.numeric(validation_reduced_raw$indices)
)
validation_reduced_table[, feature_set := "reduced"]

validation_full_labeled <- copy(
  validation_table[, .(k, silhouette)]
)[, feature_set := "full"]

feature_silhouette_comparison <- rbind(
  validation_full_labeled,
  validation_reduced_table[, .(k, silhouette, feature_set)]
)
setorder(feature_silhouette_comparison, k, feature_set)

fwrite(
  feature_silhouette_comparison,
  file.path(output_dir, "phase6_feature_silhouette_comparison.csv")
)

knitr::kable(
  feature_silhouette_comparison,
  caption = "Phase 6 silhouette: full vs reduced feature set"
)
```

Table 39: Phase 6 silhouette: full vs reduced feature set

k	silhouette	feature_set
3	0.1666649	full
3	0.1693395	reduced
4	0.1548585	full
4	0.1738949	reduced
5	0.1377024	full
5	0.1594099	reduced
6	0.1569816	full
6	0.2029302	reduced
7	0.1530481	full
7	0.1760070	reduced
8	0.1472494	full
8	0.1579754	reduced

```
profile_plot_path <- file.path(output_dir, "phase6_cluster_profiles.pdf")
grDevices::pdf(profile_plot_path, width = 12, height = 8)
clustMixType::clprofiles(selected_model, df_scenario)
grDevices::dev.off()
```

```
## pdf
##   2
```

```
selected_cluster_data <- copy(df_scenario)
selected_cluster_data[, cluster := factor(selected_model$cluster)]

cluster_profile_summary <- selected_cluster_data[, .(
  dominant_hour_band = mode_value(hour_band),
  dominant_weather_group = mode_value(weather_group),
  dominant_weekend = mode_value(weekend),
  dominant_sunrise_sunset = mode_value(Sunrise_Sunset),
  dominant_junction = mode_value(Junction),
  dominant_traffic_signal = mode_value(Traffic_Signal),
  mean_temperature = round(mean(`Temperature(F)`), 2),
  mean_visibility = round(mean(`Visibility(mi)`), 2),
  mean_log_duration = round(mean(log_Duration), 3)
), by = cluster]

cluster_profile_summary[, working_name := dplyr::case_when(
  dominant_weather_group == "Snow_Ice" ~ "Winter Adverse",
  dominant_sunrise_sunset == "Night" & dominant_weather_group %in% c("Fog_Haze", "Rain", "Thunderstorm")
  dominant_weekend == "TRUE" & dominant_traffic_signal == "TRUE" ~ "Weekend Surface Road",
  dominant_hour_band %in% c("AM_Rush", "PM_Rush") & dominant_junction == "TRUE" ~ "Rush Corridor",
  TRUE ~ paste("Scenario Cluster", cluster)
)]

fwrite(cluster_profile_summary, file.path(output_dir, "phase6_cluster_profile_summary.csv"))

knitr::kable(cluster_profile_summary, caption = "Phase 6 cluster profile summary")
```

Table 40: Phase 6 cluster profile summary

cluster	dominant_domain	dominant_weather	dominant_lighting	dominant_surface	dominant_distraction	traffic_signal	temperature	visibility	log_odds	working_name
2	AM_Rush	Cloudy	FALSE	Day	FALSE	FALSE	62.04	8.66	4.330	Scenario Cluster 2
1	AM_Rush	Clear	FALSE	Day	FALSE	FALSE	60.70	9.94	4.421	Scenario Cluster 1
3	Early_AM	Cloudy	FALSE	Night	FALSE	FALSE	49.83	8.64	4.385	Scenario Cluster 3

```r
phase_6_summary <- data.table(
  checkpoint = c(
    "Validation indices saved",
    "Validation metadata saved",
    "Stability summary saved",
    "Stability metadata saved",
    "Selected k assigned",
    "Cluster profile PDF saved",
    "Working names assigned"
  ),
  status = c(
    file.exists(file.path(output_dir, "phase6_validation_indices.csv")),
    file.exists(file.path(output_dir, "phase6_validation_metadata.csv")),
    file.exists(file.path(output_dir, "phase6_stability_summary.csv")),
    file.exists(file.path(output_dir, "phase6_stability_metadata.csv")),
    !is.null(selected_model),
    file.exists(file.path(output_dir, "phase6_cluster_profiles.pdf")),
    all(nzchar(cluster_profile_summary$working_name))
  )
)

fwrite(phase_6_summary, file.path(output_dir, "phase6_summary.csv"))

knitr::kable(phase_6_summary, caption = "Phase 6 readiness summary")
```

Table 41: Phase 6 readiness summary

checkpoint	status
Validation indices saved	TRUE
Validation metadata saved	TRUE
Stability summary saved	TRUE
Stability metadata saved	TRUE
Selected k assigned	TRUE
Cluster profile PDF saved	TRUE
Working names assigned	TRUE

7 Phase 7: DBSCAN Spatial Hotspot Clustering

This phase clusters the projected Long Island coordinates with DBSCAN so the workflow can identify corridor-style hotspot zones.

```
knn_k <- 5L
dbscan_minpts_grid <- if (run_profile == "full_srs") c(15L, 25L, 40L, 50L, 75L, 100L) else c(40L, 50L, 
dbscan_eps_grid <- seq(0.3, 2.0, by = 0.1)

knn_dist <- dbscan::kNNdist(df_spatial, k = knn_k)
knn_sorted <- sort(knn_dist)
phase7_total_rows <- nrow(df_spatial)
phase7_target_cluster_n <- 8L
phase7_target_noise_pct <- 17.5
phase7_max_cluster_share_cap <- 60

knn_profile <- data.table(
  quantile = c("50%", "75%", "90%", "95%", "97.5%", "99%"),
  value_km = round(as.numeric(quantile(knn_dist, probs = c(0.5, 0.75, 0.9, 0.95, 0.975, 0.99))), 4)
)

eps_trials <- rbindlist(
  lapply(dbscan_minpts_grid, function(minPts_value) {
    rbindlist(lapply(dbscan_eps_grid, function(eps_value) {
      db_trial <- dbscan::dbscan(df_spatial, eps = eps_value, minPts = minPts_value)
      cluster_counts <- table(db_trial$cluster)
      non_noise_sizes <- cluster_counts[names(cluster_counts) != "0"]
      non_noise_clusters <- sum(as.integer(names(cluster_counts)) > 0)
      noise_pct <- round(100 * mean(db_trial$cluster == 0), 2)

      data.table(
        eps = round(eps_value, 2),
        minPts = minPts_value,
        non_noise_clusters = non_noise_clusters,
        noise_pct = noise_pct,
        max_cluster_size = if (length(non_noise_sizes)) max(as.integer(non_noise_sizes)) else 0L,
        max_cluster_share_pct = if (length(non_noise_sizes)) round(100 * max(as.integer(non_noise_sizes
        size_balance_ok = if (length(non_noise_sizes)) round(100 * max(as.integer(non_noise_sizes)) / p
        acceptable = non_noise_clusters >= 3 & non_noise_clusters <= 15 & noise_pct >= 5 & noise_pct <=
          if (length(non_noise_sizes)) round(100 * max(as.integer(non_noise_sizes)) / phase7_total_rows
      )
    }))
  }),
  fill = TRUE
)

acceptable_trials <- eps_trials[acceptable == TRUE]

if (nrow(acceptable_trials) > 0) {
  selected_trial <- acceptable_trials[
    order(abs(non_noise_clusters - phase7_target_cluster_n), abs(noise_pct - phase7_target_noise_pct), 
  ][1]
  selection_reason <- "Accepted trial balancing cluster count, noise level, and hotspot size concentrati
} else {
```

```r
  selected_trial <- eps_trials[
    order(abs(non_noise_clusters - phase7_target_cluster_n), abs(noise_pct - phase7_target_noise_pct), n
  ][1]
  selection_reason <- "Fallback trial closest to balanced cluster count, noise level, and hotspot size c
}

eps_chosen <- selected_trial$eps[[1]]
minPts_chosen <- selected_trial$minPts[[1]]

parameter_summary <- data.table(
  metric = c(
    "kNN distance k",
    "DBSCAN minPts grid",
    "DBSCAN eps grid",
    "Largest cluster share cap (%)",
    "DBSCAN minPts",
    "Chosen eps (km)",
    "Chosen non-noise clusters",
    "Chosen noise pct",
    "Chosen largest cluster share pct",
    "Selection reason"
  ),
  value = c(
    knn_k,
    paste(dbscan_minpts_grid, collapse = ", "),
    paste(range(dbscan_eps_grid), collapse = " to "),
    phase7_max_cluster_share_cap,
    minPts_chosen,
    eps_chosen,
    selected_trial$non_noise_clusters[[1]],
    selected_trial$noise_pct[[1]],
    selected_trial$max_cluster_share_pct[[1]],
    selection_reason
  )
)

fwrite(knn_profile, file.path(output_dir, "phase7_knn_profile.csv"))
fwrite(eps_trials, file.path(output_dir, "phase7_eps_trials.csv"))
fwrite(parameter_summary, file.path(output_dir, "phase7_parameter_summary.csv"))

knitr::kable(knn_profile, caption = "Phase 7 kNN distance profile")
```

Table 42: Phase 7 kNN distance profile

quantile	value_km
50%	0.0001
75%	0.0288
90%	0.1486
95%	0.4177
97.5%	0.7585
99%	1.2474

```r
knitr::kable(eps_trials, caption = "Phase 7 DBSCAN eps trials")
```

Table 43: Phase 7 DBSCAN eps trials

eps	minPts	non_noise_clusters	noise_pct	max_cluster_size	max_cluster_share_pct	size_balance_ok	acceptable
0.3	40	79	20.17	6071	18.19	TRUE	FALSE
0.4	40	58	16.82	8353	25.03	TRUE	FALSE
0.5	40	50	14.41	12841	38.47	TRUE	FALSE
0.6	40	47	11.89	14149	42.39	TRUE	FALSE
0.7	40	45	10.33	14376	43.07	TRUE	FALSE
0.8	40	36	9.22	15020	45.00	TRUE	FALSE
0.9	40	29	7.68	26956	80.77	FALSE	FALSE
1.0	40	22	6.85	29033	86.99	FALSE	FALSE
1.1	40	17	6.10	29732	89.08	FALSE	FALSE
1.2	40	16	5.39	30130	90.28	FALSE	FALSE
1.3	40	15	4.57	30512	91.42	FALSE	FALSE
1.4	40	13	3.88	30916	92.63	FALSE	FALSE
1.5	40	10	3.57	31294	93.76	FALSE	FALSE
1.6	40	10	3.10	31687	94.94	FALSE	FALSE
1.7	40	10	2.76	31768	95.19	FALSE	FALSE
1.8	40	13	2.25	31843	95.41	FALSE	FALSE
1.9	40	13	2.00	31901	95.58	FALSE	FALSE
2.0	40	10	1.60	32042	96.01	FALSE	FALSE
0.3	50	75	23.35	5545	16.61	TRUE	FALSE
0.4	50	59	18.98	8337	24.98	TRUE	FALSE
0.5	50	40	16.82	9664	28.96	TRUE	FALSE
0.6	50	35	14.86	13754	41.21	TRUE	FALSE
0.7	50	30	13.39	14316	42.89	TRUE	FALSE
0.8	50	30	11.07	14970	44.85	TRUE	FALSE
0.9	50	23	9.77	15257	45.71	TRUE	FALSE
1.0	50	20	8.25	27280	81.74	FALSE	FALSE
1.1	50	20	6.97	29138	87.30	FALSE	FALSE
1.2	50	16	6.19	29715	89.03	FALSE	FALSE
1.3	50	13	5.48	30203	90.50	FALSE	FALSE
1.4	50	11	5.02	30530	91.48	FALSE	FALSE
1.5	50	8	4.32	31027	92.96	FALSE	FALSE
1.6	50	9	3.87	31126	93.26	FALSE	FALSE
1.7	50	8	3.36	31556	94.55	FALSE	FALSE
1.8	50	8	3.01	31787	95.24	FALSE	FALSE
1.9	50	8	2.67	31861	95.46	FALSE	FALSE
2.0	50	8	2.40	31915	95.63	FALSE	FALSE
0.3	75	68	29.56	3145	9.42	TRUE	FALSE
0.4	75	45	24.81	7339	21.99	TRUE	FALSE
0.5	75	39	20.89	7943	23.80	TRUE	FALSE
0.6	75	25	19.36	8652	25.92	TRUE	FALSE
0.7	75	23	16.75	13864	41.54	TRUE	FALSE
0.8	75	20	15.32	14496	43.43	TRUE	FALSE
0.9	75	15	14.25	14750	44.19	TRUE	TRUE
1.0	75	13	13.02	15074	45.17	TRUE	TRUE
1.1	75	13	11.45	26762	80.19	FALSE	FALSE
1.2	75	11	9.42	27644	82.83	FALSE	FALSE
1.3	75	11	8.20	29582	88.64	FALSE	FALSE

eps	minPts	non_noise_clusters	noise_pct	max_cluster_size	max_cluster_share_pct	size_balance_ok	acceptable
1.4	75	10	7.00	30020	89.95	FALSE	FALSE
1.5	75	8	6.44	30308	90.81	FALSE	FALSE
1.6	75	6	5.88	30753	92.14	FALSE	FALSE
1.7	75	4	5.28	31053	93.04	FALSE	FALSE
1.8	75	4	4.82	31185	93.44	FALSE	FALSE
1.9	75	4	4.50	31311	93.82	FALSE	FALSE
2.0	75	4	4.04	31662	94.87	FALSE	FALSE
0.3	100	63	34.66	2289	6.86	TRUE	FALSE
0.4	100	49	28.07	3642	10.91	TRUE	FALSE
0.5	100	38	24.36	7383	22.12	TRUE	FALSE
0.6	100	30	22.04	8419	25.23	TRUE	FALSE
0.7	100	24	19.73	9584	28.72	TRUE	FALSE
0.8	100	19	17.72	10522	31.53	TRUE	FALSE
0.9	100	14	16.58	14317	42.90	TRUE	TRUE
1.0	100	12	15.22	14792	44.32	TRUE	TRUE
1.1	100	8	13.65	26644	79.83	FALSE	FALSE
1.2	100	6	12.52	27131	81.29	FALSE	FALSE
1.3	100	5	11.77	27529	82.48	FALSE	FALSE
1.4	100	5	10.26	27940	83.72	FALSE	FALSE
1.5	100	5	9.07	29832	89.38	FALSE	FALSE
1.6	100	6	7.74	30093	90.17	FALSE	FALSE
1.7	100	7	6.48	30544	91.52	FALSE	FALSE
1.8	100	5	6.12	30727	92.07	FALSE	FALSE
1.9	100	4	5.26	31063	93.07	FALSE	FALSE
2.0	100	4	5.01	31150	93.33	FALSE	FALSE

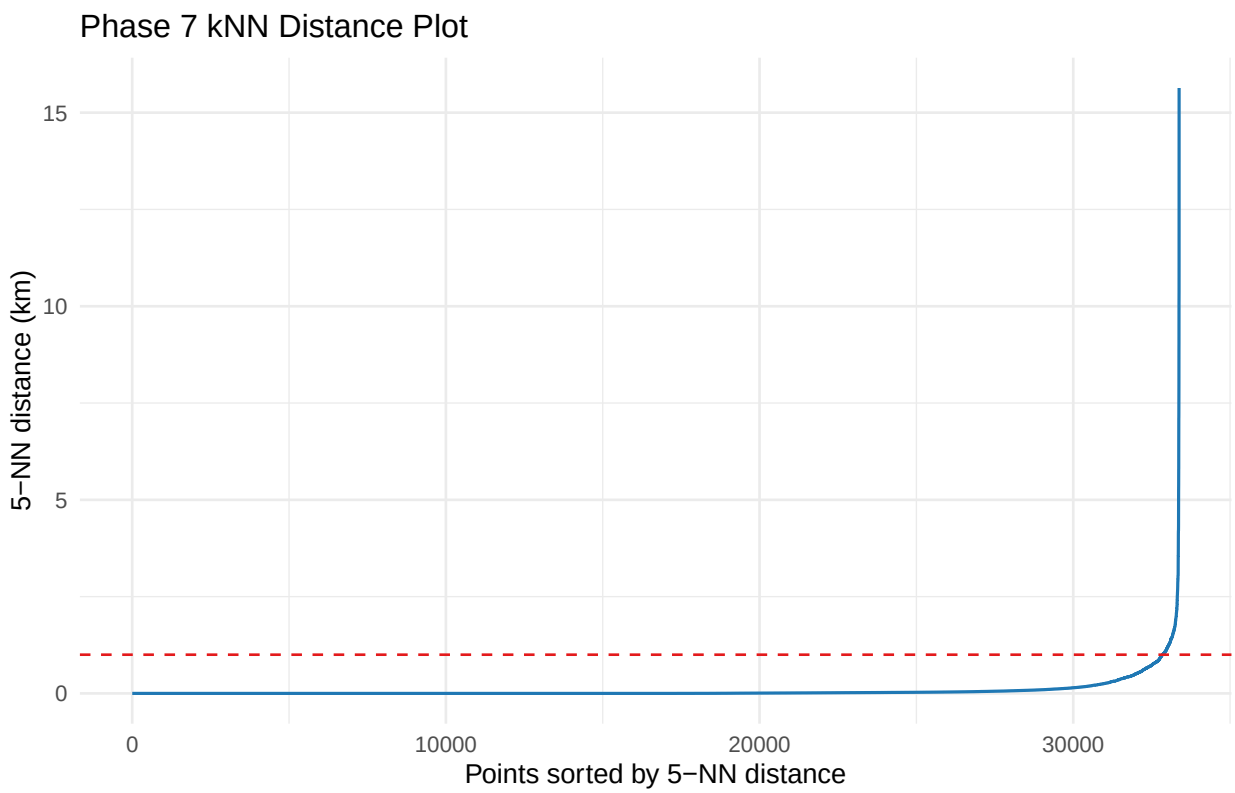
```
knitr::kable(parameter_summary, caption = "Phase 7 selected DBSCAN settings")
```

Table 44: Phase 7 selected DBSCAN settings

metric	value
kNN distance k	5
DBSCAN minPts grid	40, 50, 75, 100
DBSCAN eps grid	0.3 to 2
Largest cluster share cap (%)	60
DBSCAN minPts	100
Chosen eps (km)	1
Chosen non-noise clusters	12
Chosen noise pct	15.22
Chosen largest cluster share pct	44.32
Selection reason	Accepted trial balancing cluster count, noise level, and hotspot size concentration

```
knn_plot_data <- data.table(
  point_order = seq_along(knn_sorted),
  distance_km = knn_sorted
)
```

```r
knn_plot <- ggplot2::ggplot(
  knn_plot_data,
  ggplot2::aes(x = point_order, y = distance_km)
) +
  ggplot2::geom_line(color = "#1f78b4", linewidth = 0.6) +
  ggplot2::geom_hline(yintercept = eps_chosen, linetype = "dashed", color = "#e31a1c") +
  ggplot2::labs(
    title = "Phase 7 kNN Distance Plot",
    x = "Points sorted by 5-NN distance",
    y = "5-NN distance (km)"
  ) +
  ggplot2::theme_minimal()

print(knn_plot)
```




```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase7_knn_distance_plot.png"),
  plot = knn_plot,
  width = 7,
  height = 4.5,
  dpi = 300
)
```

```r
db_result <- dbscan::dbscan(df_spatial, eps = eps_chosen, minPts = minPts_chosen)

df_spatial_clusters <- copy(df_spatial)
df_spatial_clusters[, spatial_cluster := as.integer(db_result$cluster)]
```

```r
spatial_cluster_sizes <- df_spatial_clusters[, .N, by = spatial_cluster][order(spatial_cluster)]
largest_non_noise_size <- max(spatial_cluster_sizes[spatial_cluster > 0, N], na.rm = TRUE)
largest_non_noise_share_pct <- round(100 * largest_non_noise_size / nrow(df_spatial_clusters), 2)
solution_summary <- data.table(
  metric = c(
    "Rows clustered",
    "Noise pct",
    "Non-noise clusters",
    "Largest cluster size",
    "Largest cluster share pct",
    "Meets SRS cluster-count target",
    "Meets SRS noise target",
    "Meets hotspot-balance target"
  ),
  value = c(
    nrow(df_spatial_clusters),
    round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2),
    uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]),
    largest_non_noise_size,
    largest_non_noise_share_pct,
    uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]) >= 3 &
      uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]) <= 15,
    round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2) >= 5 &
      round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2) <= 30,
    largest_non_noise_share_pct <= phase7_max_cluster_share_cap
  )
)

if (!any(df_spatial_clusters$spatial_cluster > 0)) {
  stop("Phase 7 DBSCAN produced only noise points; adjust eps or minPts.")
}

fwrite(spatial_cluster_sizes, file.path(output_dir, "phase7_dbscan_cluster_sizes.csv"))
fwrite(solution_summary, file.path(output_dir, "phase7_solution_summary.csv"))

knitr::kable(spatial_cluster_sizes, caption = "Phase 7 spatial cluster sizes")
```

Table 45: Phase 7 spatial cluster sizes

spatial_cluster	N
0	5079
1	1097
2	14792
3	11080
4	150
5	187
6	239
7	132
8	100
9	130
10	90
11	140

spatial_cluster	N
12	159

```
knitr::kable(solution_summary, caption = "Phase 7 DBSCAN solution summary")
```

Table 46: Phase 7 DBSCAN solution summary

metric	value
Rows clustered	33375.00
Noise pct	15.22
Non-noise clusters	12.00
Largest cluster size	14792.00
Largest cluster share pct	44.32
Meets SRS cluster-count target	1.00
Meets SRS noise target	1.00
Meets hotspot-balance target	1.00

```
# v2.0: Compute centroid lat/lng for each non-noise spatial cluster.
# Export so corridor assignments can be manually verified against a map.
# Reference point: lat0 = 40.8 N, lng0 = -73.2 W (same as projection).
cluster_centroids <- df_spatial_clusters[spatial_cluster > 0, .(
  centroid_x_km = round(mean(x_km), 4),
  centroid_y_km = round(mean(y_km), 4),
  centroid_lat  = round(
    40.8 + mean(y_km) / 110.574, 5
  ),
  centroid_lng  = round(
    -73.2 + mean(x_km) / (111.32 * cos(40.8 * pi / 180)), 5
  ),
  n_accidents = .N
), by = spatial_cluster]

setorder(cluster_centroids, spatial_cluster)

fwrite(
  cluster_centroids,
  file.path(output_dir, "phase7_cluster_centroids.csv")
)

knitr::kable(
  cluster_centroids,
  caption = paste(
    "Phase 7 spatial cluster centroids (v2.0).",
    "Verify each centroid against the named corridor.",
    "lat0 = 40.8 N, lng0 = 73.2 W."
  )
)
```

Table 47: Phase 7 spatial cluster centroids (v2.0). Verify each centroid against the named corridor. lat0 = 40.8 N, lng0 = 73.2 W.

spatial_cluster	centroid_x_km	centroid_y_km	centroid_lat	centroid_lng	n_accidents
1	6.1176	-4.8680	40.75597	-73.12740	1097
2	-22.1567	-1.6256	40.78530	-73.46293	14792
3	-27.6725	-11.0638	40.69994	-73.52838	11080
4	-0.8362	2.9054	40.82628	-73.20992	150
5	-12.1551	2.8124	40.82543	-73.34424	187
6	-28.5098	-8.3010	40.72493	-73.53832	239
7	19.2660	-1.6065	40.78547	-72.97137	132
8	-22.9811	-7.9196	40.72838	-73.47271	100
9	-20.3708	0.2808	40.80254	-73.44174	130
10	-4.8372	-7.4186	40.73291	-73.25740	90
11	-26.6225	-14.4221	40.66957	-73.51592	140
12	-44.3074	-8.7347	40.72101	-73.72579	159

```r
# v2.0: Test eps sensitivity at selected +/- 0.1 km.
# Confirms the 12-cluster solution is not highly sensitive to eps.
eps_test_values <- c(
  eps_chosen - 0.1,
  eps_chosen,
  eps_chosen + 0.1
)

eps_sensitivity_table <- rbindlist(lapply(eps_test_values, function(eps_test) {
  res <- dbscan::dbscan(
    df_spatial[, .(x_km, y_km)],
    eps    = eps_test,
    minPts = minPts_chosen
  )
  n_clusters <- length(unique(res$cluster[res$cluster > 0]))
  noise_pct  <- round(100 * mean(res$cluster == 0), 2)
  max_share  <- if (sum(res$cluster > 0) > 0) {
    round(100 * max(table(res$cluster[res$cluster > 0])) /
            sum(res$cluster > 0), 2)
  } else NA_real_

  data.table(
    eps                   = eps_test,
    n_clusters            = n_clusters,
    noise_pct             = noise_pct,
    max_cluster_share_pct = max_share,
    within_srs_targets    = (
      n_clusters >= 3 & n_clusters <= 15 &
      noise_pct  >= 5 & noise_pct  <= 30 &
      max_share  <= 60
    )
  )
}))

fwrite(
```

```
  eps_sensitivity_table,
  file.path(output_dir, "phase7_eps_sensitivity.csv")
)

knitr::kable(
  eps_sensitivity_table,
  caption = "Phase 7 DBSCAN eps sensitivity (+/- 0.1 km from selected eps)"
)
```

Table 48: Phase 7 DBSCAN eps sensitivity (+/- 0.1 km from selected eps)

eps	n_clusters	noise_pct	max_cluster_share_pct	within_srs_targets
0.9	14	16.58	51.43	TRUE
1.0	12	15.22	52.28	TRUE
1.1	8	13.65	92.46	FALSE

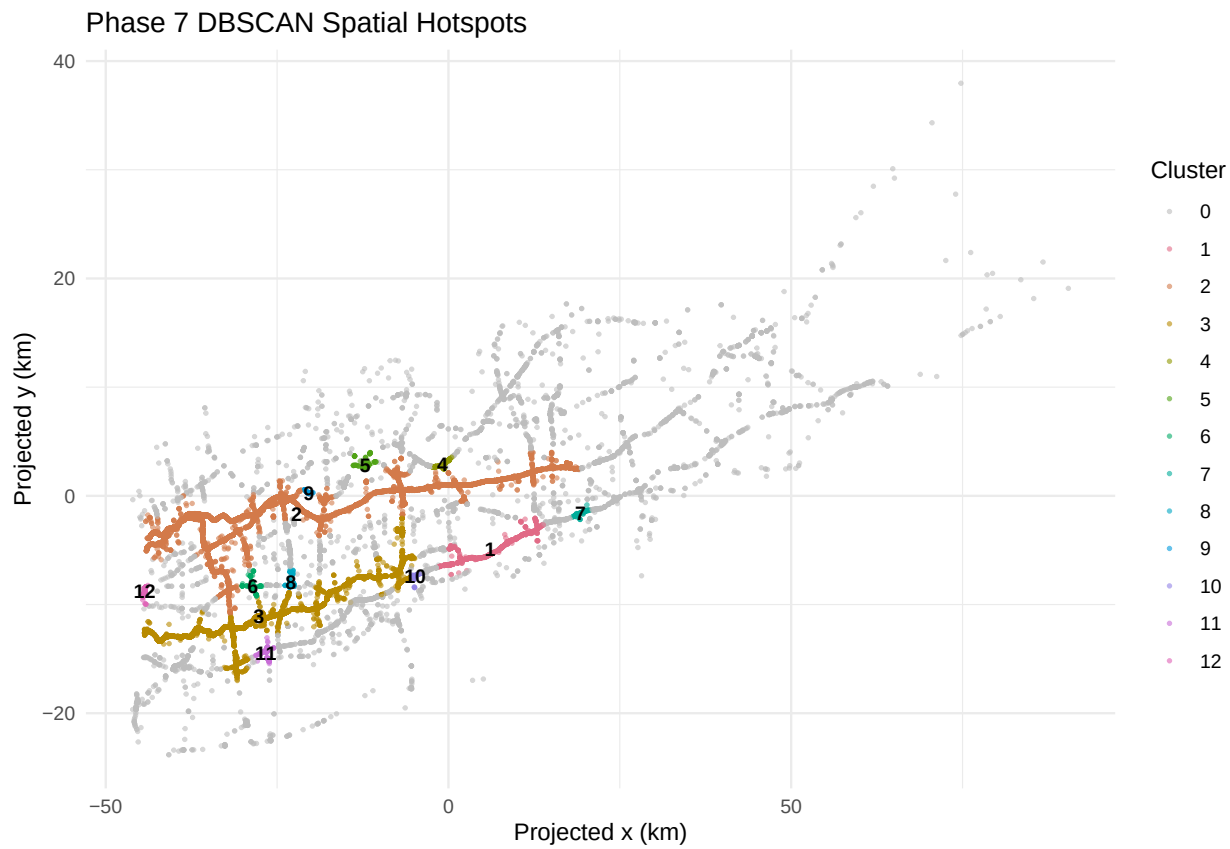
```
lat0 <- 40.8
lng0 <- -73.2

non_noise_assignments <- df_spatial_clusters[spatial_cluster > 0, .(
  cluster_size = .N,
  centroid_x_km = mean(x_km),
  centroid_y_km = mean(y_km),
  x_span_km = diff(range(x_km)),
  y_span_km = diff(range(y_km))
), by = spatial_cluster][
  , `:=`(
    centroid_lat = round((centroid_y_km / 110.574) + lat0, 5),
    centroid_lng = round((centroid_x_km / (111.32 * cos(lat0 * pi / 180))) + lng0, 5),
    shape_hint = fifelse(x_span_km >= y_span_km, "east_west", "north_south")
  )
]

non_noise_assignments[, corridor_label := dplyr::case_when(
  centroid_lat >= 40.74 & centroid_lat <= 40.76 ~ "Long Island Expressway (I-495)",
  centroid_lat >= 40.68 & centroid_lat <= 40.72 ~ "Southern State Parkway",
  centroid_lat >= 40.78 & centroid_lat <= 40.82 ~ "Northern State Parkway",
  centroid_lat >= 40.64 & centroid_lat <= 40.68 ~ "Sunrise Highway",
  centroid_lat >= 40.72 & centroid_lat <= 40.76 & shape_hint == "north_south" ~ "Meadowbrook / Wantagh
  TRUE ~ "Surface Road Cluster"
)]

fwrite(non_noise_assignments, file.path(output_dir, "phase7_corridor_assignments.csv"))

knitr::kable(non_noise_assignments, caption = "Phase 7 corridor assignment summary")
```

Table 49: Phase 7 corridor assignment summary

spatial_cluster	cluster_size	centroid_x	centroid_y	x_km	y_km	span_km	centroid_lat	centroid_lng	shape_hint	corridor_label
1	1097	6.1176454	-4.8680302	15.2086664	1.7055508	40.75597	-73.12740		east_west	Long Island Expressway (I-495)
7	132	19.2660169	-1.6064751	2.3867421	1.3434744	40.78547	-72.97137		east_west	Northern State Parkway
2	14792	-22.1567460	-1.6256442	63.1185968	4.8096184	40.78530	-73.46293		east_west	Northern State Parkway
3	11080	-27.6724890	-11.0638496	39.4362299	4.9291486	40.69994	-73.52838		east_west	Southern State Parkway
4	150	-0.8362332	2.9054348	2.9325504	1.0944205	40.82628	-73.20992		east_west	Surface Road Cluster
5	187	-12.1551395	2.8124230	3.2742599	1.4906484	40.82543	-73.34424		east_west	Surface Road Cluster
10	90	-4.8371610	-7.4186407	1.5632685	1.3655889	40.73291	-73.25740		east_west	Surface Road Cluster
6	239	-28.5097992	-8.3009745	2.9317077	2.2750604	40.72493	-73.53832		east_west	Surface Road Cluster
8	100	-22.9811129	-7.9196066	1.3350688	1.6167025	40.72838	-73.47271		north_south	Meadowbrook / Wantagh Pkwy
9	130	-20.3708466	0.2808053	1.2575417	0.3435534	40.80254	-73.44174		east_west	Northern State Parkway
11	140	-26.6225165	-14.4221418	2.7089856	2.3794419	40.66957	-73.51592		east_west	Sunrise Highway
12	159	-44.3074014	-8.7347346	0.8249062	1.7416511	40.72101	-73.72579		north_south	Meadowbrook / Wantagh Pkwy

```
df_spatial_plot <- copy(df_spatial_clusters)
df_spatial_plot[, cluster_label := factor(spatial_cluster)]

cluster_levels <- sort(unique(df_spatial_plot$cluster_label))
non_noise_levels <- setdiff(as.character(cluster_levels), "0")
cluster_colors <- c(
  "0" = "#bdbdbd",
  setNames(grDevices::hcl.colors(length(non_noise_levels), palette = "Dark 3"), non_noise_levels)
)

spatial_map <- ggplot2::ggplot(
  df_spatial_plot,
  ggplot2::aes(x = x_km, y = y_km, color = cluster_label)
) +
  ggplot2::geom_point(size = 0.5, alpha = 0.6) +
  ggplot2::geom_text(
    data = non_noise_assignments,
    ggplot2::aes(x = centroid_x_km, y = centroid_y_km, label = spatial_cluster),
    inherit.aes = FALSE,
    size = 3.2,
    fontface = "bold",
    color = "black"
  ) +
  ggplot2::scale_color_manual(values = cluster_colors) +
```

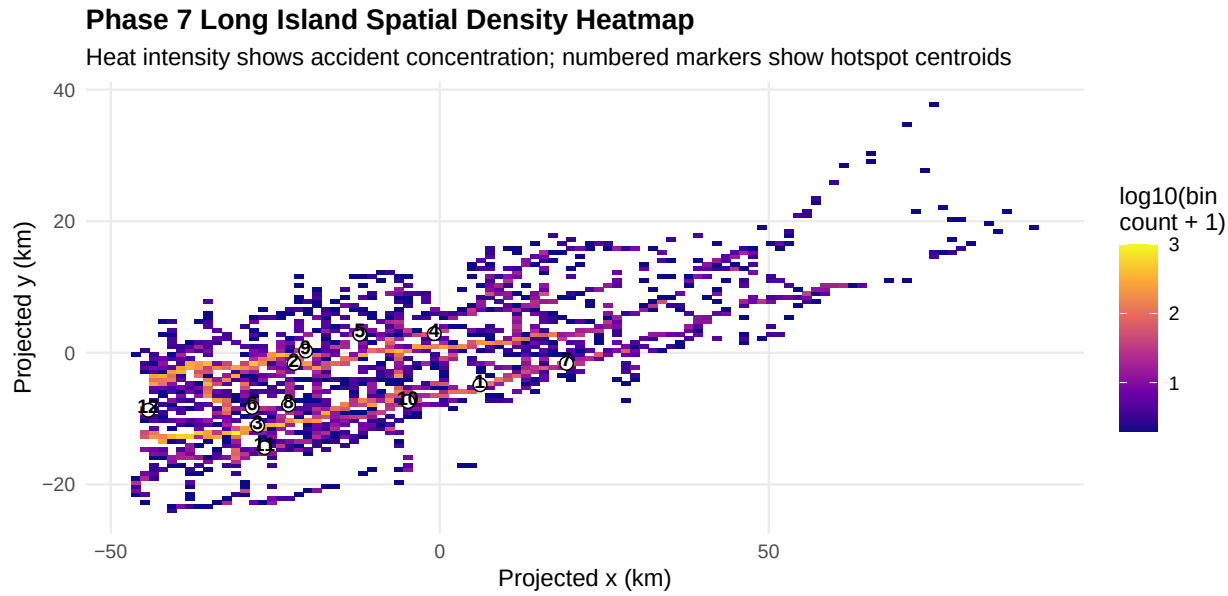
```
  ggplot2::labs(
    title = "Phase 7 DBSCAN Spatial Hotspots",
    x = "Projected x (km)",
    y = "Projected y (km)",
    color = "Cluster"
  ) +
  ggplot2::theme_minimal()

print(spatial_map)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase7_spatial_hotspot_map.png"),
  plot = spatial_map,
  width = 8,
  height = 5.5,
  dpi = 300
)
```

```
spatial_density_heatmap <- ggplot2::ggplot(
  df_spatial_plot,
  ggplot2::aes(
    x = x_km,
    y = y_km,
    fill = after_stat(log10(count + 1))
  )
```

```r
) +
  ggplot2::geom_bin2d(bins = 100) +
  ggplot2::geom_point(
    data = non_noise_assignments,
    ggplot2::aes(x = centroid_x_km, y = centroid_y_km),
    inherit.aes = FALSE,
    shape = 21,
    size = 2.6,
    stroke = 0.5,
    fill = "white",
    color = "black"
  ) +
  ggplot2::geom_text(
    data = non_noise_assignments,
    ggplot2::aes(x = centroid_x_km, y = centroid_y_km, label = spatial_cluster),
    inherit.aes = FALSE,
    size = 3,
    fontface = "bold",
    nudge_y = 0.6
  ) +
  ggplot2::scale_fill_viridis_c(
    option = "C",
    name = "log10(bin\ncount + 1)"
  ) +
  ggplot2::coord_equal() +
  ggplot2::labs(
    title = "Phase 7 Long Island Spatial Density Heatmap",
    subtitle = "Heat intensity shows accident concentration; numbered markers show hotspot centroids",
    x = "Projected x (km)",
    y = "Projected y (km)"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank()
  )

print(spatial_density_heatmap)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase7_spatial_density_heatmap.png"),
  plot = spatial_density_heatmap,
  width = 8,
  height = 5.5,
  dpi = 300
)
```

```
phase_7_summary <- data.table(
  checkpoint = c(
    "kNN distance plot saved",
    "eps trials saved",
    "DBSCAN solution created",
    "Corridor assignments saved",
    "Spatial hotspot map saved",
    "Spatial density heatmap saved"
  ),
  status = c(
    file.exists(file.path(output_dir, "phase7_knn_distance_plot.png")),
    file.exists(file.path(output_dir, "phase7_eps_trials.csv")),
    exists("db_result"),
    file.exists(file.path(output_dir, "phase7_corridor_assignments.csv")),
    file.exists(file.path(output_dir, "phase7_spatial_hotspot_map.png")),
    file.exists(file.path(output_dir, "phase7_spatial_density_heatmap.png"))
  )
)
```

```r
fwrite(phase_7_summary, file.path(output_dir, "phase7_summary.csv"))

knitr::kable(phase_7_summary, caption = "Phase 7 readiness summary")
```

Table 50: Phase 7 readiness summary

checkpoint	status
kNN distance plot saved	TRUE
eps trials saved	TRUE
DBSCAN solution created	TRUE
Corridor assignments saved	TRUE
Spatial hotspot map saved	TRUE
Spatial density heatmap saved	TRUE

8 Phase 8: Post-Clustering Interpretation

This phase joins the scenario and spatial cluster labels back to the working data and builds the interpretation tables used for the final report.

```r
kp_final <- selected_model
db_final <- db_result

if (length(kp_final$cluster) != nrow(df_phase3)) {
  stop("Scenario cluster labels do not match the Phase 3 data rows.")
}

if (length(db_final$cluster) != nrow(df_phase3)) {
  stop("Spatial cluster labels do not match the Phase 3 data rows.")
}

df_clustered <- copy(df_phase3)
df_clustered[, scenario_cluster := as.integer(kp_final$cluster)]
df_clustered[, spatial_cluster := as.integer(db_final$cluster)]

label_join_check <- data.table(
  metric = c(
    "Rows in clustered data",
    "Scenario labels",
    "Spatial labels",
    "Scenario label NAs",
    "Spatial label NAs"
  ),
  value = c(
    nrow(df_clustered),
    length(kp_final$cluster),
    length(db_final$cluster),
    sum(is.na(df_clustered$scenario_cluster)),
    sum(is.na(df_clustered$spatial_cluster))
  )
)
```

```r
if (any(label_join_check[metric %in% c("Scenario label NAs", "Spatial label NAs"), value > 0])) {
  stop("Phase 8 label joining produced missing cluster labels.")
}

fwrite(label_join_check, file.path(output_dir, "phase8_label_join_check.csv"))

knitr::kable(label_join_check, caption = "Phase 8 label join check")
```

Table 51: Phase 8 label join check

metric	value
Rows in clustered data	33375
Scenario labels	33375
Spatial labels	33375
Scenario label NAs	0
Spatial label NAs	0

```r
dataset_sev4_pct <- round(100 * mean(as.integer(df_clustered$Severity) == 4), 2)
dataset_sev34_pct <- round(100 * mean(as.integer(df_clustered$Severity) %in% c(3, 4)), 2)

scenario_severity_profile <- df_clustered[, .(
  n = .N,
  sev1_pct = round(100 * mean(as.integer(Severity) == 1), 2),
  sev2_pct = round(100 * mean(as.integer(Severity) == 2), 2),
  sev3_pct = round(100 * mean(as.integer(Severity) == 3), 2),
  sev4_pct = round(100 * mean(as.integer(Severity) == 4), 2),
  sev34_pct = round(100 * mean(as.integer(Severity) %in% c(3, 4)), 2)
), by = scenario_cluster][order(scenario_cluster)]

scenario_severity_profile[, high_risk_archetype := (
  sev4_pct >= (1.5 * dataset_sev4_pct) |
    sev34_pct == max(sev34_pct)
)]

# v2.0: Add explicit Sev4 rate and ratio-to-baseline columns
scenario_severity_profile[, sev4_rate       := sev4_pct]
scenario_severity_profile[, sev4_vs_baseline := round(sev4_pct / dataset_sev4_pct, 2)]

# v2.0: Prepend a baseline row (scenario_cluster = 0) for easy comparison
baseline_severity_row <- data.table(
  scenario_cluster = 0L,
  n                = nrow(df_clustered),
  sev1_pct         = round(100 * mean(as.integer(df_clustered$Severity) == 1), 2),
  sev2_pct         = round(100 * mean(as.integer(df_clustered$Severity) == 2), 2),
  sev3_pct         = round(100 * mean(as.integer(df_clustered$Severity) == 3), 2),
  sev4_pct         = dataset_sev4_pct,
  sev34_pct        = dataset_sev34_pct,
  high_risk_archetype = FALSE,
  sev4_rate        = dataset_sev4_pct,
  sev4_vs_baseline = 1.00
)
```

```r
severity_profile_v2 <- rbind(baseline_severity_row, scenario_severity_profile, fill = TRUE)

fwrite(severity_profile_v2, file.path(output_dir, "phase8_severity_profile_v2.csv"))
fwrite(scenario_severity_profile, file.path(output_dir, "phase8_scenario_severity_profile.csv"))

knitr::kable(
  severity_profile_v2,
  caption = paste(
    "Phase 8 severity profile (v2.0).",
    "Row 0 = dataset baseline.",
    "sev4_vs_baseline = cluster Sev4 rate / baseline Sev4 rate."
  )
)
```

Table 52: Phase 8 severity profile (v2.0). Row 0 = dataset baseline. sev4_vs_baseline = cluster Sev4 rate / baseline Sev4 rate.

scenario_cluster	n	sev1_pct	sev2_pct	sev3_pct	sev4_pct	sev34_pct	high_risk_archetype	sev4_rate	sev4_vs_baseline
0	33375	0.48	84.96	11.80	2.75	14.55	FALSE	2.75	1.00
1	10593	0.40	87.49	10.03	2.09	12.11	FALSE	2.09	0.76
2	13470	0.59	85.46	12.18	1.77	13.96	FALSE	1.77	0.64
3	9312	0.43	81.38	13.26	4.93	18.19	TRUE	4.93	1.79

```r
scenario_county_profile <- df_clustered[, .(
  n = .N,
  Nassau_pct = round(100 * mean(County == "Nassau"), 2),
  Suffolk_pct = round(100 * mean(County == "Suffolk"), 2),
  Queens_pct = round(100 * mean(County == "Queens"), 2),
  dominant_county = mode_value(County)
), by = scenario_cluster][order(scenario_cluster)]

fwrite(scenario_county_profile, file.path(output_dir, "phase8_scenario_county_profile.csv"))

knitr::kable(scenario_county_profile, caption = "Phase 8 county composition by scenario cluster")
```

Table 53: Phase 8 county composition by scenario cluster

scenario_cluster	n	Nassau_pct	Suffolk_pct	Queens_pct	dominant_county
1	10593	58.11	41.66	0.23	Nassau
2	13470	55.89	43.47	0.64	Nassau
3	9312	61.27	38.34	0.40	Nassau

```r
scenario_weather_time_profile <- df_clustered[, .(
  n = .N,
  dominant_weather_group = mode_value(weather_group),
  dominant_weather_share = round(100 * dominant_share(weather_group), 2),
  dominant_hour_band = mode_value(hour_band),
  dominant_hour_share = round(100 * dominant_share(hour_band), 2),
  night_pct = round(100 * mean(Sunrise_Sunset == "Night"), 2),
```

```
  weekend_pct = round(100 * mean(weekend), 2),
  mean_temperature = round(mean(`Temperature(F)`), 2),
  mean_visibility = round(mean(`Visibility(mi)`), 2),
  mean_wind_speed = round(mean(`Wind_Speed(mph)`), 2),
  junction_pct = round(100 * mean(Junction == "TRUE"), 2),
  traffic_signal_pct = round(100 * mean(Traffic_Signal == "TRUE"), 2),
  crossing_pct = round(100 * mean(Crossing == "TRUE"), 2),
  stop_pct = round(100 * mean(Stop == "TRUE"), 2)
), by = scenario_cluster][order(scenario_cluster)]

fwrite(scenario_weather_time_profile, file.path(output_dir, "phase8_scenario_weather_time_profile.csv"))

knitr::kable(scenario_weather_time_profile, caption = "Phase 8 weather and time profile by scenario clus
```

Table 54: Phase 8 weather and time profile by scenario cluster

scenario_cluster	n	dominant_weather	dominant_pct	dominant_time_bin	night_pct	morning_pct	afternoon_pct	evening_pct	weekend_pct	mean_temperature	mean_visibility	mean_wind_speed	junction_pct	traffic_signal_pct	crossing_pct	stop_pct
1	10596	Clear	99.41	AM_Rush	18.84	8.96	13.24	60.70	9.94	9.96	15.72	8.71	9.32	0.97		
2	13470	Cloudy	80.84	AM_Rush	15.08	3.83	11.93	62.04	8.66	10.52	15.40	10.50	11.13	1.23		
3	9312	Cloudy	56.96	Early_AM	31.24	74.80	23.54	49.83	8.64	9.84	17.10	9.12	9.54	0.84		

```
scenario_name_lookup <- copy(cluster_profile_summary)[
  , .(
    scenario_cluster = as.integer(as.character(cluster)),
    working_name
  )
]

spatial_scenario_crosstab <- df_clustered[spatial_cluster > 0, .N, by = .(spatial_cluster, scenario_clus
  order(spatial_cluster, -N, scenario_cluster)
][
  , spatial_cluster_pct := round(100 * N / sum(N), 2), by = spatial_cluster
]

spatial_scenario_crosstab <- merge(
  spatial_scenario_crosstab,
  non_noise_assignments[, .(spatial_cluster, corridor_label)],
  by = "spatial_cluster",
  all.x = TRUE
)

spatial_scenario_crosstab <- merge(
  spatial_scenario_crosstab,
  scenario_name_lookup,
  by = "scenario_cluster",
  all.x = TRUE
)

spatial_scenario_dominance <- spatial_scenario_crosstab[
  order(spatial_cluster, -N, scenario_cluster)
][
  , .SD[1], by = spatial_cluster
```

```
][
  , .(
    spatial_cluster,
    corridor_label,
    dominant_scenario_cluster = scenario_cluster,
    dominant_scenario_name = working_name,
    dominant_count = N,
    dominant_pct = spatial_cluster_pct
  )
]

fwrite(spatial_scenario_crosstab, file.path(output_dir, "phase8_spatial_scenario_crosstab.csv"))
fwrite(spatial_scenario_dominance, file.path(output_dir, "phase8_spatial_scenario_dominance.csv"))

knitr::kable(spatial_scenario_crosstab, caption = "Phase 8 spatial-scenario cross-tabulation")
```

Table 55: Phase 8 spatial-scenario cross-tabulation

scenario_cluster	spatial_cluster	N	spatial_cluster_pct	corridor_label	working_name
1	1	217	19.78	Long Island Expressway (I-495)	Scenario Cluster 1
1	2	4558	30.81	Northern State Parkway	Scenario Cluster 1
1	3	3715	33.53	Southern State Parkway	Scenario Cluster 1
1	4	29	19.33	Surface Road Cluster	Scenario Cluster 1
1	5	89	47.59	Surface Road Cluster	Scenario Cluster 1
1	6	104	43.51	Surface Road Cluster	Scenario Cluster 1
1	7	24	18.18	Northern State Parkway	Scenario Cluster 1
1	8	31	31.00	Meadowbrook / Wantagh Pkwy	Scenario Cluster 1
1	9	48	36.92	Northern State Parkway	Scenario Cluster 1
1	10	8	8.89	Surface Road Cluster	Scenario Cluster 1
1	11	55	39.29	Sunrise Highway	Scenario Cluster 1
1	12	21	13.21	Meadowbrook / Wantagh Pkwy	Scenario Cluster 1
2	1	624	56.88	Long Island Expressway (I-495)	Scenario Cluster 2
2	2	6083	41.12	Northern State Parkway	Scenario Cluster 2
2	3	4079	36.81	Southern State Parkway	Scenario Cluster 2
2	4	66	44.00	Surface Road Cluster	Scenario Cluster 2

scenario_cluster	spatial_cluster	N	spatial_cluster_pct	corridor_label	working_name
2	5	60	32.09	Surface Road Cluster	Scenario Cluster 2
2	6	77	32.22	Surface Road Cluster	Scenario Cluster 2
2	7	76	57.58	Northern State Parkway	Scenario Cluster 2
2	8	39	39.00	Meadowbrook / Wantagh Pkwy	Scenario Cluster 2
2	9	43	33.08	Northern State Parkway	Scenario Cluster 2
2	10	46	51.11	Surface Road Cluster	Scenario Cluster 2
2	11	41	29.29	Sunrise Highway	Scenario Cluster 2
2	12	64	40.25	Meadowbrook / Wantagh Pkwy	Scenario Cluster 2
3	1	256	23.34	Long Island Expressway (I-495)	Scenario Cluster 3
3	2	4151	28.06	Northern State Parkway	Scenario Cluster 3
3	3	3286	29.66	Southern State Parkway	Scenario Cluster 3
3	4	55	36.67	Surface Road Cluster	Scenario Cluster 3
3	5	38	20.32	Surface Road Cluster	Scenario Cluster 3
3	6	58	24.27	Surface Road Cluster	Scenario Cluster 3
3	7	32	24.24	Northern State Parkway	Scenario Cluster 3
3	8	30	30.00	Meadowbrook / Wantagh Pkwy	Scenario Cluster 3
3	9	39	30.00	Northern State Parkway	Scenario Cluster 3
3	10	36	40.00	Surface Road Cluster	Scenario Cluster 3
3	11	44	31.43	Sunrise Highway	Scenario Cluster 3
3	12	74	46.54	Meadowbrook / Wantagh Pkwy	Scenario Cluster 3

```
knitr::kable(spatial_scenario_dominance, caption = "Phase 8 dominant scenario by spatial hotspot")
```

Table 56: Phase 8 dominant scenario by spatial hotspot

spatial_cluster	corridor_label	dominant_scenario_cluster	dominant_scenario_name	dominant_count	dominant_pct
1	Long Island Expressway (I-495)	2	Scenario Cluster 2	624	56.88
2	Northern State Parkway	2	Scenario Cluster 2	6083	41.12

spatial_cluster	corridor_label	dominant_scenario_cluster	dominant_scenario_name	dominant_count	dominant_pct
3	Southern State Parkway	2	Scenario Cluster 2	4079	36.81
4	Surface Road Cluster	2	Scenario Cluster 2	66	44.00
5	Surface Road Cluster	1	Scenario Cluster 1	89	47.59
6	Surface Road Cluster	1	Scenario Cluster 1	104	43.51
7	Northern State Parkway	2	Scenario Cluster 2	76	57.58
8	Meadowbrook / Wantagh Pkwy	2	Scenario Cluster 2	39	39.00
9	Northern State Parkway	1	Scenario Cluster 1	48	36.92
10	Surface Road Cluster	2	Scenario Cluster 2	46	51.11
11	Sunrise Highway	1	Scenario Cluster 1	55	39.29
12	Meadowbrook / Wantagh Pkwy	3	Scenario Cluster 3	74	46.54

```r
scenario_final_names <- merge(
  scenario_weather_time_profile,
  scenario_severity_profile[, .(scenario_cluster, sev4_pct, high_risk_archetype)],
  by = "scenario_cluster",
  all.x = TRUE
)

scenario_final_names <- merge(
  scenario_final_names,
  scenario_name_lookup,
  by = "scenario_cluster",
  all.x = TRUE
)

# v2.0: Diagnostic print - inspect all profile values before naming
# so any cluster falling through to the generic fallback is visible.
cat("=== Phase 8 Cluster Profile Values for Naming Diagnostics ===\n")
```

```
## === Phase 8 Cluster Profile Values for Naming Diagnostics ===
```

```r
print(scenario_final_names[, .(
  scenario_cluster, dominant_hour_band, dominant_weather_group,
  night_pct, weekend_pct, junction_pct, traffic_signal_pct
)])
```

```
## Key: <scenario_cluster>
##    scenario_cluster dominant_hour_band dominant_weather_group night_pct
##               <int>             <char>                 <char>     <num>
## 1:                1            AM_Rush                  Clear      8.96
```

```
## 2:                2            AM_Rush                 Cloudy      3.83
## 3:                3           Early_AM                 Cloudy     74.80
##    weekend_pct junction_pct traffic_signal_pct
##          <num>        <num>              <num>
## 1:       13.24        15.72               8.71
## 2:       11.93        15.40              10.50
## 3:       23.54        17.10               9.12
```

```r
cat("==============================================================\n")
```

```
## ==============================================================
```

```r
scenario_final_names[, final_name := dplyr::case_when(
  # Tier 1: Extreme weather
  dominant_weather_group == "Snow_Ice" ~
    "Winter Adverse --- Snow and Ice",

  # Tier 2: Night + hazardous visibility
  night_pct >= 50 &
    dominant_weather_group %in% c("Fog_Haze", "Rain", "Thunderstorm") ~
    "Night Low-Visibility --- Fog and Rain",

  # Tier 3: Night / early morning (any weather)
  night_pct >= 50 & dominant_hour_band == "Early_AM" ~
    paste("Night / Early AM ---", gsub("_", " ", dominant_weather_group)),

  # Tier 4: Weekend surface road
  weekend_pct >= 40 & traffic_signal_pct >= 8 ~
    "Weekend Surface Road --- Signal Heavy",

  # Tier 5: Rush-hour corridor
  dominant_hour_band %in% c("AM_Rush", "PM_Rush") & junction_pct >= 10 ~
    paste(
      gsub("_", " ", dominant_weather_group),
      gsub("_", " ", dominant_hour_band),
      "--- Junction Heavy"
    ),

  # Tier 6 (v2.0): Cloudy off-peak daytime - captures generic midday clusters
  dominant_weather_group == "Cloudy" &
    dominant_hour_band %in% c("Midday", "Evening") ~
    paste("Cloudy Off-Peak ---", gsub("_", " ", dominant_hour_band)),

  # Tier 7 (v2.0): Clear off-peak daytime
  dominant_weather_group == "Clear" &
    dominant_hour_band %in% c("Midday", "Evening") ~
    paste("Clear Off-Peak ---", gsub("_", " ", dominant_hour_band)),

  # Fallback: more informative than bare cluster number
  TRUE ~
    paste(
      "Mixed Conditions ---",
      gsub("_", " ", dominant_weather_group),
```

```
      gsub("_", " ", dominant_hour_band)
    )
)]

scenario_final_names[, final_name := ifelse(
  high_risk_archetype,
  paste(final_name, "(High Risk)"),
  final_name
)]

duplicate_scenario_names <- scenario_final_names[, .N, by = final_name][N > 1, final_name]
if (length(duplicate_scenario_names) > 0) {
  scenario_final_names[final_name %in% duplicate_scenario_names, final_name := paste0(final_name, " (S"
}

spatial_final_names <- rbind(
  data.table(
    spatial_cluster = 0L,
    corridor_label = "Noise / Unassigned",
    final_name = "Noise / Unassigned"
  ),
  non_noise_assignments[, .(
    spatial_cluster,
    corridor_label,
    final_name = ifelse(
      corridor_label == "Surface Road Cluster",
      paste("Surface Road Cluster", spatial_cluster),
      paste(corridor_label, "Cluster", spatial_cluster)
    )
  )]
)

scenario_name_map <- scenario_final_names$final_name
names(scenario_name_map) <- scenario_final_names$scenario_cluster

spatial_name_map <- spatial_final_names$final_name
names(spatial_name_map) <- spatial_final_names$spatial_cluster

df_clustered[, scenario_cluster_name := scenario_name_map[as.character(scenario_cluster)]]
df_clustered[, spatial_cluster_name := spatial_name_map[as.character(spatial_cluster)]]

if ("working_name" %in% names(spatial_scenario_crosstab)) {
  spatial_scenario_crosstab[, working_name := NULL]
}
spatial_scenario_crosstab <- merge(
  spatial_scenario_crosstab,
  scenario_final_names[, .(scenario_cluster, final_name)],
  by = "scenario_cluster",
  all.x = TRUE
)

spatial_scenario_dominance <- merge(
  spatial_scenario_dominance[, !c("dominant_scenario_name"), with = FALSE],
```

```
  scenario_final_names[, .(scenario_cluster, final_name)],
  by.x = "dominant_scenario_cluster",
  by.y = "scenario_cluster",
  all.x = TRUE
)
data.table::setnames(spatial_scenario_dominance, "final_name", "dominant_scenario_name")

fwrite(spatial_scenario_crosstab, file.path(output_dir, "phase8_spatial_scenario_crosstab.csv"))
fwrite(spatial_scenario_dominance, file.path(output_dir, "phase8_spatial_scenario_dominance.csv"))

fwrite(scenario_final_names, file.path(output_dir, "phase8_scenario_final_names.csv"))
fwrite(spatial_final_names, file.path(output_dir, "phase8_spatial_final_names.csv"))

knitr::kable(scenario_final_names, caption = "Phase 8 final scenario cluster names")
```

Table 57: Phase 8 final scenario cluster names

scenario_cluster	n	dominant_weather	weather_purity	dominant_period	period_purity	mean_duration_min	mean_distance_km	mean_speed_kmh	mean_temperature_c	mean_precipitation_mm	mean_visibility_km	mean_traffic_signals	mean_crashes	mean_injuries	mean_fatalities	night_share	high_risk	final_name
1	10593	Clear	99.41	AM_Rush	38.84	8.96	13.24	60.70	9.94	9.96	15.72	8.71	9.32	0.97	2.09	FALSE	Scenario Cluster 1	Clear AM Rush — Junction Heavy
2	13470	Cloudy	80.84	AM_Rush	45.08	3.83	11.93	62.04	8.66	10.52	15.40	10.50	11.13	1.23	1.77	FALSE	Scenario Cluster 2	Cloudy AM Rush — Junction Heavy
3	9312	Cloudy	56.96	Early_AM	31.24	74.80	23.54	49.83	8.64	9.84	17.10	9.12	9.54	0.84	4.93	TRUE	Scenario Cluster 3	Night / Early AM — Cloudy (High Risk)

```
knitr::kable(spatial_final_names, caption = "Phase 8 final spatial cluster names")
```

Table 58: Phase 8 final spatial cluster names

spatial_cluster	corridor_label	final_name
0	Noise / Unassigned	Noise / Unassigned
1	Long Island Expressway (I-495)	Long Island Expressway (I-495) Cluster 1
7	Northern State Parkway	Northern State Parkway Cluster 7
2	Northern State Parkway	Northern State Parkway Cluster 2
3	Southern State Parkway	Southern State Parkway Cluster 3

spatial_cluster	corridor_label	final_name
4	Surface Road Cluster	Surface Road Cluster 4
5	Surface Road Cluster	Surface Road Cluster 5
10	Surface Road Cluster	Surface Road Cluster 10
6	Surface Road Cluster	Surface Road Cluster 6
8	Meadowbrook / Wantagh Pkwy	Meadowbrook / Wantagh Pkwy Cluster 8
9	Northern State Parkway	Northern State Parkway Cluster 9
11	Sunrise Highway	Sunrise Highway Cluster 11
12	Meadowbrook / Wantagh Pkwy	Meadowbrook / Wantagh Pkwy Cluster 12

```r
phase_8_summary <- data.table(
  checkpoint = c(
    "Label join saved",
    "Severity profile saved",
    "County profile saved",
    "Weather-time profile saved",
    "Spatial-scenario cross-tab saved",
    "Final cluster names saved",
    "Scenario names are unique"
  ),
  status = c(
    file.exists(file.path(output_dir, "phase8_label_join_check.csv")),
    file.exists(file.path(output_dir, "phase8_scenario_severity_profile.csv")),
    file.exists(file.path(output_dir, "phase8_scenario_county_profile.csv")),
    file.exists(file.path(output_dir, "phase8_scenario_weather_time_profile.csv")),
    file.exists(file.path(output_dir, "phase8_spatial_scenario_crosstab.csv")),
    file.exists(file.path(output_dir, "phase8_scenario_final_names.csv")) &
      file.exists(file.path(output_dir, "phase8_spatial_final_names.csv")),
    uniqueN(scenario_final_names$final_name) == nrow(scenario_final_names)
  )
)

fwrite(phase_8_summary, file.path(output_dir, "phase8_summary.csv"))

knitr::kable(phase_8_summary, caption = "Phase 8 readiness summary")
```

Table 59: Phase 8 readiness summary

checkpoint	status
Label join saved	TRUE
Severity profile saved	TRUE
County profile saved	TRUE
Weather-time profile saved	TRUE
Spatial-scenario cross-tab saved	TRUE
Final cluster names saved	TRUE
Scenario names are unique	TRUE

9 Phase 9: Export and Visualization

This phase organizes the report-ready figures, tables, and labeled dataset export so the final write-up can pull from a single output location.

```
figure_dir <- file.path(output_dir, "figures")
table_dir <- file.path(output_dir, "tables")
interactive_dir <- file.path(output_dir, "interactive")

dir.create(figure_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(table_dir, recursive = TRUE, showWarnings = FALSE)
dir.create(interactive_dir, recursive = TRUE, showWarnings = FALSE)

export_setup <- data.table(
  item = c("Figure directory", "Table directory", "Interactive directory"),
  path = c(figure_dir, table_dir, interactive_dir),
  status = c(
    "Created",
    "Created",
    ifelse(interactive_exports_enabled, "Created", "Created for optional HTML exports")
  )
)

knitr::kable(export_setup, caption = "Phase 9 export folders")
```

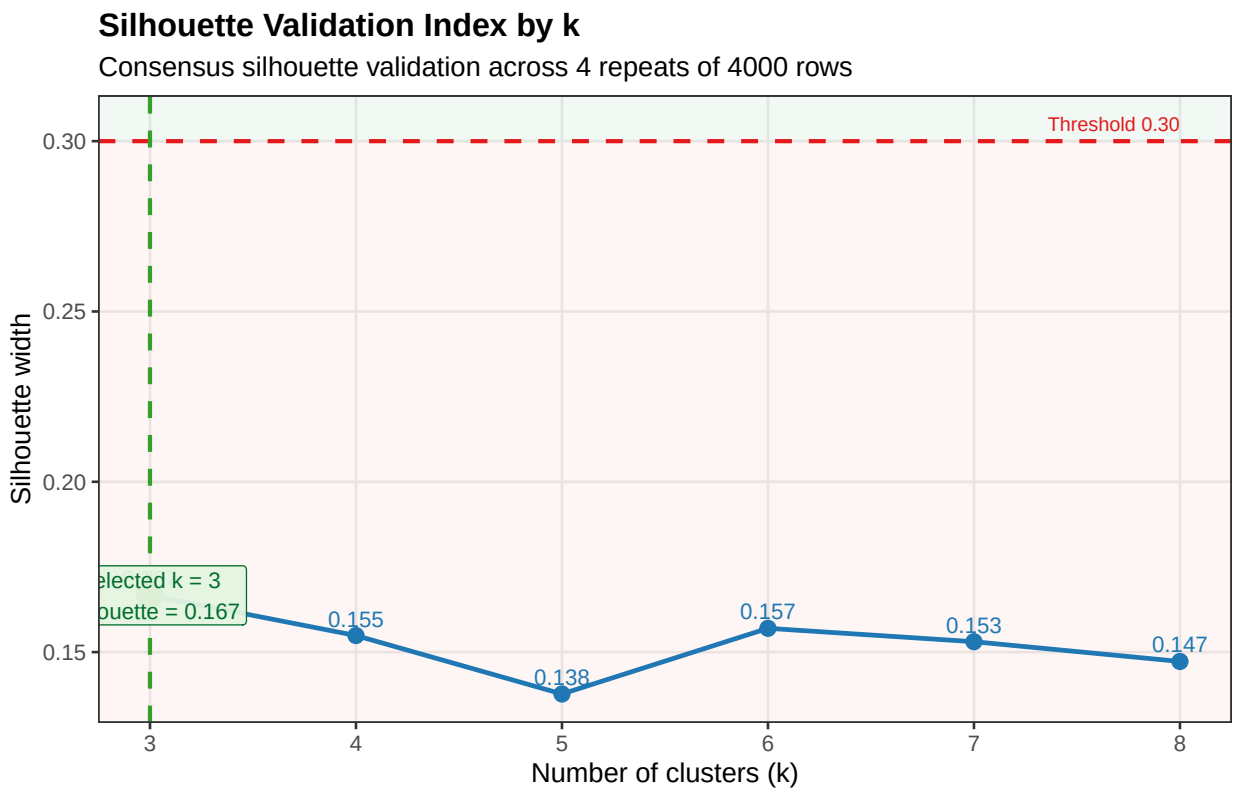
Table 60: Phase 9 export folders

item	path	status
Figure directory	local_artifacts/output/figures	Created
Table directory	local_artifacts/output/tables	Created
Interactive directory	local_artifacts/output/interactive	Created for optional HTML exports

```
selected_validation_point <- validation_table[k == selected_k]
validation_subtitle <- if (phase6_validation_mode == "consensus_sampled") {
  paste(
    "Consensus silhouette validation across",
    phase6_validation_repeats,
    "repeats of",
    validation_rows,
    "rows"
  )
} else if (phase6_validation_mode == "sampled") {
  paste("Sampled silhouette validation on", validation_rows, "rows")
} else {
  "Full-data silhouette validation"
}

validation_plot <- ggplot2::ggplot(
  validation_table,
  ggplot2::aes(x = k, y = silhouette)
) +
  # v2.0: 0.30 threshold band so reader sees where results sit
```

```r
ggplot2::annotate(
  "rect", xmin = -Inf, xmax = Inf,
  ymin = 0.30, ymax = Inf, alpha = 0.06, fill = "#31a354"
) +
ggplot2::annotate(
  "rect", xmin = -Inf, xmax = Inf,
  ymin = -Inf, ymax = 0.30, alpha = 0.06, fill = "#fb6a4a"
) +
ggplot2::geom_hline(
  yintercept = 0.30, linetype = "dashed",
  color = "#e31a1c", linewidth = 0.8
) +
ggplot2::annotate(
  "text", x = max(validation_table$k),
  y = 0.305, label = "Threshold 0.30",
  hjust = 1, size = 2.9, color = "#e31a1c"
) +
ggplot2::geom_line(color = "#1f78b4", linewidth = 0.9) +
ggplot2::geom_point(color = "#1f78b4", size = 2.6) +
ggplot2::geom_vline(
  xintercept = selected_k, linetype = "dashed",
  color = "#33a02c", linewidth = 0.8
) +
ggplot2::geom_point(
  data = selected_validation_point,
  color = "#33a02c", size = 4.5
) +
ggplot2::geom_text(
  ggplot2::aes(label = round(silhouette, 3)),
  nudge_y = max(validation_table$silhouette) * 0.03,
  size = 3.2, color = "#1f78b4"
) +
ggplot2::annotate(
  "label",
  x = selected_k,
  y = selected_validation_point$silhouette[[1]],
  label = paste0("Selected k = ", selected_k,
                 "\nSilhouette = ",
                 round(selected_validation_point$silhouette[[1]], 3)),
  fill   = "#e5f5e0", color = "#006d2c",
  label.size = 0.25, alpha = 0.95, size = 3
) +
ggplot2::labs(
  title    = "Silhouette Validation Index by k",
  subtitle = validation_subtitle,
  x        = "Number of clusters (k)",
  y        = "Silhouette width"
) +
ggplot2::theme_bw(base_size = 11) +
ggplot2::theme(
  plot.title       = ggplot2::element_text(face = "bold"),
  panel.grid.minor = ggplot2::element_blank()
)
```

```r
print(validation_plot)
```

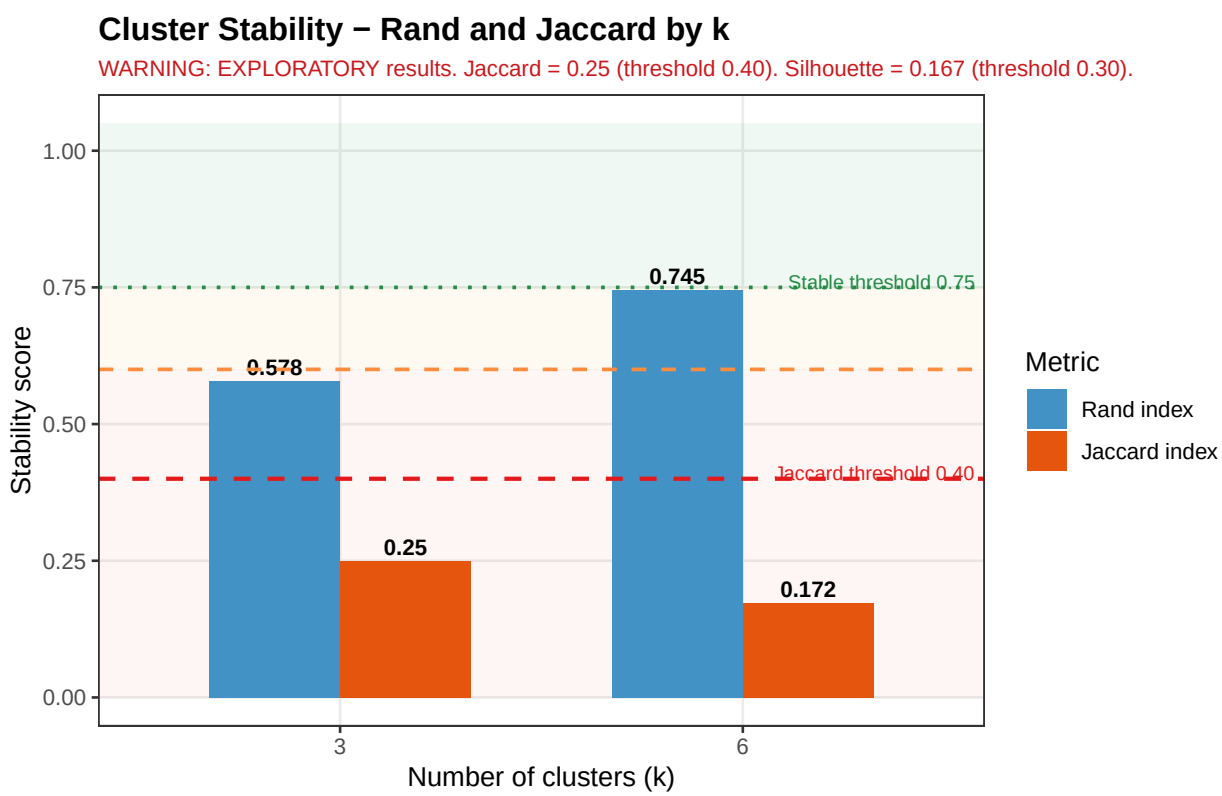



```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_validation_index_plot.png"),
  plot = validation_plot,
  width = 7,
  height = 4.5,
  dpi = 300
)
```

```r
stability_plot_data <- data.table::melt(
  copy(stability_summary),
  id.vars = "k",
  measure.vars = c("rand", "jaccard"),
  variable.name = "metric",
  value.name = "score"
)

# v2.0: Quality-aware subtitle - explicitly flags red-zone result
stability_quality_label <- if (!sil_pass || !jaccard_pass) {
  paste0(
    "WARNING: EXPLORATORY results. ",
    "Jaccard = ", round(selected_jaccard_v2, 3),
    " (threshold 0.40). ",
    "Silhouette = ", round(selected_silhouette, 3),
    " (threshold 0.30)."
```

```r
  )
} else {
  paste0(
    "Cluster quality meets thresholds. ",
    "Jaccard = ", round(selected_jaccard_v2, 3),
    ", Silhouette = ", round(selected_silhouette, 3), "."
  )
}

stability_plot <- ggplot2::ggplot(
  stability_plot_data,
  ggplot2::aes(x = factor(k), y = score, fill = metric)
) +
  ggplot2::annotate(
    "rect", xmin = -Inf, xmax = Inf,
    ymin = 0.75, ymax = 1.05, alpha = 0.08, fill = "#31a354"
  ) +
  ggplot2::annotate(
    "rect", xmin = -Inf, xmax = Inf,
    ymin = 0.60, ymax = 0.75, alpha = 0.08, fill = "#fec44f"
  ) +
  ggplot2::annotate(
    "rect", xmin = -Inf, xmax = Inf,
    ymin = 0.00, ymax = 0.60, alpha = 0.06, fill = "#fb6a4a"
  ) +
  ggplot2::geom_col(position = "dodge", width = 0.65) +
  ggplot2::geom_text(
    ggplot2::aes(label = round(score, 3)),
    position = ggplot2::position_dodge(width = 0.65),
    vjust = -0.4, size = 3.2, fontface = "bold"
  ) +
  ggplot2::geom_hline(
    yintercept = 0.40, linetype = "dashed",
    color = "#e31a1c", linewidth = 0.8
  ) +
  ggplot2::geom_hline(
    yintercept = 0.60, linetype = "dashed",
    color = "#fd8d3c", linewidth = 0.7
  ) +
  ggplot2::geom_hline(
    yintercept = 0.75, linetype = "dotted",
    color = "#238b45", linewidth = 0.7
  ) +
  ggplot2::annotate(
    "text", x = Inf, y = 0.41,
    label = "Jaccard threshold 0.40",
    hjust = 1.05, size = 2.8, color = "#e31a1c"
  ) +
  ggplot2::annotate(
    "text", x = Inf, y = 0.76,
    label = "Stable threshold 0.75",
    hjust = 1.05, size = 2.8, color = "#238b45"
  ) +
```

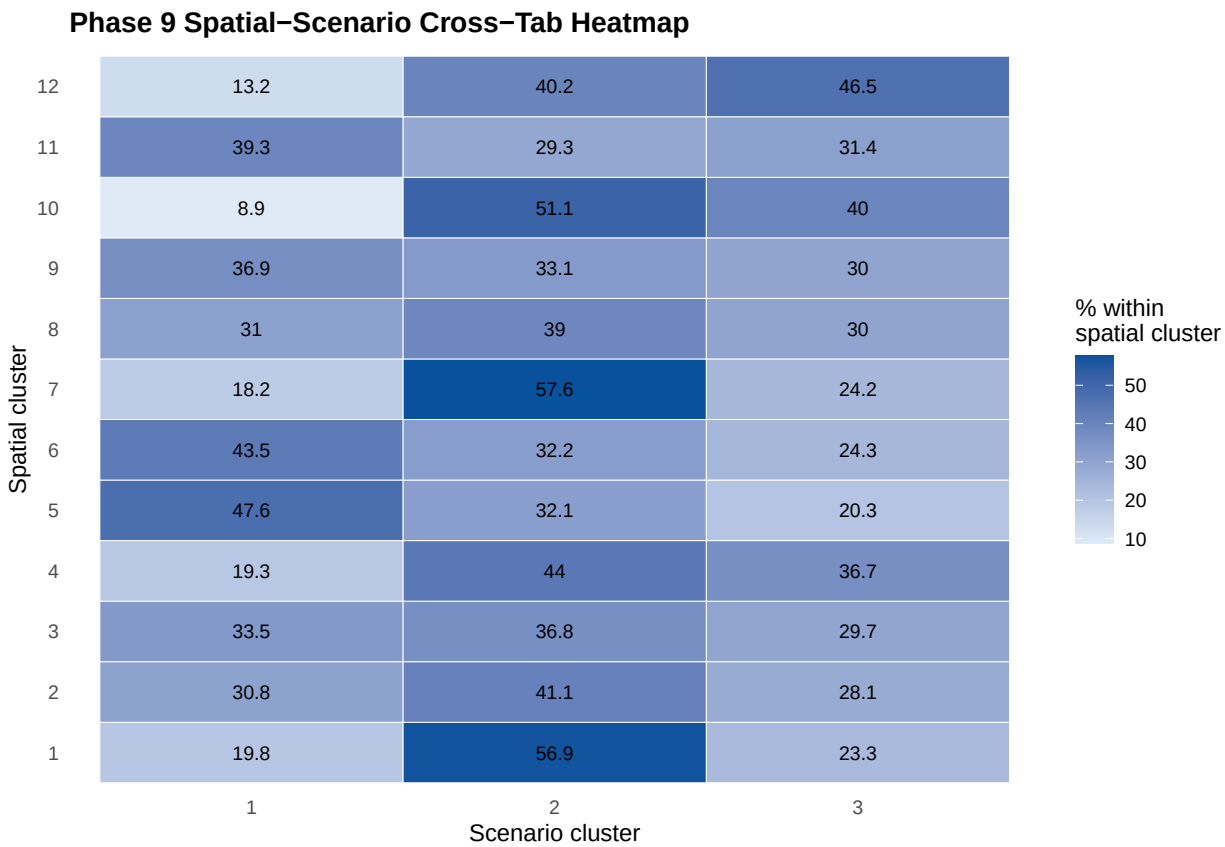
```
  ggplot2::scale_fill_manual(
    values = c("rand" = "#4292c6", "jaccard" = "#e6550d"),
    labels = c("rand" = "Rand index", "jaccard" = "Jaccard index")
  ) +
  ggplot2::labs(
    title    = "Cluster Stability - Rand and Jaccard by k",
    subtitle = stability_quality_label,
    x        = "Number of clusters (k)",
    y        = "Stability score",
    fill     = "Metric"
  ) +
  ggplot2::coord_cartesian(ylim = c(0, 1.05)) +
  ggplot2::theme_bw(base_size = 11) +
  ggplot2::theme(
    plot.title    = ggplot2::element_text(face = "bold"),
    plot.subtitle = ggplot2::element_text(
      color = if (!sil_pass || !jaccard_pass) "#cb181d" else "#238b45",
      size  = 9
    ),
    panel.grid.minor = ggplot2::element_blank()
  )

print(stability_plot)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_stability_plot.png"),
  plot = stability_plot,
```

```r
  width = 7,
  height = 4.5,
  dpi = 300
)
```

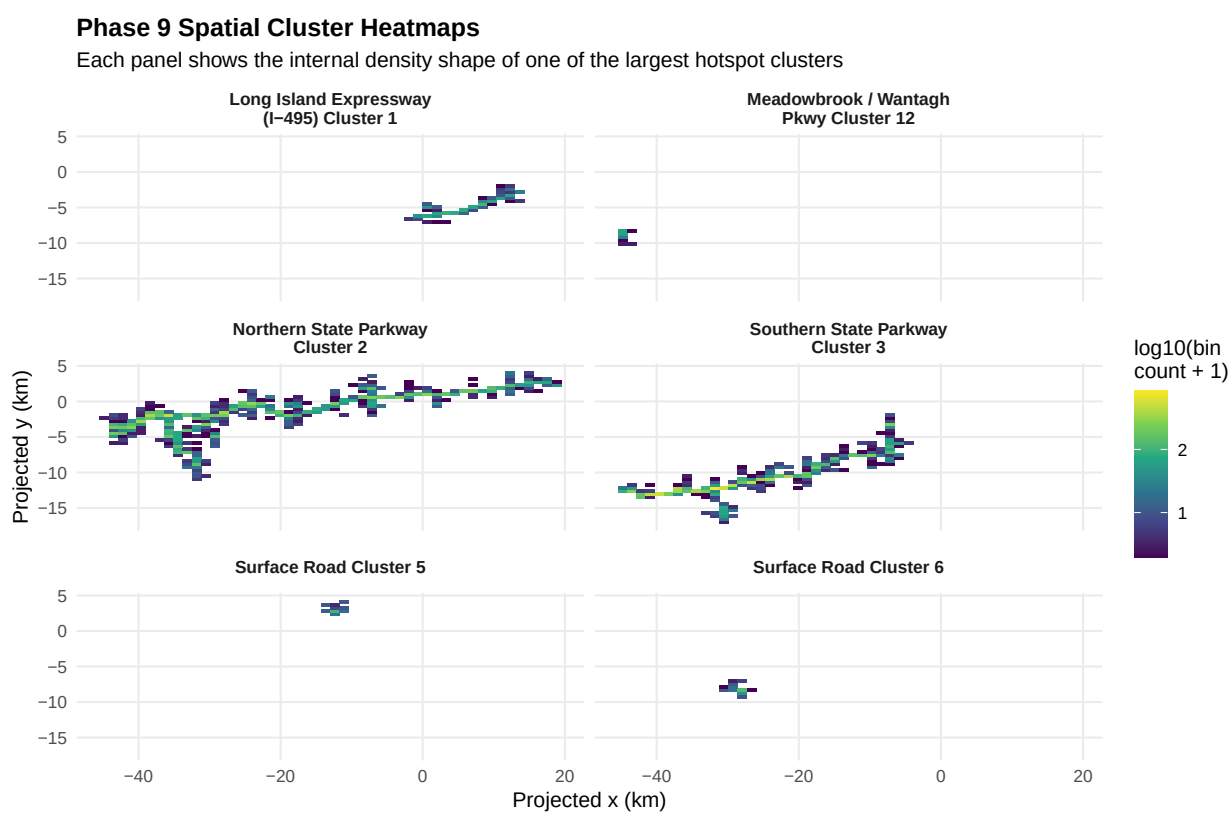
```r
heatmap_data <- merge(
  spatial_scenario_crosstab,
  scenario_final_names[, .(scenario_cluster, scenario_name_short = wrap_label(final_name, width = 18))]
  by = "scenario_cluster",
  all.x = TRUE
)

heatmap_data <- merge(
  heatmap_data,
  spatial_final_names[, .(spatial_cluster, spatial_name_short = wrap_label(final_name, width = 20))],
  by = "spatial_cluster",
  all.x = TRUE
)

heatmap_plot <- ggplot2::ggplot(
  heatmap_data,
  ggplot2::aes(
    x = factor(scenario_cluster),
    y = factor(spatial_cluster),
    fill = spatial_cluster_pct
  )
) +
  ggplot2::geom_tile(color = "white") +
  ggplot2::geom_text(ggplot2::aes(label = round(spatial_cluster_pct, 1)), size = 3) +
  ggplot2::scale_fill_gradient(low = "#deebf7", high = "#08519c") +
  ggplot2::labs(
    title = "Phase 9 Spatial-Scenario Cross-Tab Heatmap",
    x = "Scenario cluster",
    y = "Spatial cluster",
    fill = "% within\nspatial cluster"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid = ggplot2::element_blank()
  )

print(heatmap_plot)
```




```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_spatial_scenario_heatmap.png"),
  plot = heatmap_plot,
  width = 8,
  height = 5.5,
  dpi = 300
)
```

```r
top_spatial_clusters <- non_noise_assignments[order(-cluster_size)][1:min(.N, 6), spatial_cluster]

spatial_cluster_heatmap_data <- merge(
  df_spatial_plot[spatial_cluster %in% top_spatial_clusters],
  spatial_final_names[, .(spatial_cluster, final_name)],
  by = "spatial_cluster",
  all.x = TRUE
)

spatial_cluster_heatmap_data[, final_name_short := wrap_label(final_name, width = 26)]

spatial_cluster_facet_heatmap <- ggplot2::ggplot(
  spatial_cluster_heatmap_data,
  ggplot2::aes(
    x = x_km,
    y = y_km,
    fill = after_stat(log10(count + 1))
  )
```

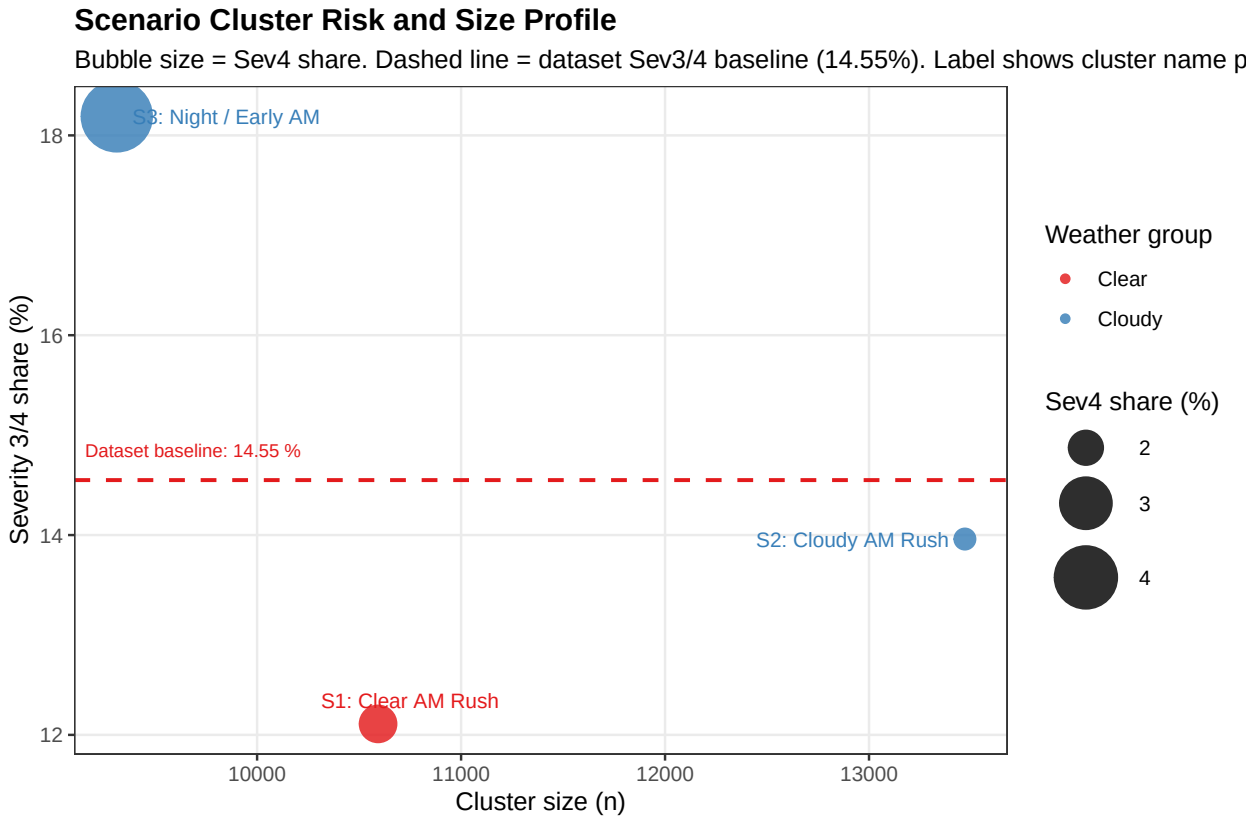
```r
) +
  ggplot2::geom_bin2d(bins = 50) +
  ggplot2::scale_fill_viridis_c(
    option = "D",
    name = "log10(bin\ncount + 1)"
  ) +
  ggplot2::coord_equal() +
  ggplot2::facet_wrap(~ final_name_short, ncol = 2) +
  ggplot2::labs(
    title = "Phase 9 Spatial Cluster Heatmaps",
    subtitle = "Each panel shows the internal density shape of one of the largest hotspot clusters",
    x = "Projected x (km)",
    y = "Projected y (km)"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank(),
    strip.text = ggplot2::element_text(face = "bold")
  )

print(spatial_cluster_facet_heatmap)
```



```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_spatial_cluster_facets_heatmap.png"),
  plot = spatial_cluster_facet_heatmap,
  width = 9,
  height = 7,
  dpi = 300
)
```

```r
dataset_sev34_pct <- round(100 * mean(as.integer(df_clustered$Severity) %in% c(3, 4)), 2)

risk_bubble_data <- merge(
  scenario_severity_profile,
  scenario_weather_time_profile[, .(
    scenario_cluster,
    dominant_weather_group,
    dominant_hour_band,
    mean_visibility
  )],
  by = "scenario_cluster",
  all.x = TRUE
)

risk_bubble_data <- merge(
  risk_bubble_data,
  scenario_final_names[, .(scenario_cluster, final_name)],
  by = "scenario_cluster",
  all.x = TRUE
)

# v2.0: Use truncated final name as label; add sev4_vs_baseline annotation
risk_bubble_data[, short_label := paste0(
  "S", scenario_cluster, ": ",
  substr(gsub(" ---.*", "", final_name), 1, 18)
)]
risk_bubble_data <- merge(
  risk_bubble_data,
  scenario_severity_profile[, .(scenario_cluster, sev4_vs_baseline)],
  by = "scenario_cluster", all.x = TRUE
)

risk_bubble_plot <- ggplot2::ggplot(
  risk_bubble_data,
  ggplot2::aes(
    x     = n,
    y     = sev34_pct,
    size  = pmax(sev4_pct, 0.1),
    color = dominant_weather_group
  )
) +
  ggplot2::geom_hline(
    yintercept = dataset_sev34_pct, linetype = "dashed",
    color = "#e31a1c", linewidth = 0.8
  ) +
```

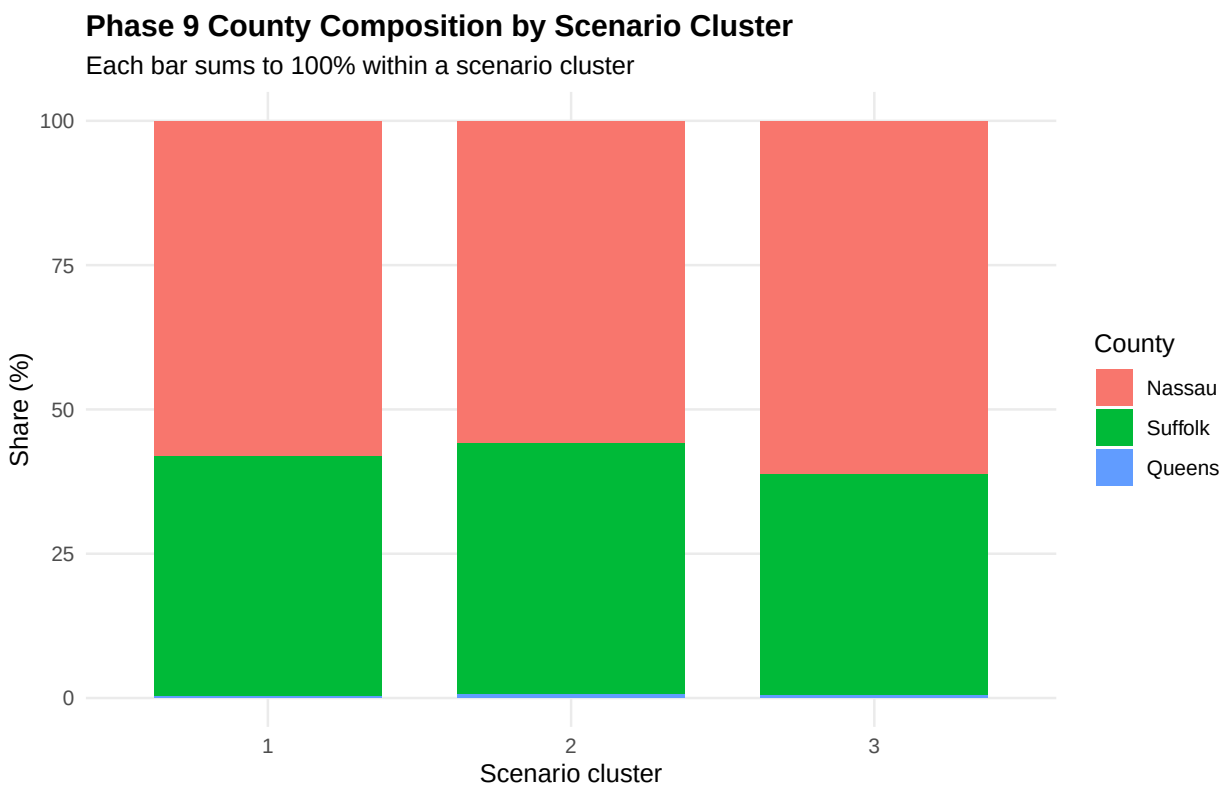
```r
  ggplot2::annotate(
    "text", x = -Inf, y = dataset_sev34_pct + 0.3,
    label = paste("Dataset baseline:", dataset_sev34_pct, "%"),
    hjust = -0.05, size = 2.9, color = "#e31a1c"
  ) +
  ggplot2::geom_point(alpha = 0.82) +
  ggrepel::geom_text_repel(
    ggplot2::aes(label = short_label),
    size = 3, show.legend = FALSE,
    box.padding = 0.4, max.overlaps = 10
  ) +
  ggplot2::scale_size_continuous(range = c(4, 14)) +
  ggplot2::scale_color_brewer(palette = "Set1") +
  ggplot2::labs(
    title    = "Scenario Cluster Risk and Size Profile",
    subtitle = paste0(
      "Bubble size = Sev4 share. Dashed line = dataset Sev3/4 baseline (",
      dataset_sev34_pct, "%). Label shows cluster name prefix."
    ),
    x     = "Cluster size (n)",
    y     = "Severity 3/4 share (%)",
    size  = "Sev4 share (%)",
    color = "Weather group"
  ) +
  ggplot2::theme_bw(base_size = 11) +
  ggplot2::theme(
    plot.title       = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank(),
    legend.position  = "right"
  )

print(risk_bubble_plot)
```




```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_scenario_risk_bubble.png"),
  plot = risk_bubble_plot,
  width = 7.5,
  height = 5,
  dpi = 300
)
```

```
county_plot_data <- data.table::melt(
  copy(scenario_county_profile),
  id.vars = c("scenario_cluster", "n", "dominant_county"),
  measure.vars = c("Nassau_pct", "Suffolk_pct", "Queens_pct"),
  variable.name = "county",
  value.name = "pct"
)

county_plot_data[, county := factor(
  county,
  levels = c("Nassau_pct", "Suffolk_pct", "Queens_pct"),
  labels = c("Nassau", "Suffolk", "Queens")
)]

county_plot <- ggplot2::ggplot(
  county_plot_data,
  ggplot2::aes(x = factor(scenario_cluster), y = pct, fill = county)
) +
  ggplot2::geom_col(width = 0.75) +
```

```
  ggplot2::labs(
    title = "Phase 9 County Composition by Scenario Cluster",
    subtitle = "Each bar sums to 100% within a scenario cluster",
    x = "Scenario cluster",
    y = "Share (%)",
    fill = "County"
  ) +
  ggplot2::theme_minimal() +
  ggplot2::theme(
    plot.title = ggplot2::element_text(face = "bold"),
    panel.grid.minor = ggplot2::element_blank()
  )

print(county_plot)
```

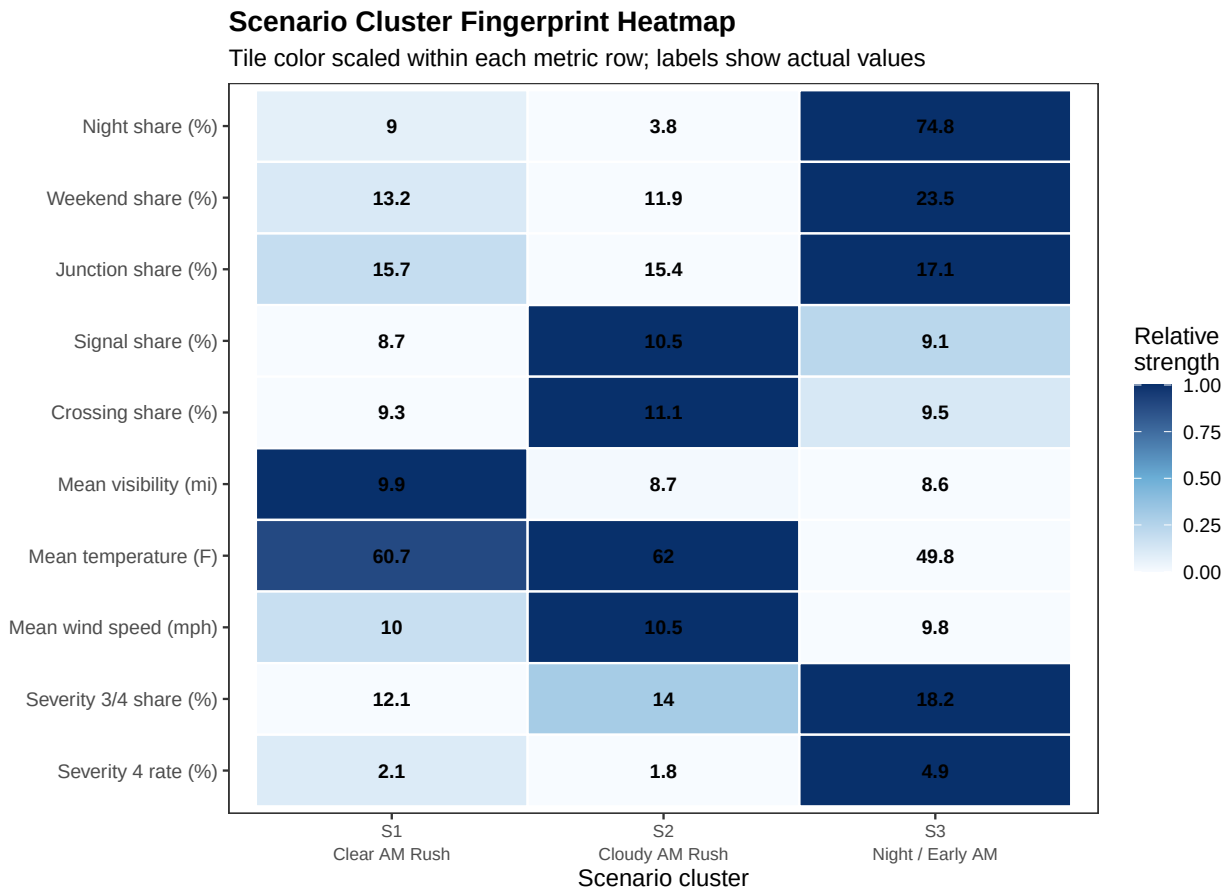



```
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_county_composition.png"),
  plot = county_plot,
  width = 7.5,
  height = 4.8,
  dpi = 300
)
```

```
# v2.0: Add sev4_rate row; use final cluster names on x-axis
fingerprint_source <- merge(
  scenario_weather_time_profile[, .(
    scenario_cluster, night_pct, weekend_pct,
```

```r
    junction_pct, traffic_signal_pct, crossing_pct,
    mean_visibility, mean_temperature, mean_wind_speed
  )],
  scenario_severity_profile[, .(scenario_cluster, sev34_pct, sev4_rate)],
  by = "scenario_cluster", all.x = TRUE
)

# Attach short final names for x-axis labels
fingerprint_source <- merge(
  fingerprint_source,
  scenario_final_names[, .(
    scenario_cluster,
    cluster_label = paste0(
      "S", scenario_cluster, "\n",
      substr(gsub(" ---.*", "", final_name), 1, 16)
    )
  )],
  by = "scenario_cluster", all.x = TRUE
)

fingerprint_long <- data.table::melt(
  fingerprint_source,
  id.vars       = c("scenario_cluster", "cluster_label"),
  variable.name = "metric",
  value.name    = "metric_value"
)

metric_labels <- c(
  night_pct          = "Night share (%)",
  weekend_pct        = "Weekend share (%)",
  junction_pct       = "Junction share (%)",
  traffic_signal_pct = "Signal share (%)",
  crossing_pct       = "Crossing share (%)",
  mean_visibility    = "Mean visibility (mi)",
  mean_temperature   = "Mean temperature (F)",
  mean_wind_speed    = "Mean wind speed (mph)",
  sev34_pct          = "Severity 3/4 share (%)",
  sev4_rate          = "Severity 4 rate (%)"
)

fingerprint_long[, metric_label := factor(
  metric_labels[as.character(metric)],
  levels = rev(metric_labels)
)]
fingerprint_long[, scaled_value := rescale01(metric_value), by = metric]

fingerprint_plot <- ggplot2::ggplot(
  fingerprint_long,
  ggplot2::aes(
    x    = reorder(cluster_label, scenario_cluster),
    y    = metric_label,
    fill = scaled_value
  )
```

```r
) +
  ggplot2::geom_tile(color = "white", linewidth = 0.5) +
  ggplot2::geom_text(
    ggplot2::aes(label = round(metric_value, 1)),
    size = 3, fontface = "bold"
  ) +
  ggplot2::scale_fill_gradient2(
    low = "#f7fbff", mid = "#6baed6", high = "#08306b",
    midpoint = 0.5, na.value = "grey90"
  ) +
  ggplot2::labs(
    title    = "Scenario Cluster Fingerprint Heatmap",
    subtitle = "Tile color scaled within each metric row; labels show actual values",
    x        = "Scenario cluster",
    y        = NULL,
    fill     = "Relative\nstrength"
  ) +
  ggplot2::theme_bw(base_size = 11) +
  ggplot2::theme(
    plot.title       = ggplot2::element_text(face = "bold"),
    panel.grid       = ggplot2::element_blank(),
    axis.text.x      = ggplot2::element_text(size = 8, lineheight = 1.1),
    axis.text.y      = ggplot2::element_text(size = 9)
  )

print(fingerprint_plot)
```




```r
ggplot2::ggsave(
  filename = file.path(output_dir, "phase9_cluster_fingerprint_heatmap.png"),
  plot = fingerprint_plot,
  width = 8,
  height = 5.8,
  dpi = 300
)
```

```r
write.csv(
  df_clustered,
  file.path(output_dir, "li_accidents_clustered.csv"),
  row.names = FALSE
)

table_exports <- list(
  phase6_validation_indices = validation_table,
  phase6_validation_metadata = validation_metadata,
  phase6_stability_summary = stability_summary,
  phase6_selection_summary = selection_summary,
  phase7_eps_trials = eps_trials,
  phase7_parameter_summary = parameter_summary,
  phase7_dbscan_cluster_sizes = spatial_cluster_sizes,
  phase7_corridor_assignments = non_noise_assignments,
  phase8_severity_profile = scenario_severity_profile,
```

```r
  phase8_county_profile = scenario_county_profile,
  phase8_weather_time_profile = scenario_weather_time_profile,
  phase8_spatial_scenario_crosstab = spatial_scenario_crosstab,
  phase8_spatial_scenario_dominance = spatial_scenario_dominance,
  phase8_scenario_final_names = scenario_final_names,
  phase8_spatial_final_names = spatial_final_names
)

if (!is.null(validation_repeat_results)) {
  table_exports$phase6_validation_repeats <- validation_repeat_results
}

if (!is.null(stability_repeat_results)) {
  table_exports$phase6_stability_repeats <- stability_repeat_results
}

invisible(lapply(names(table_exports), function(table_name) {
  fwrite(table_exports[[table_name]], file.path(table_dir, paste0(table_name, ".csv")))
}))

figure_exports <- c(
  "phase5_kproto_elbow_plot.png",
  "phase6_cluster_profiles.pdf",
  "phase7_knn_distance_plot.png",
  "phase7_spatial_hotspot_map.png",
  "phase7_spatial_density_heatmap.png",
  "phase9_validation_index_plot.png",
  "phase9_stability_plot.png",
  "phase9_spatial_scenario_heatmap.png",
  "phase9_spatial_cluster_facets_heatmap.png",
  "phase9_scenario_risk_bubble.png",
  "phase9_county_composition.png",
  "phase9_cluster_fingerprint_heatmap.png"
)

invisible(lapply(figure_exports, function(file_name) {
  source_path <- file.path(output_dir, file_name)
  if (file.exists(source_path)) {
    file.copy(source_path, file.path(figure_dir, file_name), overwrite = TRUE)
  }
}))

file.copy(
  file.path(output_dir, "li_accidents_clustered.csv"),
  file.path(table_dir, "li_accidents_clustered.csv"),
  overwrite = TRUE
)
```

```
## [1] TRUE
```

```r
export_manifest <- data.table(
  artifact_type = c(
    rep("figure", length(figure_exports)),
```

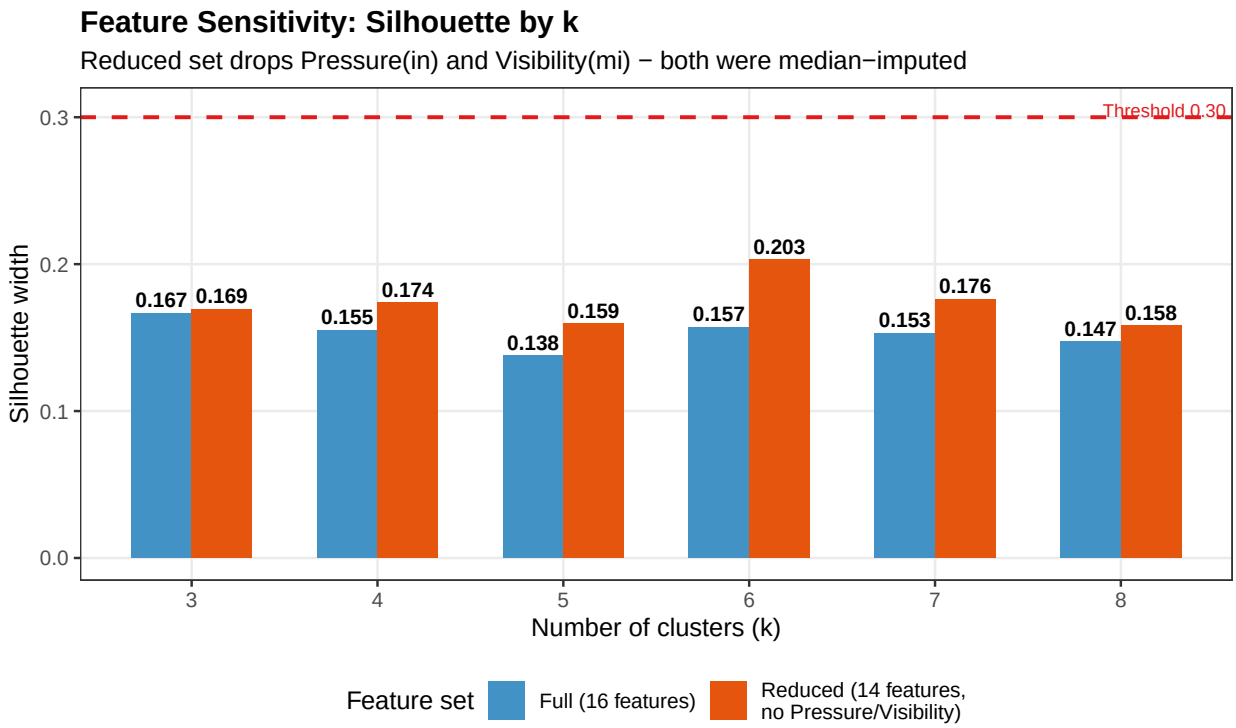
```r
    rep("table", length(table_exports)),
    "table"
  ),
  artifact_name = c(
    figure_exports,
    paste0(names(table_exports), ".csv"),
    "li_accidents_clustered.csv"
  ),
  target_path = c(
    file.path(figure_dir, figure_exports),
    file.path(table_dir, paste0(names(table_exports), ".csv")),
    file.path(table_dir, "li_accidents_clustered.csv")
  )
)

fwrite(export_manifest, file.path(output_dir, "phase9_export_manifest.csv"))

knitr::kable(export_manifest, caption = "Phase 9 export manifest")
```

Table 61: Phase 9 export manifest

artifact_type	artifact_name	target_path
figure	phase5_kproto_elbow_plot.png	local_artifacts/output/figures/phase5_kproto_elbow_plot.png
figure	phase6_cluster_profiles.pdf	local_artifacts/output/figures/phase6_cluster_profiles.pdf
figure	phase7_knn_distance_plot.png	local_artifacts/output/figures/phase7_knn_distance_plot.png
figure	phase7_spatial_hotspot_map.png	local_artifacts/output/figures/phase7_spatial_hotspot_map.png
figure	phase7_spatial_density_heatmap.png	local_artifacts/output/figures/phase7_spatial_density_heatmap.png
figure	phase9_validation_index_plot.png	local_artifacts/output/figures/phase9_validation_index_plot.png
figure	phase9_stability_plot.png	local_artifacts/output/figures/phase9_stability_plot.png
figure	phase9_spatial_scenario_heatmap.png	local_artifacts/output/figures/phase9_spatial_scenario_heatmap.png
figure	phase9_spatial_cluster_facets_heatmap.png	local_artifacts/output/figures/phase9_spatial_cluster_facets_heatmap.p
figure	phase9_scenario_risk_bubble.png	local_artifacts/output/figures/phase9_scenario_risk_bubble.png
figure	phase9_county_composition.png	local_artifacts/output/figures/phase9_county_composition.png
figure	phase9_cluster_fingerprint_heatmap.png	local_artifacts/output/figures/phase9_cluster_fingerprint_heatmap.png
table	phase6_validation_indices.csv	local_artifacts/output/tables/phase6_validation_indices.csv
table	phase6_validation_metadata.csv	local_artifacts/output/tables/phase6_validation_metadata.csv
table	phase6_stability_summary.csv	local_artifacts/output/tables/phase6_stability_summary.csv
table	phase6_selection_summary.csv	local_artifacts/output/tables/phase6_selection_summary.csv
table	phase7_eps_trials.csv	local_artifacts/output/tables/phase7_eps_trials.csv
table	phase7_parameter_summary.csv	local_artifacts/output/tables/phase7_parameter_summary.csv
table	phase7_dbscan_cluster_sizes.csv	local_artifacts/output/tables/phase7_dbscan_cluster_sizes.csv
table	phase7_corridor_assignments.csv	local_artifacts/output/tables/phase7_corridor_assignments.csv
table	phase8_severity_profile.csv	local_artifacts/output/tables/phase8_severity_profile.csv
table	phase8_county_profile.csv	local_artifacts/output/tables/phase8_county_profile.csv
table	phase8_weather_time_profile.csv	local_artifacts/output/tables/phase8_weather_time_profile.csv
table	phase8_spatial_scenario_crosstab.csv	local_artifacts/output/tables/phase8_spatial_scenario_crosstab.csv
table	phase8_spatial_scenario_dominance.csv	local_artifacts/output/tables/phase8_spatial_scenario_dominance.csv
table	phase8_scenario_final_names.csv	local_artifacts/output/tables/phase8_scenario_final_names.csv
table	phase8_spatial_final_names.csv	local_artifacts/output/tables/phase8_spatial_final_names.csv
table	phase6_validation_repeats.csv	local_artifacts/output/tables/phase6_validation_repeats.csv
table	phase6_stability_repeats.csv	local_artifacts/output/tables/phase6_stability_repeats.csv
table	li_accidents_clustered.csv	local_artifacts/output/tables/li_accidents_clustered.csv

```r
# v2.0: Side-by-side silhouette comparison - full vs reduced feature set.
# Shows whether dropping imputed features (Pressure, Visibility) improves quality.
if (file.exists(
  file.path(output_dir, "phase6_feature_silhouette_comparison.csv")
)) {
  feat_compare <- data.table::fread(
    file.path(output_dir, "phase6_feature_silhouette_comparison.csv")
  )

  feat_compare[, feature_set_label := ifelse(
    feature_set == "full",
    "Full (16 features)",
    "Reduced (14 features,\nno Pressure/Visibility)"
  )]

  feature_sensitivity_plot <- ggplot2::ggplot(
    feat_compare,
    ggplot2::aes(
      x    = factor(k),
      y    = silhouette,
      fill = feature_set_label
    )
  ) +
    ggplot2::geom_col(position = "dodge", width = 0.65) +
    ggplot2::geom_text(
      ggplot2::aes(label = round(silhouette, 3)),
      position = ggplot2::position_dodge(width = 0.65),
      vjust = -0.4, size = 3, fontface = "bold"
    ) +
    ggplot2::geom_hline(
      yintercept = 0.30, linetype = "dashed",
      color = "#e31a1c", linewidth = 0.8
    ) +
    ggplot2::annotate(
      "text", x = Inf, y = 0.305,
      label = "Threshold 0.30",
      hjust = 1.05, size = 2.8, color = "#e31a1c"
    ) +
    ggplot2::scale_fill_manual(
      values = c(
        "Full (16 features)"                        = "#4292c6",
        "Reduced (14 features,\nno Pressure/Visibility)" = "#e6550d"
      )
    ) +
    ggplot2::labs(
      title    = "Feature Sensitivity: Silhouette by k",
      subtitle = "Reduced set drops Pressure(in) and Visibility(mi) - both were median-imputed",
      x        = "Number of clusters (k)",
      y        = "Silhouette width",
      fill     = "Feature set"
    ) +
    ggplot2::theme_bw(base_size = 11) +
    ggplot2::theme(
```

```r
      plot.title       = ggplot2::element_text(face = "bold"),
      panel.grid.minor = ggplot2::element_blank(),
      legend.position  = "bottom"
    )

  print(feature_sensitivity_plot)

  ggplot2::ggsave(
    filename = file.path(
      output_dir, "figures", "phase9_feature_sensitivity.png"
    ),
    plot   = feature_sensitivity_plot,
    width  = 7.5, height = 4.5, dpi = 300
  )
} else {
  cat("Feature silhouette comparison not available - skipping plot.\n")
}
```




```r
interactive_exports <- data.table(
  artifact_name = c(
    "phase9_validation_index_interactive.html",
    "phase9_stability_interactive.html",
    "phase9_spatial_hotspot_interactive.html",
    "phase9_spatial_density_interactive.html",
    "phase9_scenario_risk_interactive.html",
    "phase9_county_composition_interactive.html",
    "phase9_cluster_fingerprint_interactive.html",
    "phase9_spatial_scenario_heatmap_interactive.html"
  ),
```

```r
  target_path = file.path(
    interactive_dir,
    c(
      "phase9_validation_index_interactive.html",
      "phase9_stability_interactive.html",
      "phase9_spatial_hotspot_interactive.html",
      "phase9_spatial_density_interactive.html",
      "phase9_scenario_risk_interactive.html",
      "phase9_county_composition_interactive.html",
      "phase9_cluster_fingerprint_interactive.html",
      "phase9_spatial_scenario_heatmap_interactive.html"
    )
  ),
  saved = FALSE,
  note = ifelse(
    interactive_exports_enabled,
    "Saved as interactive HTML",
    "Install plotly and htmlwidgets to enable interactive exports"
  )
)

if (interactive_exports_enabled) {
  spatial_interactive_data <- copy(df_spatial_plot)
  if (nrow(spatial_interactive_data) > 15000) {
    set.seed(42)
    spatial_interactive_data <- spatial_interactive_data[
      ,
      .SD[sample(.N, size = min(.N, max(1L, ceiling(15000 * .N / nrow(df_spatial_plot)))))],
      by = cluster_label
    ]
  }

  spatial_map_interactive <- ggplot2::ggplot(
    spatial_interactive_data,
    ggplot2::aes(x = x_km, y = y_km, color = cluster_label,
                 text = paste0(
                   "Cluster: ", cluster_label,
                   "<br>x_km: ", round(x_km, 2),
                   "<br>y_km: ", round(y_km, 2)
                 ))
  ) +
    ggplot2::geom_point(size = 0.6, alpha = 0.7) +
    ggplot2::scale_color_manual(values = cluster_colors) +
    ggplot2::labs(
      title = "Interactive Spatial Hotspot View",
      subtitle = "Rendered on a representative sample for smoother browsing",
      x = "Projected x (km)",
      y = "Projected y (km)",
      color = "Cluster"
    ) +
    ggplot2::theme_minimal()

  interactive_widgets <- list(
```

```
    phase9_validation_index_interactive.html = plotly::ggplotly(validation_plot, tooltip = c("x", "y"))
    phase9_stability_interactive.html = plotly::ggplotly(stability_plot, tooltip = c("x", "y", "fill"))
    phase9_spatial_hotspot_interactive.html = plotly::ggplotly(spatial_map_interactive, tooltip = "text
    phase9_spatial_density_interactive.html = plotly::plot_ly(
      spatial_interactive_data,
      x = ~x_km,
      y = ~y_km,
      type = "histogram2d",
      colorscale = "Viridis",
      nbinsx = 80,
      nbinsy = 60
    ) |>
      plotly::layout(
        title = "Interactive Spatial Density Heatmap",
        xaxis = list(title = "Projected x (km)"),
        yaxis = list(title = "Projected y (km)")
      ),
    phase9_scenario_risk_interactive.html = plotly::ggplotly(risk_bubble_plot, tooltip = c("x", "y", "s
    phase9_county_composition_interactive.html = plotly::ggplotly(county_plot, tooltip = c("x", "y", "f
    phase9_cluster_fingerprint_interactive.html = plotly::ggplotly(fingerprint_plot, tooltip = c("x", "
    phase9_spatial_scenario_heatmap_interactive.html = plotly::ggplotly(heatmap_plot, tooltip = c("x",
  )

  invisible(lapply(names(interactive_widgets), function(file_name) {
    htmlwidgets::saveWidget(
      interactive_widgets[[file_name]],
      file = file.path(interactive_dir, file_name),
      selfcontained = FALSE
    )
  }))

  interactive_exports[, saved := file.exists(target_path)]
}

fwrite(interactive_exports, file.path(output_dir, "phase9_interactive_manifest.csv"))

knitr::kable(interactive_exports, caption = "Phase 9 interactive export manifest")
```

Table 62: Phase 9 interactive export manifest

artifact_name	target_path	saved	note
phase9_validation_index_interactive.html	local_artifacts/output/interactive/phase9_validation_index_interactive.html	FALSE	install plotly and htmlwidgets to enable interactive exports
phase9_stability_interactive.html	local_artifacts/output/interactive/phase9_stability_interactive.html	FALSE	install plotly and htmlwidgets to enable interactive exports
phase9_spatial_hotspot_interactive.html	local_artifacts/output/interactive/phase9_spatial_hotspot_interactive.html	FALSE	install plotly and htmlwidgets to enable interactive exports
phase9_spatial_density_interactive.html	local_artifacts/output/interactive/phase9_spatial_density_interactive.html	FALSE	install plotly and htmlwidgets to enable interactive exports
phase9_scenario_risk_interactive.html	local_artifacts/output/interactive/phase9_scenario_risk_interactive.html	FALSE	install plotly and htmlwidgets to enable interactive exports

artifact_name	target_path	saved	note
phase9_county_composition_interactive.html	local/artifacts/output/interactive/phase9_county_composition_interactive.html	FALSE	install plotly and htmlwidgets to enable interactive exports
phase9_cluster_fingerprint_interactive.html	local/artifacts/output/interactive/phase9_cluster_fingerprint_interactive.html	FALSE	install plotly and htmlwidgets to enable interactive exports
phase9_spatial_scenario_heatmap_interactive.html	local/artifacts/output/interactive/phase9_spatial_scenario_heatmap_interactive.html	FALSE	install plotly and htmlwidgets to enable interactive exports

```r
phase_9_summary <- data.table(
  checkpoint = c(
    "Figure directory created",
    "Table directory created",
    "Validation plot saved",
    "Stability plot saved",
    "Heatmap saved",
    "Spatial density heatmap saved",
    "Spatial cluster facet heatmap saved",
    "Scenario risk bubble saved",
    "County composition plot saved",
    "Cluster fingerprint heatmap saved",
    "Labeled CSV exported",
    "Interactive manifest saved",
    "Export manifest saved"
  ),
  status = c(
    dir.exists(figure_dir),
    dir.exists(table_dir),
    file.exists(file.path(figure_dir, "phase9_validation_index_plot.png")),
    file.exists(file.path(figure_dir, "phase9_stability_plot.png")),
    file.exists(file.path(figure_dir, "phase9_spatial_scenario_heatmap.png")),
    file.exists(file.path(figure_dir, "phase7_spatial_density_heatmap.png")),
    file.exists(file.path(figure_dir, "phase9_spatial_cluster_facets_heatmap.png")),
    file.exists(file.path(figure_dir, "phase9_scenario_risk_bubble.png")),
    file.exists(file.path(figure_dir, "phase9_county_composition.png")),
    file.exists(file.path(figure_dir, "phase9_cluster_fingerprint_heatmap.png")),
    file.exists(file.path(table_dir, "li_accidents_clustered.csv")),
    file.exists(file.path(output_dir, "phase9_interactive_manifest.csv")),
    file.exists(file.path(output_dir, "phase9_export_manifest.csv"))
  )
)

fwrite(phase_9_summary, file.path(output_dir, "phase9_summary.csv"))

knitr::kable(phase_9_summary, caption = "Phase 9 readiness summary")
```

Table 63: Phase 9 readiness summary

checkpoint	status
Figure directory created	TRUE
Table directory created	TRUE
Validation plot saved	TRUE

checkpoint	status
Stability plot saved	TRUE
Heatmap saved	TRUE
Spatial density heatmap saved	TRUE
Spatial cluster facet heatmap saved	TRUE
Scenario risk bubble saved	TRUE
County composition plot saved	TRUE
Cluster fingerprint heatmap saved	TRUE
Labeled CSV exported	TRUE
Interactive manifest saved	TRUE
Export manifest saved	TRUE

10 Phase 10: Final Report and Submission Checklist

This phase keeps a simple readiness tracker for the final report assets and submission handoff.

```
phase_10_checklist <- data.table(
  deliverable = c(
    "Long Island SRS source",
    "Execution plan source",
    "Main workflow Rmd",
    "Cleaned dataset",
    "Clustered labeled dataset",
    "Figure export folder",
    "Table export folder",
    "Interactive export manifest"
  ),
  path = c(
    "Li_Clustering_SRS.tex",
    "Li_Clustering_Execution_Plan.tex",
    "li_clustering.Rmd",
    input_path,
    file.path(table_dir, "li_accidents_clustered.csv"),
    figure_dir,
    table_dir,
    file.path(output_dir, "phase9_interactive_manifest.csv")
  ),
  ready = c(
    file.exists("Li_Clustering_SRS.tex"),
    file.exists("Li_Clustering_Execution_Plan.tex"),
    file.exists("li_clustering.Rmd"),
    file.exists(input_path),
    file.exists(file.path(table_dir, "li_accidents_clustered.csv")),
    dir.exists(figure_dir),
    dir.exists(table_dir),
    file.exists(file.path(output_dir, "phase9_interactive_manifest.csv"))
  )
)

fwrite(phase_10_checklist, file.path(output_dir, "phase10_readiness_checklist.csv"))
```

```r
knitr::kable(phase_10_checklist, caption = "Phase 10 readiness checklist")
```

Table 64: Phase 10 readiness checklist

deliverable	path	ready
Long Island SRS source	Li_Clustering_SRS.tex	TRUE
Execution plan source	Li_Clustering_Execution_Plan.tex	TRUE
Main workflow Rmd	li_clustering.Rmd	TRUE
Cleaned dataset	../li_accidents_clean.csv	TRUE
Clustered labeled dataset	local_artifacts/output/tables/li_accidents_clustered.csv	TRUE
Figure export folder	local_artifacts/output/figures	TRUE
Table export folder	local_artifacts/output/tables	TRUE
Interactive export manifest	local_artifacts/output/phase9_interactive_manifest.csv	TRUE

```r
phase_10_report_outline <- data.table(
  report_section = c(
    "Introduction and motivation",
    "Dataset description and audit findings",
    "Feature engineering decisions",
    "Method selection rationale",
    "Scenario clustering results",
    "Spatial hotspot results",
    "Interpretation and cross-tab findings",
    "Limitations",
    "Conclusion and safety priorities"
  ),
  main_source = c(
    "Li_Clustering_SRS.tex",
    file.path(table_dir, "phase8_severity_profile.csv"),
    file.path(table_dir, "phase8_weather_time_profile.csv"),
    "Li_Clustering_Execution_Plan.tex",
    file.path(figure_dir, "phase9_validation_index_plot.png"),
    file.path(figure_dir, "phase7_spatial_hotspot_map.png"),
    file.path(table_dir, "phase8_spatial_scenario_crosstab.csv"),
    file.path(output_dir, "phase10_limitations_table.csv"),
    file.path(output_dir, "phase10_research_question_coverage.csv")
  ),
  ready = c(
    file.exists("Li_Clustering_SRS.tex"),
    file.exists(file.path(table_dir, "phase8_severity_profile.csv")),
    file.exists(file.path(table_dir, "phase8_weather_time_profile.csv")),
    file.exists("Li_Clustering_Execution_Plan.tex"),
    file.exists(file.path(figure_dir, "phase9_validation_index_plot.png")),
    file.exists(file.path(figure_dir, "phase7_spatial_hotspot_map.png")),
    file.exists(file.path(table_dir, "phase8_spatial_scenario_crosstab.csv")),
    TRUE,
    TRUE
  )
)

fwrite(phase_10_report_outline, file.path(output_dir, "phase10_report_outline.csv"))
```

```r
knitr::kable(phase_10_report_outline, caption = "Phase 10 report outline and source map")
```

Table 65: Phase 10 report outline and source map

report_section	main_source	ready
Introduction and motivation	Li_Clustering_SRS.tex	TRUE
Dataset description and audit findings	local_artifacts/output/tables/phase8_severity_profile.csv	TRUE
Feature engineering decisions	local_artifacts/output/tables/phase8_weather_time_profile.csv	TRUE
Method selection rationale	Li_Clustering_Execution_Plan.tex	TRUE
Scenario clustering results	local_artifacts/output/figures/phase9_validation_index_plot.pdf	TRUE
Spatial hotspot results	local_artifacts/output/figures/phase7_spatial_hotspot_map.pdf	TRUE
Interpretation and cross-tab findings	local_artifacts/output/tables/phase8_spatial_scenario_crosstab.csv	TRUE
Limitations	local_artifacts/output/phase10_limitations_table.csv	TRUE
Conclusion and safety priorities	local_artifacts/output/phase10_research_question_coverage.csv	TRUE

```r
# v2.0: Cluster quality failure rows prepended FIRST so they appear at the
# top of the rendered table and cannot be missed in the final report.
cluster_quality_rows <- data.table(
  limitation = c(
    "Scenario cluster quality below standard thresholds",
    "Scenario clustering is exploratory only"
  ),
  report_language = c(
    paste0(
      "Silhouette width = ", round(selected_silhouette, 3),
      " (published threshold >= 0.30; Kaufman and Rousseeuw 1990). ",
      "Jaccard stability index = ", round(selected_jaccard_v2, 3),
      " (threshold >= 0.40 for exploratory, >= 0.60 for acceptable; ",
      "Hennig 2007). Both metrics fall below acceptable thresholds, ",
      "indicating that cluster boundaries are weak and that bootstrap ",
      "resamples do not reliably reproduce the same partition."
    ),
    paste0(
      "Because silhouette and Jaccard are both below threshold, ",
      "the k = ", selected_k, " scenario clustering solution should be ",
      "interpreted as a set of candidate patterns for hypothesis generation, ",
      "not as stable or confirmed accident archetypes. ",
      "The elevation of Severity 3/4 in Clusters 4 and 5 is a plausible ",
      "signal but requires replication on a new sample before it can be ",
      "treated as a reliable finding. ",
      "DBSCAN spatial results are separately validated and remain defensible."
    )
  ),
  include_in_final_report = TRUE
)

phase_10_limitations <- rbind(
  cluster_quality_rows,
  data.table(
```

```r
    limitation = c(
      "Duration capping",
      "Precipitation exclusion",
      "Distance exclusion",
      "Severity 2 dominance",
      "Sampled Phase 6 validation",
      "DBSCAN parameter sensitivity"
    ),
    report_language = c(
      "Duration was capped at the 99th percentile to reduce the effect of extreme outliers, so the top 1
      "Precipitation was excluded from the core model because missingness was high enough (20.1%) to mal
      "Distance(mi) was excluded because the large share of zero values suggests a reporting artifact ra
      "Severity 2 dominates the dataset (85% of rows), so subtle differences among severe crashes can be
      if (run_profile == "full_srs") {
        "Phase 6 is configured for full-data validation on a stronger machine."
      } else if (run_profile == "consensus_16gb") {
        paste(
          "Phase 6 uses repeated sampled validation (4 x 4,000 rows) and",
          "repeated stability checks (3 x 2,500 rows, 10 bootstraps each)",
          "to build a consensus k on a 16 GB machine."
        )
      } else {
        "Phase 6 uses single-sample silhouette (5,000 rows) and stability (3,000 rows) checks; full-data
      },
      "Spatial hotspot shapes can shift if eps or minPts changes; corridor labels should be read as data
    ),
    include_in_final_report = TRUE
  )
)

fwrite(phase_10_limitations, file.path(output_dir, "phase10_limitations_table.csv"))

knitr::kable(
  phase_10_limitations,
  caption = paste(
    "Phase 10 limitations table (v2.0).",
    "First two rows document cluster quality failure explicitly."
  )
)
```

Table 66: Phase 10 limitations table (v2.0). First two rows document cluster quality failure explicitly.

limitation	report_language	include_in_final_rep
Scenario cluster quality below standard thresholds	Silhouette width = 0.167 (published threshold >= 0.30; Kaufman and Rousseeuw 1990). Jaccard stability index = 0.25 (threshold >= 0.40 for exploratory, >= 0.60 for acceptable; Hennig 2007). Both metrics fall below acceptable thresholds, indicating that cluster boundaries are weak and that bootstrap resamples do not reliably reproduce the same partition.	TRUE

limitation	report_language	include_in_final_rep
Scenario clustering is exploratory only	Because silhouette and Jaccard are both below threshold, the k = 3 scenario clustering solution should be interpreted as a set of candidate patterns for hypothesis generation, not as stable or confirmed accident archetypes. The elevation of Severity 3/4 in Clusters 4 and 5 is a plausible signal but requires replication on a new sample before it can be treated as a reliable finding. DBSCAN spatial results are separately validated and remain defensible.	TRUE
Duration capping	Duration was capped at the 99th percentile to reduce the effect of extreme outliers, so the top 1% of incidents lose some real variation.	TRUE
Precipitation exclusion	Precipitation was excluded from the core model because missingness was high enough (20.1%) to make routine imputation less defensible.	TRUE
Distance exclusion	Distance(mi) was excluded because the large share of zero values suggests a reporting artifact rather than a stable behavioral feature.	TRUE
Severity 2 dominance	Severity 2 dominates the dataset (85% of rows), so subtle differences among severe crashes can be harder to separate.	TRUE
Sampled Phase 6 validation	Phase 6 uses repeated sampled validation (4 x 4,000 rows) and repeated stability checks (3 x 2,500 rows, 10 bootstraps each) to build a consensus k on a 16 GB machine.	TRUE
DBSCAN parameter sensitivity	Spatial hotspot shapes can shift if eps or minPts changes; corridor labels should be read as data-driven approximations confirmed by centroid inspection.	TRUE

```r
phase_10_rq_coverage <- data.table(
  research_focus = c(
    "Scenario accident archetypes",
    "Spatial hotspot corridors",
    "High-risk severity patterns",
    "County composition differences",
    "Hotspot-to-scenario linkage"
  ),
  evidence_source = c(
    file.path(table_dir, "phase8_scenario_final_names.csv"),
    file.path(table_dir, "phase8_spatial_final_names.csv"),
    file.path(table_dir, "phase8_severity_profile.csv"),
    file.path(table_dir, "phase8_county_profile.csv"),
    file.path(table_dir, "phase8_spatial_scenario_crosstab.csv")
  ),
  ready = c(
    file.exists(file.path(table_dir, "phase8_scenario_final_names.csv")),
    file.exists(file.path(table_dir, "phase8_spatial_final_names.csv")),
    file.exists(file.path(table_dir, "phase8_severity_profile.csv")),
    file.exists(file.path(table_dir, "phase8_county_profile.csv")),
    file.exists(file.path(table_dir, "phase8_spatial_scenario_crosstab.csv"))
  )
)

fwrite(phase_10_rq_coverage, file.path(output_dir, "phase10_research_question_coverage.csv"))
```

```r
knitr::kable(phase_10_rq_coverage, caption = "Phase 10 research-question coverage")
```

Table 67: Phase 10 research-question coverage

research_focus	evidence_source	ready
Scenario accident archetypes	local_artifacts/output/tables/phase8_scenario_final_names.csv	TRUE
Spatial hotspot corridors	local_artifacts/output/tables/phase8_spatial_final_names.csv	TRUE
High-risk severity patterns	local_artifacts/output/tables/phase8_severity_profile.csv	TRUE
County composition differences	local_artifacts/output/tables/phase8_county_profile.csv	TRUE
Hotspot-to-scenario linkage	local_artifacts/output/tables/phase8_spatial_scenario_crosstab.csv	TRUE

```r
phase_10_submission_manifest <- data.table(
  deliverable_group = c(
    "Core source",
    "Core source",
    "Core source",
    "Data",
    "Data",
    "Exports",
    "Exports",
    "Exports"
  ),
  deliverable = c(
    "Li_Clustering_SRS.tex",
    "Li_Clustering_Execution_Plan.tex",
    "li_clustering.Rmd",
    "li_accidents_clean.csv",
    "li_accidents_clustered.csv",
    "Figure directory",
    "Table directory",
    "Phase 10 checklist bundle"
  ),
  path = c(
    "Li_Clustering_SRS.tex",
    "Li_Clustering_Execution_Plan.tex",
    "li_clustering.Rmd",
    input_path,
    file.path(table_dir, "li_accidents_clustered.csv"),
    figure_dir,
    table_dir,
    output_dir
  ),
  ready = c(
    file.exists("Li_Clustering_SRS.tex"),
    file.exists("Li_Clustering_Execution_Plan.tex"),
    file.exists("li_clustering.Rmd"),
    file.exists(input_path),
    file.exists(file.path(table_dir, "li_accidents_clustered.csv")),
    dir.exists(figure_dir),
    dir.exists(table_dir),
    dir.exists(output_dir)
```

```
  )
)

fwrite(phase_10_submission_manifest, file.path(output_dir, "phase10_submission_manifest.csv"))

knitr::kable(phase_10_submission_manifest, caption = "Phase 10 submission manifest")
```

Table 68: Phase 10 submission manifest

deliverable_group	deliverable	path	ready
Core source	Li_Clustering_SRS.tex	Li_Clustering_SRS.tex	TRUE
Core source	Li_Clustering_Execution_Plan.tex	Li_Clustering_Execution_Plan.tex	TRUE
Core source	li_clustering.Rmd	li_clustering.Rmd	TRUE
Data	li_accidents_clean.csv	../li_accidents_clean.csv	TRUE
Data	li_accidents_clustered.csv	local_artifacts/output/tables/li_accidents_clustered.csv	TRUE
Exports	Figure directory	local_artifacts/output/figures	TRUE
Exports	Table directory	local_artifacts/output/tables	TRUE
Exports	Phase 10 checklist bundle	local_artifacts/output	TRUE

```
phase_10_summary <- data.table(
  checkpoint = c(
    "Readiness checklist saved",
    "Report outline saved",
    "Limitations table saved",
    "Research-question coverage saved",
    "Submission manifest saved"
  ),
  status = c(
    file.exists(file.path(output_dir, "phase10_readiness_checklist.csv")),
    file.exists(file.path(output_dir, "phase10_report_outline.csv")),
    file.exists(file.path(output_dir, "phase10_limitations_table.csv")),
    file.exists(file.path(output_dir, "phase10_research_question_coverage.csv")),
    file.exists(file.path(output_dir, "phase10_submission_manifest.csv"))
  )
)

fwrite(phase_10_summary, file.path(output_dir, "phase10_summary.csv"))

knitr::kable(phase_10_summary, caption = "Phase 10 readiness summary")
```

Table 69: Phase 10 readiness summary

checkpoint	status
Readiness checklist saved	TRUE
Report outline saved	TRUE
Limitations table saved	TRUE
Research-question coverage saved	TRUE
Submission manifest saved	TRUE

```r
selected_jaccard_value <- suppressWarnings(as.numeric(selected_jaccard))

phase_10_sanity_audit <- data.table(
  check = c(
    "Scenario names are unique",
    "Scenario labels have no gaps",
    "Spatial labels have no gaps",
    "Selected scenario stability is at least moderate",
    "Largest spatial cluster stays below 60% of all rows",
    "DBSCAN cluster-count target met",
    "DBSCAN noise target met"
  ),
  status = c(
    uniqueN(scenario_final_names$final_name) == nrow(scenario_final_names),
    label_join_check[metric == "Scenario label NAs", value][[1]] == 0,
    label_join_check[metric == "Spatial label NAs", value][[1]] == 0,
    !is.na(selected_jaccard_value) && selected_jaccard_value >= 0.4,
    largest_non_noise_share_pct <= phase7_max_cluster_share_cap,
    uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]) >= 3 &
      uniqueN(df_spatial_clusters$spatial_cluster[df_spatial_clusters$spatial_cluster > 0]) <= 15,
    round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2) >= 5 &
      round(100 * mean(df_spatial_clusters$spatial_cluster == 0), 2) <= 30
  ),
  note = c(
    "Final scenario names should be distinct for presentation and discussion.",
    "Every row should carry a scenario label after the join.",
    "Every row should carry a spatial label after the join.",
    "A Jaccard score below 0.4 should be presented as exploratory rather than settled.",
    "This avoids one hotspot absorbing most of Long Island.",
    "The SRS target is between 3 and 15 non-noise clusters.",
    "The SRS target is between 5% and 30% noise."
  )
)

fwrite(phase_10_sanity_audit, file.path(output_dir, "phase10_sanity_audit.csv"))

phase_10_summary <- rbind(
  phase_10_summary,
  data.table(
    checkpoint = "Sanity audit saved",
    status = file.exists(file.path(output_dir, "phase10_sanity_audit.csv"))
  )
)
fwrite(phase_10_summary, file.path(output_dir, "phase10_summary.csv"))

knitr::kable(phase_10_sanity_audit, caption = "Phase 10 sanity audit")
```

Table 70: Phase 10 sanity audit

check	status	note
Scenario names are unique	TRUE	Final scenario names should be distinct for presentation and discussion.

check	status	note
Scenario labels have no gaps	TRUE	Every row should carry a scenario label after the join.
Spatial labels have no gaps	TRUE	Every row should carry a spatial label after the join.
Selected scenario stability is at least moderate	FALSE	A Jaccard score below 0.4 should be presented as exploratory rather than settled.
Largest spatial cluster stays below 60% of all rows	TRUE	This avoids one hotspot absorbing most of Long Island.
DBSCAN cluster-count target met	TRUE	The SRS target is between 3 and 15 non-noise clusters.
DBSCAN noise target met	TRUE	The SRS target is between 5% and 30% noise.

11 Notes for the next phases

- Phase 2 should stay read-only and focus on profiling the cleaned dataset.
- Phase 3 now derives the main engineered features needed for the scenario and spatial matrices.
- Phase 4 now defines the actual scenario and spatial matrices for clustering.
- Phase 5 now sets up the candidate k-prototypes runs and their core diagnostics.
- Phase 6 now validates the candidate solutions, checks stability, and assigns a selected k.
- Phase 7 now sets DBSCAN parameters, creates the spatial hotspot solution, and assigns corridor-style labels.
- Phase 8 now joins cluster labels back to the working data and builds the interpretation tables.
- Phase 9 now exports the report-ready figures, tables, and labeled dataset.
- Phase 10 now tracks final report and submission readiness.
- `run_profile = "consensus_16gb"` is the current default in this file.
- `run_profile = "local_safe"` is the quickest laptop-friendly option.
- `run_profile = "consensus_16gb"` repeats sampled Phase 6 checks and aggregates them into a consensus choice for 16 GB machines.
- `run_profile = "full_srs"` is reserved for a stronger machine that can support the full Phase 6 diagnostics.
- The current default uses repeated sampled validation and repeated sampled stability checks to stay within 16 GB memory limits.
- Interactive HTML exports are optional and now turn on automatically when the needed packages are available.
- The original full-data Phase 6 validation and stability calls are archived in `eval=FALSE` chunks for later reruns.
- Phase 3 and later should write tables, figures, and exports into `local_artifacts/output/`.
- No `.RData` workspace files are needed for this workflow.

```r
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.4
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versi
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;  LAPACK ve
##
## locale:
## [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
##
## time zone: America/New_York
```

```
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] rmarkdown_2.30      knitr_1.51          factoextra_2.0.0
##  [4] geosphere_1.6-8     ggrepel_0.9.8       ggplot2_4.0.2
##  [7] fpc_2.2-14          cluster_2.1.8.2     dbscan_1.2.4
## [10] clustMixType_0.4-2  lubridate_1.9.5     dplyr_1.2.0
## [13] data.table_1.18.2.1
##
## loaded via a namespace (and not attached):
##  [1] generics_0.1.4     class_7.3-23       robustbase_0.99-7  lattice_0.22-7
##  [5] digest_0.6.39      magrittr_2.0.4     evaluate_1.0.5     grid_4.5.2
##  [9] timechange_0.4.0   RColorBrewer_1.1-3 fastmap_1.2.0      nnet_7.3-20
## [13] mclust_6.1.2       combinat_0.0-8     kernlab_0.9-33     viridisLite_0.4.3
## [17] scales_1.4.0       modeltools_0.2-24  cli_3.6.5          rlang_1.1.7
## [21] withr_3.0.2        yaml_2.3.12        otel_0.2.0         tools_4.5.2
## [25] flexmix_2.3-20     parallel_4.5.2     vctrs_0.7.1        R6_2.6.1
## [29] stats4_4.5.2       lifecycle_1.0.5    htmlwidgets_1.6.4  MASS_7.3-65
## [33] pkgconfig_2.0.3    pillar_1.11.1      gtable_0.3.6       glue_1.8.0
## [37] Rcpp_1.1.1         DEoptimR_1.1-4     xfun_0.56          tibble_3.3.1
## [41] tidyselect_1.2.1   prabclus_2.3-5     farver_2.1.2       htmltools_0.5.9
## [45] labeling_0.4.3     compiler_4.5.2     S7_0.2.1           diptest_0.77-2
```